Oral and Poster Presentations

Successful factors contributed to malaria control and key challenges in Vietnam

Huy Nam Vu[1], Hiroshi Ohara[2]
[1]General Planning Department,
National Institute of Malariology Parasitology and Entomology, Hanoi, Vietnam.,
[2]Bureau of International Cooperation, National Center for Global Health and Medicine, Tokyo, Japan

In Vietnam malaria has been a major health issue with high priority in control strategy. However due to the great efforts of the government to implement control measures, malaria has been successfully controlled. In this study, successful factors in malaria control were analyzed, particularly in relation to health system.

Information obtained from the document reviews, interviews, and field surveys were analyzed from the viewpoint of interface between control program and the health system in accordance with the 6 building blocks of health system (Leadership and Governance, Service delivery, Workforce, Information system, Medical products and technology, and Financing) with special emphasis on good practices and challenges in implementing malaria control program.

Leading good practices include: 1)Strong government commitment for malaria control, 2)National strategy for rural development and intensified education for residents, 3)Effective vertical system from national to village level for malaria surveillance and service delivery, 4)Domestic antimalarial production and high coverage of control measures, 5)Strengthening the capacity of health workers along with mobilization of mass organizations, and 6)Support from international organizations.

Key challenges include: 1)Large number of population at risk in remote and mountainous areas, 2)Uncontrolled seasonal migrants and cross border movement, 3)Lack of human resources for malaria network, and 4)Increasing drug resistance of malaria parasite.

Effective implementation under the strong leadership of National Steering Committee utilizing the existing health system was outstanding. Besides, strengthening of the vertical health program appeared to have a good impact on the general health system, particularly at the primary level.

Prevalence of chloroquine resistant *Plasmodium falciparum* in North Sulawesi, Indonesia

Yamagishi Junya[1], Josef Tuda[2], Ilham Habib Djarkoni[2], Inka Sukarno[2], Visia Vrisca[2], George Eduardo Jacobs[2], Jane Angela Kalangi[2], Patrick Reteng[2], Kyohko Kameyama[3], Ryuichiro Maeda[3], Yutaka Suzuki[4]
[1]Research Center for Zoonosis Control, Hokkaidou University, Sapporo, Japan,
[2]faculty of medicine, Sam Ratulangi University, [3]Obihiro University of Agriculture and Veterinary Medicine,
[4]Graduate School of Frontier Sciences, The University of Tokyo

Introduction: North Sulawesi which locates east part of Indonesia is known as malaria endemic region. Since 2004, in exchange of Chloroquine (CQ) and sulphadoxine-prymethamine, artemisinin combination therapy is used to treat Malaria in Indonesia. For CQ resistance, polymorphism of pfcrt 76T and pfmdr1 86Y were involved. Before the withdrawal of CQ, their prevalence in North Sulawesi were 94% and 95%, respectively. A study in Malawi, it is reported that population of CQ resistant *Plasmodium falciparum* (CQR) was decreased after withdrawal of CQ. Therefore we examined the polymorphisms after the withdrawal of CQ in North Sulawesi.

Outline: We collected 95 peripheral blood from malaria infected patients in 2010. Their genotypes were examined by PCR restriction fragment length polymorphism. The specimens were examined by local medical staffs by microscopy in advance then 55, 29 and 3 were described as Pf, Pv and mix infection, respectively. For *pfcrt* gene, 52 out of the 95 were demonstrated as 76T. No amplicon was detected from the others. For *pfmdr1* gene, 58 were demonstrated as 86Y, 6 were 86N and 31 were no signal.

Discussion: The prevalence of CQ resistance genes was still high in 6 years after the withdrawal of CQ. Unauthorized usage of CQ and/or short period after the withdrawal may affect the observed high prevalence.

Conclusion: Parasites with the CQ resistance polymorphism were 100% in *pfcrt* and 90.6% in *pfmdr1*. Continuous molecular surveillances are highly recommended to understand geographical and temporal dynamics of CQR distribution.

Factors Associated with Non-compliance with Anti-malarial Treatment among Malaria Patients in Palawan, the Philippines

Pilarita T. Rivera, Maria Sonia S. Salamat, Ofelia P. Saniel, Analigaya R. Agoncillo, Kristine Ayessa Elaine B. Coronacion, Julienne Theresa T. Dagdag, Ma. Stephanie C. Matira, Nina Kashka E. Pamintuan, Charles Sherwin M. Soriano
College of Public Health, University of the Philippines, Manila, the Philippines

Background Early diagnosis and treatment remains to be the primary strategy for malaria control and elimination. However, its effectiveness has been threatened by the development of drug resistance, inadequate treatment and non-compliance to the prescribed treatment. This study was done in Palawan, the first of five provinces still having more than 1,000 malaria cases per year, to determine the association of factors with non-compliance to anti-malaria treatment.
Methods Using the Health Belief Model in identifying factors that affect compliance to treatment, a structured questionnaire was developed, and administered to 320 patients diagnosed and treated for malaria in January to October 2010. Descriptive statistics were used to estimate the proportion of non-compliance and associated factors, while logistic regression models were constructed to derive the likelihood of non-compliance.
Results The rate of non-compliance to anti-malaria treatment was 17% (95% CI 12.1-21.2%). Symptom perception as a cue to action and forgetfulness as a perceived barrier to treatment compliance were found to be significantly associated with non-compliance to treatment. The odds of non-compliance were three times higher for individuals who perceived that an improvement in symptoms implied cure of malaria. An individual who forgot to take at least one dose of medication was 17 times more likely to be non-compliant with treatment compared to someone who did not forget to take a single dose.
Conclusion More effective ways of ensuring compliance with anti-malaria treatment may be explored e.g., doing directly observed treatment, utilizing treatment partners and patient education about the consequences of non-compliance.

The potential of *Culicinomyces clavisporus* a mosquito killing fungus to control malaria vector *Anopheles flavirostris* in the Philippines

Elena A. Villacorte[1], Christopher Panter[2]
[1]College of Public Health, University of the Philippines, Manila, the Philippines,
[2]Monash University, Melbourne, Australia

Background In the Philippines, malaria is transmitted by female *Anopheles flavirostris* breeding in slow flowing fresh streams. Besides using insecticide treated bed nets, biological control is another alternative for malaria control. Some biological agents that have been studied and tested for control of mosquitoes include predator fish, bacterial pathogens, a nematode, aquatic plant and fungal pathogens. *Ceolomomyces, Lagenidium, Culicinomyces clavisporus* and *Metarrhizium* are fungi that have been studied as pathogens of mosquito larvae and other dipterans. *Culicinomyces* invades the body cavity of mosquito larvae. Infection of larvae is through ingestion of spores. The hyphae grow in the interior of the larvae and penetrate through the outer integument.
Method Conidiospores of the fungal pathogen of mosquitoes, *Culicinomyces clavisporus* were produced on selective agar. These were tested against larvae of the malaria vector *Anopheles flavirostris* in its breeding sites in streams in Kapalong, Davao del Norte and Angono, Rizal of known malaria endemic provinces. *C. clavisporus* spores of 10^{5}-10^{6} spores/ml were applied in breeding sites and larval mortality was daily monitored.
Result There was clear evidence that the fungus was able to kill anopheline larvae, but the kill was only about 10% of the larval population. Performance of the fungus could have been improved with a formulation of rice husk with glucan binder that could enable the sinking spores to float and be available to the surface feeding larvae.
Conclusion *C. clavisporus* spores in floating formulation can has the potential biocontrol agent against Anopheles mosquitoes.

The new safe NP formulated DNA vaccine encoding PYGPI8 Transamidase related protein induce protection by increasing DCs but not B cell population.

Mahamoud Sama Cherif[1], Mohammed Nasir Shuaibu[1], Yukinobu Kodama[2], Evaristus Chibunna Mbanefo[1], Shusaku Mizukami[1], Akbari Massoud[4], Mihoko Kikuchi[1], Tetsuo Yanagi[3], Hitoshi Sasaki[2], Eduardo Jose Kawasaki[5], Nguyen Tien Huy[5], Juntra Karbwang[5], Kenji Hirayama[1]
[1]Department of Immunogenetics, Institute of Tropical Medicine (NEKKEN), Nagasaki University, Japan, [2]Department of Hospital Pharmacy, Nagasaki University, Japan, [3]National Bio-Resource Center for Tropical Medicine, Nagasaki, Japan., [4]4Division of Immunology, Department of Molecular Microbiology and Immunology, Graduate School of Biomedical Sciences, Nagasaki University, Japan, [5]5Department of Product Development, Institute of Tropical Medicine, Nagasaki University, Nagasaki, Japan

Malaria is a life-threatening tropical disease caused by Plasmodium. According to the World Health Organization (WHO), malaria affected 219 million people around the world in 2010, causing over 600,000 deaths.Compared to natural vaccine vectors, including engineered viruses or attenuated pathogens, nanoscale DNA vaccine formulations are safer, more controlled, and have the potential to be more effective. Previously, we found the augmented protective vaccine effect by PyTAM DNA by using Polyethyleimine(PEI) and &gammaPGA complex as a delivery system. In this study, The new NP containing Dendri Grafted polyLysine (DGL) and &gammaPGA complex, formulated- malaria antigen showed an induction of protective immune responses with significant enhancement of survival and reduction of parasitemia. However the mechanism still not well elucidated. On the other hand, we have observed APC derived cytokines such as IL-12 and TNF&alpha and Th1 derived cytokines (IFN&gamma) were significantly increased as compared to non-coated DNA antigen.Dendritic cells are essential for the proper functioning of the innate and adaptive immune systems through their role in antigen presentation. The new formulation showed a significant increase of both DC populations as compared to mice immunized with NP-formulated empty plasmid DNA. Despite of significant production of specific IgG subtypes, we did not observe any increase of B cell population in the spleen.To conclude, the new NP formulation could significantly raise the protective effect of PyTAM DNA vaccine in murine model, and this enhancement was associated with the increased of IFN&gamma production and increased DC populations.

Timeliness of malaria case reporting in Japan: from first consultation to reports to healthcare centers

ATSUHIRO KANAYAMA[1,2], TAMANO MATSUI[3], KAZUNORI OISHI[3]
[1]Field Epidemiology Training Program, National Institute of Infectious Diseases, Tokyo, Japan, [2]Department of Global Infectious Diseases and Tropical Medicine, National Defense Medical College, [3]Infectious Disease Surveillance Center, National Institute of Infectious Diseases, Tokyo, Japan

[Background] Malaria is a potentially life-threatening disease and should be diagnosed immediately and be reported timely to health-care centers. However, little is known about the timeliness of this process. [Method] In Japan malaria is a notifiable disease requiring an immediate report to health-care centers after diagnosis. Notification criteria include demographic data, type of parasites, dates of initial diagnosis, reporting to health-care centers, and acceptance of the reports by health-care centers. We collected and analyzed data regarding 525 cases reported to the national surveillance system between 2006 and 2014. [Results] Among 525 cases reported between the observation periods, 57% was *Plasmodium falciparum*, 30% was *P. vivax*, 4% was *P. ovale*, and 2% was *P. malariae*. The periods required for diagnosis of 90% of 525 cases were 5 days; 2 days for P. falciparum, 7 days for *P. ovale*, 8 days for *P. vivax*, and 12 days for *P. malariae*. After diagnosis, 90% of the cases were reported to health-care centers within 3 days, and were accepted by health-care centers within additional 3 days. Only 80% were reported immediately, i.e. within one day after diagnosis. [Conclusion] Diagnosis for *P. malariae*, *P. vivax*, and *P. ovale* required more time than that for *P. falciparum*, while some cases were diagnosed as *P. falciparum* more than 3 days after the first consultation. Ideally, these cases should have been diagnosed earlier and more cases should be reported earlier.

More Skills Less Pills: Community-Driven Adoption of Healthy Practices

HIKARI MORIKAWA[1], Masahiro Morikawa[2]
[1]Mercado Global, Panajachel, Guatemala,
[2]University Hospitals Case Medical Center/Case Western Reserve University, Cleveland, USA

Introduction: Mercado Global, a social enterprise NGO in New York, provides business training to Mayan women in Panajachel, Guatemala. The improved skills have helped increase their income through global market and enabled to invest for future. Case Western Reserve University, Dept. of Family Medicine, was asked to help the indigenous women stay healthy to continue working. Overview: The interviews with the artisans and physical examinations revealed: 1. their belief on their health conditions does not necessarily match with the result of health examinations, 2. Any aches and pains are amenable to simple physical exercises and behavioral modifications even though they strongly believe pills can take care of them, and 3. Some of the pains and anxieties are manifestations of pervasive psycho-social stresses and trauma. In response to the specific needs of the population, we have implemented a program to promote community-driven adoption of healthy practice, entitled *More Skills Less Pills* in two pilot villages since 2012.Discussion: The program contains 3 components: 1) Pain Away: a set of exercises for Musculoskeletal pain, repetitive wrist motion and dysfunction, 2) Power To Change: 10-step activities for group problem-solving skills and positive mind, and 3) Do-It-Together Health Education: hands-on learning activities to acquire healthy behaviors. Learning1. Poverty reduction and health promotion are indispensable to each other.2. Health programs should be developed based on health examination and observation of the most prevalent problems in the community.3. Health care professionals should be able to develop learning programs specific to the local context.

Impact of introduction of rural community water supply in Western Kenya: Who receives the benefits?

KAYAKO SAKISAKA[1,2], Evans Asena Chadeka[3], sachiyo Nagi[4], Dorcas SShambi Mwandembo[5], Masamine Jimba[2]
[1]Faculty of Policy Studies, Organization for Common Education, Chuo University, Tokyo, Japan,
[2]Department of Community and Global Health, Graduate School of Medicine, the University of Tokyo,
[3]Nagasaki University Kenya Research Station, Nairobi, Kenya,
[4]Graduate School of Biomedical Science Medicine, Nagasaki University, Nagasaki, Japan,
[5] Department of Social Services, Ministry of Gender, Children and Social Development, Nairobi, Kenya

Background: Access to safe water and sanitation is poor in rural Kenya. We examined the impact of introducing community tube wells in rural Western Kenya, where a cholera outbreak was reported in 2007 and 2008.Methods: We conducted a pre (2009) and post intervention (2011), cross-sectional study using structured questionnaires for mothers.Results: In total, 1391 (576 in 2009 and 815 in 2011) mothers participated. The introduction of wells changed the participants' choice of drinking-water source from river water to ground water. Distance to the water source and collection time reduced 40% to50% (median distance from 500.0 to 300.0 m; median proximity from 30.0 to 15.0 minutes); water consumption per household increased approximately 20% (from 82.6 to 99.0 L); non-treatment of water before drinking decreased about 10%. The prevalence of child diarrhoea may also have been reduced. Regression analyses revealed mothers with higher monthly incomes had the highest satisfaction levels (beta = 0.167, $p < 0.001$). Positive impacts were strongest among wealthier groups.Conclusions:The small-scale community water supply dramatically affected users' hygiene behaviours and daily life. Water supply levels improved from 'No Access' to 'Basic Access'. Household connections to a water supply and hygiene education are the next steps to enhancing community health.

Human capital and social capital among people in a community-based health promotion project in La Paz, Bolivia

MOTOYUKI YUASA[1], YOSHIHISA SHIRAYAMA[1], KEIICHI OSATO[2],
TOSHIHIRO NAKAJIMA[3], FREDDY ILUSCO[3], SONIA LUJAN[3]
[1]Department of Public Health, Juntendo University Graduate School of Medicine, Tokyo, Japan,
[2]Human Development Department, Japan International Cooperation Agency, Tokyo, Japan,
[3]JICA project of FORSA La Paz, La Paz, Bolivia

[Introduction] In 2012, the Japan International Cooperation Agency launched a project to strengthen the health system in La Paz, Bolivia. The project has technically supported the implementation of community-based health promotion. This study aimed to examine the association of participation in the health promotion activities with human capital and social capital among people at the project and control sites.[Method]We randomly selected 200 participants from 20 communities where the project has been carried out as an intervention arm and 200 participants from 20 communities in non-project sites as a control arm. We interviewed both groups using a structured questionnaire. We scored general self efficacy (GSE) as human capital; and frequency of participation in community activities, general trust, and perceived social solidarity as social capital. Analyses were performed by a student t-test and a chi-square test. A protocol of the study was approved by ethical committees of Juntendo University and La Paz Medical Association. [Results]We analyzed a total of 400 participants and found that there were no significant differences between the intervention and control arms in teams of a mean age, sex and educational attainment. The study revealed that GSE in the project arm was significantly higher than that in the control arm, and social capital among people in the intervention arm were also significantly different from those among people in the control arm.[Conclusion]These findings suggest that human and social capital of people at the intervention sites be associated with participation in the health promotion activities.

The status and problems of occupational therapy toward foreigners in Japan

YUKIKO KAMIMURA[1], RYUJI KOBAYASHI[2], ISAMU MATSUDA[3]
[1]Department of Occupational Therapy, Graduate School of Health Science Studies, Kibi International University, Okayama, Japan., [2]Div.of Occupational Therapy, Faculty of Health Sciences,Tokyo Metropolitan University, Tokyo,Japan, [3] Graduate School of Health Science, Kibi International University, Okayama, Japan.

Background; The purpose of this study is to clarify the current status and problems on the occupational therapy toward foreigners in Japan. Methods; An anonymous survey was conducted for the occupational therapists in 169 institutions (municipality hospitals and home visit nursing stations) where the number of foreign residents are over 10000 people based on statistics of the Immigration Breau.Results;The survey was obtained from 180 occupational therapists, among whom 148 occupational therapists have provided experience with occupational therapy toward 220 cases and 32 nationalities. At the time of providing occupational therapy which they feel about any problems, what is important when they perform occupational therapy, required knowledge, language and non-verbal communication).Moreover, the reply was obtained that we could get knowledge about we think their goal which demand by patients are different as each national character, regard for consciousness, the position of the rehabilitation at each countries, and how is the rehabilitation at each countries. Conclusion;It was related to language as the most common problem. Practical linguistic talent would be future subjects in medical front. Actually, 80 percent occupational therapists have provides experience as survey and to focus the occupation, which treats in consideration of the structure and the meaning of work, occupational therapists must be consideration for the cultures of each patients while comparing with occupational therapist's culture, how devise a method of patient's culture in order, the significance of disease is given in what kind of from. The problem, we need more investigation and understanding the process.

Team rehabilitation for landmine survivors in Colombia

Kyo Takahashi[1,2], Masamine Jimba[1], Tsutomu Iwaya[2], Yoshiko Tobimatsu[2], Satoshi Nakadomari[2]
[1]Department of Community and Global Health, Graduate School of Medicine, The University of Tokyo, Tokyo, Japan.,
[2]National Rehabilitation Center for Persons with Disabilities, Saitama, Japan.

Introduction: Assistance to landmine survivors is a challenge in Colombia. From 2008 to 2012, JICA conducted the project 'Strengthening the integral rehabilitation for persons with disabilities, especially for landmine survivors' in Valle del Cauca and Antioquia. We report the project's achievements and challenges.
Outline: The purpose of project was improving the quality of rehabilitation for persons with disabilities including landmine survivors. Under the cooperation with governmental organizations, prefectural health offices, university hospitals and medical NGOs, professionals' capacities were developed and rehabilitation guidelines were standardized. In particular, more than 100 professionals were given training in team rehabilitation which aimed to improve activities of daily living, and 26 among them were dispatched to Japan for deeper understanding. Trained professionals diffused what they learned about team rehabilitation to their colleagues and over 1,500 professionals in community-based institutions in Valle del Cauca and Antioquia.
Discussion: Thanks to trained professionals' commitment, a series of trainings in team rehabilitation are continuously provided in Valle del Cauca and Antioquia. Team rehabilitation is expected to spread out all over Colombia with the support by governmental organizations. On the other hand, the project did not conduct sufficient activities in welfare assistance after rehabilitation. Community-based welfare assistance focusing on education, livelihood, social and empowerment is still insufficient for landmine survivors' social participation.
Conclusion: Through the project, team rehabilitation for landmine survivors spread out in Valle del Cauca and Antioquia. Diffusing team rehabilitation to all over Colombia and strengthening the welfare assistance after rehabilitation are required.

Integrating STI/HIV indicators into general health information system in Madagascar

TOSHIHARU OKAYASU[1,2], Jean Luis Razafimahatratra[3], Harisoa Julie Norovahangy[3],
Jacky Rahamefy[4], Saori Kashima[5], Kae Furukawa[6]
[1]JICA Benin, Cotonou, Benin, [2]Department of Healthcare Policy and Management, Graduate School of Comprehensive Human Sciences, University of Tsukuba, [3]Health Statistics Service, Ministry of Public Health, Antananarivo, Madagascar, [4]National AIDS Program, Ministry of Public Health, Antananarivo, Madagascar, [5]Department of Public Health and Health Policy, Institute of School of Biomedical and Health Science, University of Hiroshima, Hiroshima, Japan, [6]JICA Burundi, Bujumbura, Burundi

[Introduction] This study aimed to introduce an approach of health system strengthening (HSS) for integrating STI/HIV indicators in Madagascar. Like other Sub Sahara African countries, STI/HIV indicators in Madagascar were independently collected by the national AIDS program of Ministry of Health up to 2009. The minimum activities package for counseling and testing for HIV was installed and integrated into other services of Basic Health Centers (CSB). Then, the number of HIV testing sites had drastically increased from 314 in 2005 to 815 in 2009 among 3,240 health facilities. The reporting proportion of HIV indicators, however, had never exceeded 47% per month. The double counting of STI/HIV indicators was also a problem.
[Outline] Since 2010, the Ministry of Health has integrated indicators of STI/HIV into the collection system through Reports Monthly Activity (RMA) in collaboration with JICA HIV Prevention Strengthening Project. In 2010 and 2011, we updated the database software for RMA and gave training to health officers in 22 regions and 112 districts and hospital statistics officers.
[Discussion] The proportions of complete and timely reporting at the level of CSB in 2011 improved to 98% and 73%, respectively from those in 2009. Those were the best figures since 1998. And those proportions in 2011 at the district/regional hospitals levels were also the best. The proportion of inconsistencies for STI/HIV indicators also declined.
[Conclusion] Although reporting completeness and timeliness much improved, the quality of PLWHIV cohort and drug stock data are yet to be improved.

HIV data quality improvement intervention study in Antananarivo Renivohitra and Antananarivo Atsimondrano districts, Madagascar

TOSHIHARU OKAYASU[1,2], YUKA IINO[3], SUSUMU TANIMURA[4], Jean Luis RAZAFIMAHATRATRA[5,]
Harisoa Julie NOROVOAHANGY[5], Jacky RAHAMEFY[5], KAE FURUKAWA[6], MASAHIDE KONDO[7]
[1]JICA benin, Cotonou, Benin, [2]Department of Healthcare Policy and Management, Graduate School of Comprehensive Human Sciences, University of Tsukuba,Tsukuba, Japan , [3]JICA Senegal, Dakar,Senegal,
[4]Department of Public Health, Hyogo College of Medicine,Hyogo, Japan,
[5]5Ministry of Public Health, Antananarivo, Madagascar, [6]JICA Burundi, Bujumbura, Burundi,
[7]Department of Health Care Policy and Health Economics, Faculty of Medicine, University of Tsukuba, Tsukuba, Japan

Background JICA HIV Prevention Strengthening project has integrated STI/HIV indicators in routine Monthly Activity Report (RMA) and updated RMA database software in 2010. As a result, the proportions of completeness and timeliness have recorded the best figures in 2010 since 1998. However, the quality of such data has not been evaluated yet. The objectives in this study were to evaluate the quality and find better approach to reduce error events in HIV and other indicators.
Methods In this intervention study, 57 health facilities(43 publics, 14 privates) providing HIV testing in two districts (Antananarivo Renivohitra and Antananarivo Atsimondrano) in Analmanga region were randomly assigned to a control group and 3 interventions groups: (G1) distribution of annotated RMA, (G2) supervision, (G3) training. For 75 indicators of STI /HIV and others indicators in RMA with three categories of errors such as omission(47), incoherence(20) and miscalculation(8), the frequency of errors in each group has been compared before (January-March 2011) and after intervention (October-December 2011).
Results We found the type of errors as follows: 22.6% for omission, 4.6% for incoherence and 5.3% for miscalculation. The most errors were observed in pharmacy stock for test kits and in counselling and testing for high risk populations. The average reduction of errors per month in G1, G2, G3, and the control group were 4.7, 3.4, -0.3 and -0.9, respectively however the difference was not statistically significant with ANOVA test.
Conclusion The distribution of annotated RMA tended to prevent errors more than other conventional methods, though the difference was not statistically significant.

Keys for successful implementation of Human Resource For Health Information Systems, Experiences from Tanzania

HISAHIRO ISHIJIMA[1], Mapunda Martin [2], Mndeme Mathew[3], Emiko Nishimura[4], Shizu Takahashi[5]
[1] Japan International Cooperation Agency (JICA) expert, assigned to HRH Development Project, Ministry of Health and Social Welfare, United Republic of Tanzania, [2]HRH Planning Section, Devision of Human Resource Development, Ministry of Health and Social Welfare, United Republic of Tanzania, [3]Department of Computer Science, University of Dar es Salaam, [4]Japan International Cooperation Agency (JICA), Operations Strategy Department, [5]Former JICA expert to HRH Development Project

Introduction: Tanzania is one of the 56 countries, which have critical shortages of human resources for health (HRH). In 2008, there was an overall shortage of 65% of human resources in Tanzanian health sector. Effective HRH information systems were needed for evidence-based human resource planning and management. Outline: Since 2008, Tanzanian Ministry of Health and Social Welfare (MOHSW), with support from the Japan International Cooperation Agency (JICA), has developed and rolled out two complementary systems to improve the HRH planning and management: HRH Information System (HRHIS) and Training Institution Information System (TIIS). As of June 2014, HRHIS has been rolled out to all the local governments and national hospitals and data are collected from both public and private health facilities. TIIS has been rolled out to all the 154 training institutions, both public and private. Data collection rate is remarkably high: 98.4% in public health facilities, 80.9% in private facilities, 100% on teachers and 75% on students in training institutions. Discussion: Key factors are identified for successful rollout of the two information systems: 1) Systems were designed to support daily routine work of the local and central government officials, 2) Local resources were fully utilized in system development, rollout and maintenance, 3) Capacity building of users on data utilization was incorporated for sustainable operation. Challenges remain in coordination with other information systems in other ministries, and human and financial resources for system maintenance.Conclusion: Tanzanian experiences may be applied to other countries in developing and rolling-out human resource information systems.

Relationship between Multicultural policy and Health Care Interpreter Services of New South Wales State, Australia

Sangnim Lee, Takashi Sawada
Services for the Health in Asian & African Regions (SHARE)

Background: In New South Wales (NSW) State, Health Care Interpreter Services (HCIS) is available free of charge in public healthcare system. This study aims to examine HCIS and Multicultural policy of NSW, and to provide implication for healthcare interpreter development in Japan
Methods: Documents about Multicultural policy and HCIS published by Commonwealth and NSW Governments were reviewed. Pub Med was used for literature search using terms including Australia, multicultural, health, policy.
Results: NSW had `Community Relations Commission (CRC) and Principles of Multiculturalism Act 2000 (hereinafter called Multicultural law)'. Its six principles included, such as respecting diversity of people and providing access to Government services for all individuals. CRC had a role in monitoring public agencies' actions under this law.
Policy Directive of Standard Procedure for HCIS published by NSW Department of Health explained the situations in which interpreters must be used for people from non-English speaking backgrounds in public healthcare system. Legal evidence was based on Multicultural law: Responsibility of the healthcare system to respond to culturally diverse people's needs, and Anti-Discrimination Act: Equal access to healthcare services.
Conclusion: Understanding Australia's interpreter services and related laws would be helpful for making Japanese healthcare services accessible to foreigners with limited Japanese. NSW's Multicultural law played a vital role in directing HCIS policy for equitable access to health services for people from Non-English speaking backgrounds. It was also identified CRC could require that public health agencies develop protocols for language assistance, and that health personnel utilize these assistance appropriately. These political approaches would help promote healthcare interpreter use.

Promoters and Hindrances to Healthy Diet and Exercise among Immigrant Brazilians: A Qualitative Study

MARIE TASHIRO[1], Masamine Jimba[2], Poudel Krishna[3]
[1]Department of Nursing, Baika Women's University,
[2]Department of Community and Global Health, Graduate School of Medicine, The University of Tokyo,
[3]School of Public Health and Health Sciences, University of Massachusetts, Amherst

Background Immigrants to Japan are one of the vulnerable groups to having chronic diseases. Healthy diet and exercise are the keys to prevent metabolic syndrome and its related diseases. This study focuses on immigrant Brazilians who are a major immigrant group in Japan. We explore promoting and hindering factors to their healthy habits in their lives. **Methods** We conducted eight focus group sessions with total 42 participants recruited from a health checkup program for immigrant Brazilian men. Verbal data were recorded and transcribed into texts. We managed the texts in Weft QDA and analyzed them by using thematic analysis. Systematic coding was done, and we sorted codes into three major categories. Promoting and hindering factors were divided into sub categories as personal or environmental factors suggested by Social Cognitive Theory model. **Results** For healthy diet and exercise, participants found personal promoting factors as attending health checkups, sense of improving health and expectation of better personal appearance. They felt environmental promoting factors as partner's support in diet. For hindrances, they described personal factors as heavy labors, living alone, feelings of discrimination in public space, and anxiety of finance. They mentioned environmental hindrance as lack of health information and lack in sense of location. **Conclusion** Some promoting factors such as health checkup participation are widely encouraged in Japan, which is not necessarily accessible for immigrants. Hindering factors including felt discrimination are specific to immigrants. We suggest community approach in collaboration with government, companies and citizens to solve hindrances to improve immigrants' health.

A study of the intercultural adaptation and mental health of Indonesians coming to Japan under the Economic Partnership Agreement

FUMIKO SATO
Senri Kinran University, Faculuty of Nursing

Purpose The present study aimed to gain an understanding of the state of intercultural adaptation and mental health of Indonesians coming to Japan under the EPA, while also clarifying factors that affected subjects' mental health. Methods From November 2012 through March 2013, a self-administered survey was mailed to medical and care facilities throughout Japan that openly accept Indonesians under the EPA. The adaptability scale by Uehara (1992) was modified and used to investigate intercultural adaptation. Mental health was measured using the General Health Questionnaire (GHQ28). Factors affecting mental health status were subjected to multiple regression analysis with the mental health score as a dependent variable and all other items as independent variables. Results The survey was distributed to 87 facilities throughout Japan. Responses were received from 71 subjects at 35 facilities (facility response rate: 40.2%). For intercultural adaptation, when organized according to subscale, interpersonal relationships were found to have the highest degree of maladjustment, followed by culture, training and research, ease of living and economics, and mental and physical health and emotions. With regard to mental health, 55 (77.5%) subjects had no particular problems, while 16 (22.5%) had some sort of problem. The results of multiple regression analysis indicated that overall intercultural adaptation score, sex, and age were significantly related ($R = 0.696$, $R2 = 0.484$). Conclusions The results revealed the state of intercultural adaptation and mental health in Indonesians who came to Japan under the EPA. We also found that overall intercultural adaptation score, sex, and age affected mental health status.

Immunogenicity and anti-fecundity effect of nanoparticle coated SjGST DNA vaccine against murine Schistosoma japonicum infection

Evaristus C. Mbanefo[1], Takashi Kumagai[2], Yukinobu Kodama[3], Rieko Furushima-Shimogawara[2], Sama M. Cherif[1], Tien-Huy Nguyen[4], Nobuo Ohta[2], Hitoshi Sasaki[3], Kenji Hirayama[1]
[1]Department of Immunogenetics, Institute of Tropical Medicine (NEKKEN) and Global COE Program, Nagasaki University, 1-12-4 Sakamoto, 852-8523, Japan, [2]Section of Environmental Parasitology, Tokyo Medical and Dental University Graduate School of Medical and Dental Science, 113-8519, Japan, [3]Department of Hospital Pharmacy, Nagasaki University Hospital, 1-7-1 Sakamoto, 852-8501, Japan, [4]Department of Clinical Product Development, Institute of Tropical Medicine (NEKKEN), Nagasaki University, 1-12-4 Sakamoto, 852-8523, Japan

There is still urgent need for a vaccine against schistosomiasis, especially in Schistosoma japonicum endemic areas where even a vaccine that will interrupt zoonotic transmission will be potentially effective as an intervention tool. We had developed a novel nanoparticle gene delivery system which has proven efficacious in gene transfection to target immune cells with complementary adjuvant effect and high protective efficacy in several diseases. Here, we have applied this nanoparticle system in combination with S. japonicum glutathione S-transferase (SjGST) DNA vaccine to show improved immunogenicity and anti-fecundity effect of the nanoparticle coated vaccine formulation against murine schistosomiasis. The nanoparticle coated DNA vaccine formulation induced desired immune responses with significantly increased humoral response, T-helper 1 polarized cytokine environment, higher proportion of IFN-&gamma producing CD4+ T-cells and the concomitant decrease in IL-4 producing CD4+ T-cells. Although there was no effect on worm burden, the proportion of immature worms was higher in the SjGST vaccinated groups with a lower proportion of mature paired worms. We recorded an unprecedented reduction in tissue egg burden. There was 71.9% decrease in liver egg burden, 71.6% decrease in intestinal egg burden, and 54.7% reduction in the fecundity of female adult worms. Our data showed that the combination of nanoparticle gene delivery system with SjGST DNA vaccination significantly improved the characteristic anti-fecundity effect of SjGST, thereby proving this DNA vaccine formulation as a promising candidate for anti-pathology and transmission blocking application.

Exploring skin diseases and their impact on children's lives of Cote d'Ivoire: baseline field study for early detection and treatment of leprosy and Buruli ulcer

RIE YOTSU[1,2], Norihisa Ishii[3], Julien Ake[4], Rigobert Abe[5]
[1]Department of Dermatology, National Center for Global Health and Medicine, Tokyo, Japan, [2]National Suruga Sanatorium; Gotenba-shi, Shizuoka, [3]Leprosy Research Center, National Institute for Infectious Diseases; Higashimurayama-shi, Tokyo, [4]MAP International West Africa; Abidjan, Cote d'Ivoire, [5]National Leprosy Control Programme, Ministry of Health, Abidjan, Cote d'Ivoire

[Introduction] Leprosy and Buruli ulcer (BU) are infectious diseases caused by mycobacteria of different species. The infections feature skin changes and ulcerations. Both diseases are endemic in Cote d'Ivoire, a developing country in sub-Saharan Africa. Early case detection and treatment are necessary to prevent severe and life-long disabilities. In BU, children aged below 15 years are particularly at highest risk. Besides the two diseases, various other skin diseases exist in children in the country ranging from fungal infection to eczema. However, its distribution is not yet well known. [Outline] School-based skin survey is a project to diagnose and treat children with skin diseases at early stage. We performed a baseline field study in May 2014 for implementing school-based skin survey in Cote d'Ivoire. [Discussion] Tinea captis, scabies, and general eczema were the major skin diseases observed in children in the study site. Due to its appearance, some children experienced stigma and discrimination. Many children were taken to traditional healers by their parents for their treatment. The level of knowledge for both leprosy and BU was low, and stigma and discrimination was significantly more than what was for other skin diseases. The study also implied on possible undiagnosed BU cases in the community.[Conclusion] The study found that the school-based skin survey has a potential to have an impact on leprosy and BU control programmes in Cote d'Ivoire, if implemented in the coming future.

Distribution of *Lutzomyia ayacuchensis*, the vector of Andean-type cutaneous leishmaniasis, at different altitudes on the Andean slope of Ecuador

HIROTOMO KATO[1], Eduardo A. Gomez[2], Tatsuyuki Mimori[3], Yoshihisa Hashiguchi[4,5,6]
[1]Laboratory of Parasitology, Graduate School of Veterinary Medicine, Hokkaido University, Japan, [2]Departamento de Medicina Tropical, Facultad de Medicina, Universidad Catolica de Guayaquil, Guayaquil, Ecuador, [3]Department of Microbiology, Faculty of Life Sciences, Graduate School of Health Sciences, Kumamoto University, Japan, [4]Centro de Biomedicina, Universidad Central del Ecuador, Ecuador, [5]Prometeo, Secretaría Nacional de Educacion Superior, Ciencia, Tecnologia e Innovacion (SENESCYT), Quito, Ecuador, [6]Department of Parasitology, Kochi Medical School, Kochi University, Japan

[Introduction] Distribution of the vector species is a major risk factor for the endemicity of leishmaniasis. In the present study, the vertical distribution of *Lutzomyia (Lu.) ayacuchensis*, the vector of *Leishmania (Leishmania) mexicana* in the Ecuadorian Andes was surveyed at different altitudes (300-2,300 m above sea level) of the Andean slope. [Outline] The vector species *Lu. ayacuchensis* was identified at an altitude of 650 m and a higher areas, and higher distribution ratio of the species was observed at higher altitudes. In addition, high ratios of *L. (L.) mexicana* infection were detected in higher areas, but none in lower populations of sand flies. Since an association between sand fly populations and vector competence is suggested in *Lu. ayacuchensis*, haplotype analysis was performed on the species from different altitudes of the study areas; however, no apparent difference was observed among populations. [Discussion and Conclusion] The present study identified a vertical distribution of sand flies through an Andean slope, and the distribution range of *Lu. ayacuchensis*, the vector of *L. (L.) mexicana* was determined. No marked genetic divergence found in Lu. ayacuchensis *Lu. ayacuchensis*populations between Andean highland and slope areas suggests that *Lu. ayacuchensis* in Andean slope areas has potential to transmit *L. (L.) mexicana*.

Visceral leishmaniasis in Bangladesh: active survey with urine samples

MAKOTO ITOH[1], BUNPEI TOJO[2], EMI OGASAWARA[2], FUMIAKI NAGAOKA[1], YASUTAKA OSADA[3], EISEI NOIRI[2]
[1]Dept. of Microbiol. and Immunol., Aichi Medical Univ., Nagakute, Japan,
[2]Department of Nephrology and Endocrinology, The University of Tokyo,
[3]Dept. of Molecular Immunol., School of Agricultural and Life Sciences, The University of Tokyo

Introduction:Leishmaniasis, a zoonotic infectious disease, is one of the NTDs (Neglected Tropical Diseases) transmitted by sand flies. At least 300,000 people suffer from the most dangerous Visceral leishmaniasis (VL) every year, and 90% of them are from Sudan, Bangladesh, India and Nepal: at least 50,000 new VL patients are estimated in Bangladesh. However the actual situation of the infection has not been clearly disclose because of little active surveys. Outline:We developed a sensitive and specific ELISA which detect *Leishmania* specific antibodies in urine samples. It is useful for surveys of young age groups as the sample collection is non-invasive. We surveyed school children in VL endemic areas at Trishal in Mymensingh, Bnagladesh and showed 10 times more antibody positives than hospital based incidence of the disease.Discussion:Our study revealed that the positive rate of antibody positives is much higher than incidence of patients. As the antibody positives were asymptomatic, they are suspected as (1) persons at a latent period, or (2) persons infected but no VL development. They may be reservoirs of *Leishmania* parasites: a target of the VL control program.Conclusion:Active surveys for VL were conducted with school children in the endemic areas in Bangladesh and higher antibody positive rate were found than hospital based incidence of VL. Follow-up study of the antibody positives will contribute to elucidate mechanisms of VL development and to find out reservoirs of the parasites.

Japanese priority and challenge: How can we contribute to achieve the 2020 goal for control/elimination/eradication of NTDs?

MIDORI MORIOKA, Fumiko Hirabayashi, Haruki Yamada
NPO DNDi Japan

Background: Neglected tropical diseases (NTDs), which is a group of diseases caused by specific virus, protozoa, helminth and bacteria, affect an estimated one billion people in countries where poverty is prevalent. These diseases induce not only health problem, but also economic and social burden of patient and community. Global health agenda lately shade light on fighting against NTDs toward 2020 goal for control/elimination/eradication of NTDs. This study aims to clarify a Japanese priority and challenge on NTDs control to align with these global efforts.
Methods: Literature review was conducted to obtain the Japanese ODA policy and global health strategy. One project was selected to analysis the evidence of Japanese effort related to NTDs control.
Results: Japanese ODA policy emphasizes on self-reliance and ownership of recipient country. Through the technical cooperation, Japanese global health strategy promotes an approach to develop human resource and enhances the problem-solving capacity of recipient country at individual and organizational level. Global effort to control parasitic diseases, which is currently integrated into the agenda of NTDs, made a progress from the first global initiative toward parasite control proposed by Japan. The Asian Center of International Parasite Control (ACIPAC) project disseminated school health approach through the capacity development of human resource and outcome was showed in some literatures.
Conclusion: Japan has a priority on enhancing the human resource to delivery health services. This priority might strengthen the current Japanese policy focused on dissemination of Japanese knowledge and technology. Further data collection and analysis is on-going.

iNTRODB: database for NTDs drug target protein selection

TAKASHI ISHIDA[1], Kozue Hasumi[2], Yutaka Akiyama[1,3]
[1]Grad School of Inform Sci and Eng, Tokyo Institute of Technology, Tokyo, Japan,
[2]Dept of Comp Science, Tokyo Institute of Technology, Tokyo, Japan,
[3]Ed Academy of Comp Life Sci, Tokyo Institute of Technology, Tokyo, Japan

We are now working on a collaborative research for drug discovery of neglected tropical diseases (NTDs) with several organizations including Astellas pharma inc., the University of Tokyo and Ngasaki University and so on. Our drug discovery employs structure-based drug design (SBDD), which uses tertiary structural information of a protein for in silico screening and drug compound optimization. SBDD enables efficient drug discovery and design but requires selection of a protein that is a target of drugs as the first process. This drug target protein selection is an important process in SBDD because we have to pay more efforts at the following processes for inappropriate targets. To help the selection of drug target proteins, we have developed an integrated database system named iNTRODB. Currently, the iNTRODB is mainly covering the NTDs caused by parasitic protozoa including leishmaniasis, Chagas' disease and Human African trypanosomiasis. The system is developed for searching target proteins suitable for SBDD, and provides structural information of the proteins obtained from the PDB as well as chemical information from ChEMBL. In this presentation, we are going to explain the details of the system, especially the novel functions about biochemical pathways, and discuss about drug target protein candidates selected by using the system.

The impact of user training and medical equipment maintenance on health infrastructure management -Lessons learnt from the Project on Improvement of Health Service through Health Infrastructure Management in the Republic of Uganda-

MASUMI OKAMOTO[1], NAOKI TAKE[2], YASUHIRO HIRUMA[3], NAOKI MIMURO[3],
SITRA MULEPO[4], AMONE JACKSON[4], SONOKO TAKAHASHI[1], ARIMI MITSUNAGA[1]
[1]Japan International Cooperation Agency, [2]Kaihatsu Management Consulting. Inc.,
[3]International Techno Center Co., Ltd., [4]Ministry of Health, the Republic of Uganda

<Introduction>Problems such as inappropriate use of medical equipment and low awareness of the importance of maintenance remain in public health facilities in Uganda. A survey conducted in 22 health facilities indicated that the percentage of medical equipment "In good condition and in use" was only 41.4%(2008). The Ministry of Health, Uganda and JICA undertook a technical cooperation project with the aim of assessing the impact of user training and medical equipment maintenance on health infrastructure management.
<Outline>16 nursing officers from target health facilities were trained six times on 26 kinds of priority medical equipment and they carried out trainings in each facility. Additionally, the Project provided training on medical equipment maintenance for technicians. As a result, the percentage of equipment "In good condition and in use" in the target health facilities raised from 43.1% (2008) to 61.9% (2014). The impact assessment of the Project showed that there was statistically significant increase in equipment "In good condition and in use" among health facilities where training of users was conducted. ($p<0.05$)
<Discussion>This study revealed that infrastructure management was improved through user training and medical equipment maintenance. It is presumed that training may have great impact on the reduction in the number of unused equipment and failure of medical equipment.
<Conclusion>Providing health workers with simple techniques for appropriate use and daily maintenance can increase functionality of medical equipment in health facilities. This result is also practical when considering the possibility of utilization of health technologies developed by Japan's health industry in developing countries.

Improvement of Staff Motivation, Waiting Time and Patient Satisfaction through 5S Practice in Hospitals of Uganda

NAOKI TAKE[1], SARAH BYAKIKA[2], HIROSHI TASEI[3], TORU YOSHIKAWA[4]
[1]Kaihatsu Management Consulting, Inc., [2]Ministry of Health, Republic of Uganda,
[3]International Techno Center Co., Ltd., [4]The Institute for Science of Labour

Introduction: JICA commenced a project in 2011 to support the improvement of quality of healthcare through 5S practice in hospitals in Uganda. The extent of improvement was assessed as a result of three-year project intervention.
Methods: A project scenario, to improve patient satisfaction as well as waiting time of patients through higher staff motivation as a result of 5S practice, was assessed. Staff motivation, waiting time and patient satisfaction were measured in 13 Regional Referral Hospitals (RRH) and 8 General Hospitals (GH) implementing 5S in February-March 2012 - 2014. 10 RRH implementing 5S were compared with 3 without implementation. As for the 8 GH, 2 GH intervened by the Project were compared with 6 without intervention.
Results: [RRH] "Satisfaction with work environment" and "commitment to work in the current hospital" were significantly higher in 10 RRH implementing 5S practice. Waiting time in the dispensary of 10 RRH implementing 5S was significantly improved. However, significant difference was not identified on patient satisfaction.
[GH] Project intervention of 5S realised significant difference of the patient satisfaction from various aspects. Waiting time was significantly improved more in OPD of 2 GH with the intervention. However, the intervention of only 5S could not sustain the staff motivation in the 2 GH.
Conclusion: The results provided two insights: (1) it takes at least 4 years to improve patient satisfaction through 5S practice in Uganda, (2) the 4th year since commencement of 5S can be a threshold to move forward to KAIZEN to maintain the staff motivation.

Improvement of IPC through KAIZEN approach, experiences from Tanzania

HISAHIRO ISHIJIMA[1], NORIYUKI MIYAMOTO2, KIWELU HAMPHREY[3]
[1]JICA Expert assigned to HRH Development Project, JICA/Ministry of Health and Social Welfare,
United Republic of Tanzania, [2]Fujita Planning Co., LTD., [3]Mbeya Consultant Hospital

[Introduction]Ministry of Health and Social Welfare (MoHSW) has been introducing Infection Prevention Control (IPC) strategy for quality improvement of health care at all levels of hospital. However, segregating and disposing of medical wastes is one of the major issues in IPC at health facility.[Outline]JICA HRH Development Project has been introducing 5S-KAIZEN-TQM Approach, Japanese style of management, into national hospitals since 2007 in collaboration with MoHSW. The purpose of introducing the approach is to improve working environment as well as quality of health care. Mbeya Consultant Hospital is a pilot hospital of KAIZEN approach, and currently 13 KAIZEN cases have been practiced. One of the KAIZEN cases from IPC section made great success of solving medical waste segregation and disposing problem. [Discussion]IPC office identified several contributing factors of the problem such as waste bin liners in wrong waste bin, waste bin without a bin liner, mixing up of waste etc. The KAIZEN case reduced the number of frequency of the contributing factors from 155 to 22. Additionally, as an intangible effect of the KAIZEN case, number of rolls procured for making bin liners was also decreased from 18 to 9, which equivalent from 12,160,440 to 6,080,220 Tanzania Shillings. The KAIZEN case succeeded 50 % reduction of the cost of waste segregation in the hospital.[Conclusion]Sharing the success of this KAIZEN case to other hospitals, and implementation of countermeasures in the public hospitals will help to improve medical waste segregation smoothly and appropriately for improvement of IPC.

Sustainable implementation mechanism of effective and efficient regional supportive supervision, experiences from Tanzania

NORIYUKI MIYAMOTO[1], HISAHIRO ISHIJIMA[2], MALUNDA JOSEPH[3]
[1]Fujita Planning Co., LTD., [2]Expert assigned to HRH Development Project, JICA/Ministry of Health and Social Welfare, United Republic of Tanzania, [3]Singida Regional Referral Hospital

[Introduction]Ministry of Health and Social Welfare (MoHSW) has been introducing several Quality Improvement Programs (QIPs) including 5S-KAIZEN-TQM Approach into all levels of hospital. Along with this situation, Monitoring and Evaluation (M&E) of QIPs is established and practiced vertically although chronic shortage of resources in Tanzania. Therefore, Regional Health Management Team (RHMT) that is responsible for the M&E has a difficult in the practice the M&E regularly and the sustainability of 5S-KAIZEN-TQM Approach.[Outline]JICA HRH Development Project has been introducing 5S-KAIZEN-TQM Approach, Japanese style of management, into national hospitals since 2007 in collaboration with MoHSW. In view of chronic shortage of resources, the project supported Singida RHMT, pilot region, to establish sustainable mechanism of regional Supportive Supervision (SS) in effective and efficient way after the project ended. Singida RHMT compared amount of resource inputs into joint implementation of regional SS with 5S M&E, separately implementation of regional SS and 5S M&E. As a result, it was defined that expenditure and labor can be reduced in the joint implementation.[Discussion]The RHMT reduced 22~36% the time consumption, 30% the number of human resources, 1~4% the labor inputs and 43~65% the total labor inputs by the joint implementation of regional SS with 5S M&E.[Conclusion]Sharing the experience of effective and efficient implementation of regional SS with 5S M&E to other regions can help to sustain implementation of regional SS by reducing resource inputs for regional SS.

Versatility Study of the Pain Assessment Tool for Older People in the UK and Malaysia

HARUNA TANAKA[1], YUMIKO MOMOSE[2], AKIRA MIZOO[3]
[1]Department of Nursing, School of Health Sciences, Gifu University of Medical Science, Seki, Japan, [2]Department of Nursing & Health, School of Nursing & Health, Aichi Prefectural University, Nagoya, Japan, [3]Japan Community Health care Organization Tokyo Shinjuku Medical Center, Tokyo, Japan

(Purpose) The Pain Checklist for Older People (P-COP) was first developed for Japanese older people, and this validity has been verified within Japan. In this study, we examined the versatility of P-COP, with the English version, while conducting pain assessments in countries outside of Japan.
(Study Methods) Malaysian and British nursing homes were chosen as subjects to examine the versatility of P-COP for this study. Subjects were chosen based on regular use of pain medication and the ability to verbally communicate their pain intensity. Nurses conducted the pain assessments on their nursing home residents, using P-COP. These same nurses also used the Verbal Rating Scale (VRS), when checking residents' pain intensity. The data collected was then analyzed using Spearman's Rank Correlation Coefficient. This study was subject to an ethical review and approved by a research ethics committee, from a university in Japan. Participation was voluntary and based on informed consent, with confidentiality and anonymity assured.
(Results) Pain assessment scenes were 200 in Malaysia and 192 in the UK. Results indicated that 15 out of 27 items from the P-COP had weak to strong positive correlation with the VRS (ρ=0.21 to 0.71) in the UK and Malaysia.
(Discussion) 15 out of 27 items suggest validity when identifying residents' pain. The remaining 12 items need to be reassessed for their reliability, when checking for pain and therefore, further studies will be conducted in other countries.
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Childcare and health of hunter-gatherer infants and caregivers in African rainforest

SAORI ITO[1], TARO YAMAUCHI[2]
[1]Graduate School of Health Sciences, Hokkaido University, Hokkaido, Japan,
[2]Faculty of Health Sciences, Hokkaido University, Hokkaido, Japan

[Background] In modern societies where the nuclear family is common, mothers often play a major role in childcare, while individuals other than the child's parents rarely participate in such care. In traditional societies, on the other hand, children are frequently cared for by the people who surround them, such as their relatives and even non-relative neighbors. We conducted field research focusing on the childcare practices of modern hunter-gatherers in African rainforests who still pursued a &quottraditional&quot lifestyle. We analyzed quantitative data in order to ascertain the actual situation regarding childcare in that society.
[Methods] The subjects were infants/children and mothers/caregivers of a Pygmy hunter-gatherer population in the Republic of Cameroon. The field research involved (1) anthropometric data, such as height and weight in adults, and length, weight, and circumferences of head, chest, and upper-arm in infants/children; (2) childcare observations, recorded from 6:00 to 18:00 daily for 45 min/h, totaling 9 h/day; (3) a dietary survey, all foods consumed by infants/children a day were measured, as well as their breastfeeding frequencies and durations, and sucking times; and (4) interviews with mothers/caregivers on breastfeeding, weaning foods, the eruption of teeth, and motor development, including crawling, standing, and walking.
[Expected outcomes] Behavioral data revealed the actual conditions of childcare in the modern hunter-gatherer society, especially the role of caregivers other than mothers of infants/children. In order to further understand the significance of allo-parenting within the society, we intend to perform assessments of its influence on the growth and development of children.

Family caregivers perception of their needs in taking care of Children with Special Health Care Needs in Rural Thailand

SATOKO YANAGISAWA[1], Amorn Swannimitr[2], Supaporn Aryamuang[2],
Kocharporn Singhala[2], Hidemi Hashimoto[3], Mariko Sakamoto[4]
[1]School of Health and Nursing, Aichi Prefectural University, [2]Faculty of Nursing, Mahasarakham University,
[3]Faculty of Nursing, Mie Prefectural College of Nursing, [4]College of Nursing, Aichi Medical University

Background Recent improvement in health indicators led the Ministry of Public Health shift its policy from mortality reduction to health promotion, disease prevention and control. However, family caregivers of CSHCN (Children with Special Health Care Needs) are left with various unmet needs. The present study aimed to explore health care needs of CSHCN caregivers in a Northeastern province in Thailand.
Methods Semistructured interviews were conducted at Mahasarakham Province from February to March in 2013. Participants were 15 caregivers who were taking care of CSHCNs. Nurse educators in Mahasarakham University interviewed them with local Thai dialect. Interviews were recorded with IC recorders. Transcriptions were modified into standard Thai language, then translated into English. Analysis was made upon the English version using NVivo 10.
Results Participants were 7 mothers, 5 grandmothers, 2 fathers and 1 grandfather. Their age ranged from 20 to 72 years old. Age of CSHCN ranged from 1 year and 8 months to 17 years old. They were 7 males and 8 females. Their chronic conditions included physical impairment, developmental disorder, mental problem and HIV. Unmet needs were found in five areas, i.e. health and medical needs, welfare needs, education needs, social needs and information needs.
Conclusion Caregivers of CSHCN were burdened with various unmet care needs. Many parents worked away from home in the study area, therefore, grandparents were left behind with CSHCNs. Special attention is necessary on these grandparents whose physical and psychological care burden could be enormous.

Physical fitness performance and body composition of urban elementary students and their rural counterparts in Central Java, Indonesia

HANAKA SETA[1], AYA YOSHIMURA[1], Maria Mexitalia[2], TARO YAMAUCHI[3]
[1]Graduate school of Health Sciences, Hokkaido University, Hokkaido, Japan,
[2]Faculty of Medicine, Diponegoro University, Indonesia, [3]Faculty of Health Sciences, Hokkaido University, Japan

Background: Decreasing in physical fitness among children, in connection with increasing obesity, has become a social issue all over the world. There are few reports assessing the physical fitness and exercise performance of children in Southeast Asian countries with a remarkable economic growth. The objective of this study is to compare the nutritional status and the physical fitness levels of elementary school children living in contrasting urban and rural areas in Indonesia. Methods: We examine the elementary school students living in the capital city of the Central Java Province (urban) and those living in the Jepara district (rural). Anthropometric measurements (height, weight, percentage of body fat) and physical fitness tests (50 m sprit, standing broad jump, flexed arm hang, sit-ups, 4×10 m shuttle run, long-distance run, forward flexion of the trunk) were conducted and the body mass index (BMI) was calculated. Results and conclusions: Urban children were taller and heavier than their rural counterparts. 38.9 % and 28.2 % of urban boys and girls, respectively, were categorized as "overweight" or "obesity", while this percentage was 2.9 % among the boys and girls children in the rural areas. Assessing physical fitness levels, the percentages of the people divided "poor" and "very poor" classes in urban boys and girls were 80.6 % and 69.6 %, respectively. On the other hand, these percentages were 13.5 % and 40.5 % among rural boys and girls, respectively. Urban subjects significantly tended to be obesity and a decline in physical fitness.

The trend of oxygen saturation for infants after birth in National Maternal and Child Health Center, Cambodia

Yuta Yokobori[1], Azusa Iwamoto[1], Mitsuaki Matsui[2], Yuriko Egami[1,3], Sotha Keth Ly[3,4]
[1]Bureau of International Medical Cooperation, National Center for Global Health and Medicine,
[2]Graduate School of International Health Development, Nagasaki University,
[3]JICA, Cambodia Project for Improving Maternal and Child Health through Midwifery Capacity Development,
[4]National Maternal and Child Health Center in Cambodia

[Background] Intrapartum hypoxia could seriously affect the lives of babies after birth, leading to asphyxia which is one of the main causes for neonatal death. In order to evaluate the oxygenation level, one of the good indicators is how quickly they reach to the adequate level of oxygen saturation (SpO2) after birth. However, there have been few researches on the SpO2 of newborn infants in developing countries including Cambodia. [Method] We collected the data of 164 newborn infants born in the National Maternal and Child Health Center in Cambodia from 2013/Oct-2014/Feb, who didn't receive any resuscitation. The SpO2 were prospectively measured on minutes-basis till 10 minutes after birth. The trend of SpO2 was summarized in a graph and compared with the previous research, by using the same method in Australia. Furthermore, risk factors which influence the time to reach adequate SpO2 levels were also analyzed.[Results] In comparison with the previous research in Australia, the time to reach adequate SpO2 levels was shorter in Cambodia. In addition, Cesarean section and uterotonic medication use did significantly affect the time to reach at 90% of SpO2, after adjusted for possible confounders by using multivariate Cox regression analysis. [Discussion] The possible explanation for shorter duration to reach adequate SpO2 in Cambodia could be some differences of birth style such as anesthetic delivery. In addition, Cesarean section and uterotonic medication could increases the risk of low SpO2. The staff for neonatal care should pay more attention for infants with such risks.

Fact finding survey about the effects of the health insurance on access to health services and financial protection for delivery and child illness in Hoa Binh Province, Viet Nam

Chieko Matsubara[1], Masahiko Doi[1], Tai Anh Nguyen[2], Kimiko Inaoka[1], Yumiko Tanaka[1], Yasuko Sato[1], Saeko Yamamoto[1], Yuta Yokobori[1], Hitoshi Murakami[1]
[1]Bureau of International Medical Cooperation, National Center for Global Health and Medicine, [2]CDA, Hoa Binh, Viet Nam

[Background] The universal health coverage (UHC) has been one of the well-discussed agenda in the context of post Millennium Development Goals (MDGs). Many efforts have been conducted, e.g. expanding health insurance (HI), however, beneficiaries tend to be limited to urban areas and formal sectors. We conducted a cross-sectional study at a rural Viet Nam, where the HI coverage vigorously expanded among the poor, children and ethnic minorities, particularly since 2012.[Methods] Data collection was conducted in January 2014 at Luong Song District, where many of the population are ethnic minority (Muong). The 316 participants who had delivered in 2013, selected by stratified and cluster sampling, were asked their socio-demographics, delivery and child illness. Participants were categorized by per capita household annual income. Out-of-pocket payment and health facility visits were dependent variables for statistical analysis. [Results] The HI coverage was 72.3%. All mothers delivered at health facilities. The out-of-pocket expenses for normal delivery were significantly different ($p \leq 0.05$) between the insured and non-insured among three groups (absolute-poor, pro-poor and others). The 60% children have experienced fever and cough at one time. There was no difference in health facility visits between the absolute-poor and others. No difference was found in out-of-pocket payment to public facilities between them, whereas there was a significant difference in payment to private clinics ($p = 0.041$).[Conclusion] The results showed the effects of the HI by the equity in access to health facilities and the protection from financial risks, which are important to achieve the UHC.

Current status and issues of health insurance in Hoa Binh Province, Viet Nam

SAEKO YAMAMOTO, YUTA YOKOBORI, CHIEKO MATSUBARA, HITOSHI MURAKAMI
Dept of International Medical Cooperation, National Center for Global Health and Medicine (NCGM), Japan

Objective The objective of this study was to identify the degree of the expansion of non out-of-pocket health financing schemes (including health insurance: HI), their design and impacts on care-seeking among health care clients in Hoa Binh Province, Viet Nam.
Methods Data were collected through semi-structured interviews with fund managers and health facility staff in charge of finance/accounting in Hoa Binh Province with particular focus on Luong Son District.
Results The study identified the following features of HI in Hoa Binh Province vis-a-vis Viet Nam: (1) social security schemes including HIs were solely administered by Viet Nam Social Security (VSS); (2) HI coverage expanded vigorously through premium subsidization for the poor, ethnic minorities and children aged <6 years; and (3) residents who had HI consistently had higher outpatient and inpatient rates per population than those who did not. Notable issues of HI included the followings: (1) low coverage and deficit spending of the voluntary HI; (2) incompliance of companies to pay mandatory HI premiums for their employees especially during bad financial times; and (3) cumbersome annual certification of the poor on the basis of revised minimum wage leading to their instable coverage.
Conclusion Japan can potentially support Viet Nam's HI in two major categories: macro-policy advises and technical assistances. The former may include how to contain health expenditures under the universal HI coverage, and the latter may include sharing experiences of the enrolment advocacy works for Japan's National HI (NHI) conducted by local governments and health volunteers.

Progress of development of the road map toward universal health coverage in Kenya

SATORU WATANABE
JICA Senior Policy and Technical Advisor on UHC/Health Financing,
Minitry of Health, the Government of Kenya, Nairobi, Kenya

[Introduction] The Kenya Demographic Health Survey 2008-09 shows declines in infant and under-five mortality from 77 to 52 and from 115 to 74 during the period from 2003 to 2007. Despite these improvements in child health, the prevalence of stunting has not changed significantly. The maternal mortality ratio remains high (around 530 per 100,000 live births). Moreover, regional and socio-economical disparities in health indicators are important issues in providing equitable health services. Therefore, the new administration in March 2013 introduced free maternal care and free user fee at primary health facilities. However, the disparity in health financing among Counties is large and more over the risk pooling mechanisms is very fragmented and covered population is limited. At the same time, of course, the quality of services is also an issue.[Outline] To respond to these situations, MOH has been finalizing the Kenya Health Policy 2012-2030, the Kenya Health Sector Strategic and Investment Plan 2013-2017 and the Strategy for Human Resource for Health (2014-2018). Based on those key documents, the Road Map to UHC in Kenya showing steps focusing on health financing with timeframe until 2030 is expected to be developed.[Discussion/Conclusion] The author is providing technical and policy advises on the development of the Road Map, especially on the definition of equity, target coverage (services and population), legal settings, financial arrangement, risk pooling mechanisms and purchasing. The session will review the progress as well as issues towards UHC in the case of Kenya as a case of low income county.

Challenging the expansion of the Universal Health Coverage(UHC) for migrants in Thailand

YUKA MAEKAWA[1,2], Jun Kobayashi[1,2]
[1]Department of Global Health, School of Health Sciences, University of the Ryukyus,
[2]NPO Japan Association for Mae Tao Clinic

[Introduction]International Organization for Migrant (IOM) reported the total number of migrant population in the world was 215 million in 2010 and would reach to 405 million by 2050. We aim to describe the present health system and figure out challenges on expansion of the Universal Health Coverage (UHC) in Thailand. [Outline]There are estimated 130,000 refugees and 3 million migrant workers living in Thailand and 2.3 million are from Myanmar. Although Myanmar government has moved to democracy, economic immigrates, which cross the border to Thailand, is not deceasing. After UHC achieved high coverage for Thai citizens, the health insurance system for migrants has been promoted for expansion of UHC. Only about 16 % of migrants in Tak, which is one of the province where Myanmar migrants live in, are insured.The utilization of health insurance (per person) among migrants remains a quarter of that among Thai citizens. [Discussion and Conclusion]The two review literatures mentioned, "Strategies to improve the structure of health system and social policies are needed."(Andres AAS 2012) and"Relevant policies of public medical insurance and assistance program should be implemented for providing affordable health care services to the migrants." (Peng Y 2010) We recommend the following two strategies in order to expand UHC to migrants. 1) Sharing information of insurance system with employers 2) Reinforcing linkage between existing community based health facility such as Mae Tao Clinic and Thai health system to remove language and cultural barriers as a feasible and effective approach.

Universal Health Coverage and Post 2015 Development Agenda

TOMOHIKO SUGISHITA
Senior Advisor, Japan International Cooperation Agency

WHO and World Bank released Framework of Universal Health Coverage and a resolution of Post 2015 Development Agenda was endorsed by World Health Assembly in May 2014. While the discussion of overall framework of Post 2015 Development Agenda has been continuously discussed by Open Working Group of Sustainable Development Goals, UN Taskforce Team and UN High Level Panel, four key words such as Equity, Fiiance, Systems and Resilience became increasingly important to articulate the future agenda setting. Concerning to our planetary boundary, some essential ideas such healthy lifestyle, transformative learning, global health governance and universalism became significant to future resilience of the health systems.

Optimization of a fluorescence- based assay for mass drug screening against and Theileria parasites

MOHAMED ABDO RIZK, SHIMAA ABDELSALAM ELSAYED, NAOAKI YOKOYAMA, IKUO IGARASHI
National Research Center for Protozoan Diseases, Obihiro University of Agriculture and Veterinary Medicine

A rapid and accurate assay for evaluating antibabesial drugs on a large scale is required for the discovery of new chemotherapy against *Babesia* parasites. In the current study, we evaluated the usefulness of a fluorescence-based assay for determining the efficacies of antibabesial drugs in in vitro cultures against bovine and equine *Babesia* parasites. Three different hematocrits (HCTs; 2.5%, 5%, and 10%) were used without daily replacement of the medium to optimize our previously developed assay for mass drug screening. The results of a high-throughput screening assay revealed that the optimal HCT was 2.5% for bovine *Babesia* parasites and 5% for equine *Babesia* and *Theileria* parasites. The calculated IC_{50}s of diminazene aceturate by fluorescence- and microcopy-based methods were not significantly different with 2.5% and 5% HCTs for bovine and equine *Babesia* and *Theileria* parasites, respectively. Likewise, there were no significant differences between the IC_{50} values of luteolin, pyronaridine tetraphosphate, nimbolide, gedunin, and enoxacin based on the two methods. These results revealed that the current fluorescence-based assay with lower HCTs and without daily replacement of the medium is suitable for mass drug screening against bovine and equine *Babesia* and *Theileria* parasites in vitro.

Zoonotic *Enterocytozoon* may circulate among animals and humans in rural areas in Thailand

HIROTAKE MORI, Aongart Mahittikorn, Yaowalark Sukthana
Department of Protozoology, Faculty of Tropical Medicine, Mahidol University , Bangkok, Thailand

Enterocytozoon bieneusi is a common opportunistic intestinal pathogen worldwide. Recently, zoonotic genotypes have been often found in humans and animals in developing countries, however, the impact of zoonotic involvement has not been fully clarified.
We conducted cross-sectional studies of *E. bieneusi* in humans and various animals in Central and Western Thailand. We investigated 180 healthy children in rural communities, 40 non-HIV immunocompromised adults in a hospital, 95 dogs and 80 cats in a Thai temple, 315 wildlife and domestic animals, 55 non-human primates. The prevalence of *E. bieneusi* was determined by nested PCR, and genotype characteristics and host specificity of the pathogen were also investigated.
Zoonotic *E. bieneusi* genotypes were wildly found in children in rural communities, cats, and non-human primates with the prevalence of 3.3%, 31.3%, and 20%. *E. bieneusi* was not found in on- HIV immunocompromised adults. Among genotypes, genotype D, the most commonly identified zoonotic genotype, was predominant. In rural areas in Thailand, *E. bieneusi* zoonotic genotypes may be transmitted among humans and various animals.

Problems with anti-Parasite Action in Lao PDR

DAISAKU URABE[1,2], HIROAKI YAMASAKI[1], TAKAO SUGIMOTO[1],
TOICHIRO ISO[1,2], KIYOMI HANDOU[2], TAKATOSI KOBAYAAKWA[2]
[1]Intrernational Cooperation Department, St.Mary's Hospital,Kurume,Japan, [2]Internationa Support and Partnership for Healht, Tokyo Japan, [3]Khammouane Provincial Health Office, Thakhek, Lao PDR

[Background]In Lao PDR, 500mg of Mebendazole (MBZ) is given to local people twice a year as to control parasite infections. To better understand the effectiveness of this practice we investigated the efficacy and the problems of current anti-parasite action in Lao.[Methods]After obtaining informed consent, we conducted stool examinations on local people in the sub-village of Donsawang, Kammuane Province in May and November 2013. We administered MBZ to helminth-positive persons and repeated medication three times if the follow-up stool tests were positive. Prazinquantel (PZQ) was administered to liver fluke-positive persons. We frequently delivered health educations, emphasizing parasite infections, based on field surveys conducted in May 2013 and February 2014, the awareness of village people about parasites was monitored.[Results]The stool tests conducted in May showed 124 out of 179 persons were infected with some parasites; 97 with helminths and 72 with liver flukes. 7 remained hookworm-positive after three doses of MBZ and 3 remained liver fluke-positive after one dose of PZQ. The test result in November showed 115 out of 166 persons were infected with some parasites; 95 with helminths and 50 with liver flukes.[Conclusion]The current practice of administering MBZ twice a year is insufficient against parasites prevalent in Laos because re-infection easily occurs even after treatment. More radical action seems necessary to disrupt the strong infection routes underlying in the environment including anti-liver fluke action which is not taken now. Recurrent educations seemed also necessary to raise people's awareness of parasites.

Effect of Ascofuranone and its derivatives on the growth of *Trypanosoma brucei brucei in vivo*

CHIAKI TSUGE[1], Yasutoshi Kido[1], Hiroyuki Saimoto[2], Kiyoshi Kita[1]
[1]Department of Biomedical Chemistry,Graduate School of Medicine, the University of Tokyo, Japan.,
[2]Department of Materials Science,Graduate School of Engineering,Tottori University,Japan

[Background] African trypanosomiasis is a parasitic infection caused by *Trypanosoma brucei*. Since current drugs for African trypanosomiasis show profound side effects, there is an urgent need for development of new drugs. As an ideal target for chemotherapy, trypanosome alternative oxidase (TAO), which is a mitochondrial enzyme that catalyzes aerobic terminal oxidase reaction, has been focused because this enzyme does not exist in mammalian host and essential for trypanosome to survive. Previously we found ascofuranone as a potent and specific inhibitor for TAO. In order to provide ascofuranone for clinical use, the effect on rTAO and parasite growth have been examined using over a hundred of its derivatives and structure activity relationship was elucidated. In addition, it is essential to examine therapeutic effect on infected animal for development of new drug. [Methods] For the *in vivo* study, mice were inoculated intraperitoneally with 10^4 trypanosomes from a previously infected mouse blood. Parasitemia was estimated from tail blood smears. Experimental treatments were done 2 days after the inoculation of trypanosomes. Treatment group mice were administered ascofuranone or its derivative solution intraperitoneally. Mice, which were negative for trypanosomes until 60 days after the drug treatment, were considered to be completely cured.[Results] Four of the compounds succeeded in curing infected mice, and one of them showed especially potent therapeutic effect. [Conclusion] These results suggested that ascofuranone, including its derivatives can be promising drug candidate for African trypanosomiasis.

Characterization of a *Toxoplasma gondii* CaMK-related kinase

KENTARO KATO, Tatsuki Sugi, Hitoshi Takemae, Ryo Takano
National Research Center for Protozoan Diseases,
Obihiro University of Agriculture and Veterinary Medicine, Hokkaido, Japan

Toxoplasmosis has been spread worldwide. The antibody-positive rate varies in each region. Toxoplasmosis is the zoonosis with a cat as a final host. The clinical problem is the first infection for pregnant woman which causes abortion and fetal encephalopathy thorough the transplacental infection for infants. We previously showed that *Plasmodium falciparum* expresses a homolog of human calcium calmodulin-dependent protein kinase (CaMK) that is important for host cell invasion. To identify novel targets for the treatment of *Toxoplasma gondii* infection (another apicomplexan parasite), we sought to identify a CaMK-like protein in the *T. gondii* genome and to characterize its role in the life cycle of this parasite. We identified a novel CaMK homolog in *T. gondii*, *T. gondii* CaMK-related kinase (TgCaMKrk). TgCaMKrk was purified as a fusion protein that was labeled with [γ-^{32}P]ATP; this labeling was then eliminated by phosphatase. Phosphorylation activity of TgCaMKrk was eliminated when its putative catalytic lysine residue was replaced with alanine. Therefore, TgCaMKrk exhibits calmodulin-independent autophosphorylation and substrate phosphorylation activity. However, calmodulin antagonists had no effect on its kinase activity. Therefore, neither Ca^{2+} nor calmodulin are required for the protein kinase activity of TgCaMKrk. The localization is observed at the apex of tachyzoite inside or outside the *T. gondii*-infected cell. The regulatory domain (regulatory domain and CaM binding domain) which is conserved in PfPK2 is not conserved in TgCaMKrk. These results suggested that another protein which regulates the phosphorylation of TgCaMKrk could be existed.

Cost Effectiveness of Tuberculosis Active Case finding

Isaac CHIKWANHA, Nobuko KUROSAKI
Medecins Sans Frontieres Japan

Cambodia is in the top 5 of the 22 high Tuberculosis (TB) burden countries in the world with an incidence rate of all forms of TB of 424/100 000, prevalence of 817 /100,000 and TB mortality of 63 /100,000 population (Global Tuberculosis Report 2012). The National Tuberculosis Prevalence survey 2011 showed a 4.2% annual decrease in smear positive TB in the ten years between 2002 and 2011 but also highlighted that more than 30% of the TB cases are not diagnosed every year mostly amongst the asymptomatic patients (less likely to seek medical treatment), amongst the smear negative cases (due to emphasis on microscopy for diagnosis) and that the prevalence of TB increased with age, with highest prevalence in the over 55 year olds.The reduction in the TB burden has been accomplished by the efforts by the Cambodia NTP and partners. Medecin Sans Frontieres (MSF) has been supporting the TB department in Kampong Chum Provincial Hospital since 2009. In an effort to increase case detection amongst the vulnerable groups highlighted in the prevalence study, increase the rate of reduction in TB prevalence and pilot a sustainable model of case detection, MSF, in collaboration with John Hopkins, will conduct a cost effectiveness study of Active Case Finding in TB and assess the impact on TB prevalence in Kampong Chum Province in Cambodia.In this presentation, we present a summary of that study.

The impact of HIV Comprehensive care center (CSI) in Antananarivo, Madagascar

TOSHIHARU OKAYASU[1,2], SHUKO NAGAI[3], KAE FURUKAWA[4], Evelyne Raherivololona[5], J
ulie Vony Soa Hanitra Harimanana[5], MASAHIDE KONDO[6]
[1]JICA Benin, Cotonou, Benin, [2]Department of Healthcare Policy and Management, Graduate School of Comprehensive Human Sciences, University of Tsukuba, Tsukuba, Japan,
[3]The Research Institute of Tuberculosis, Tokyo, Japan,
[4]JICA Burundi, Bujumbura, Burundi, [5]Ministry of Health, Antananarivo, Madagascar,
[6]Department of Health Care Policy and Health Economics, Faculty of Medicine, University of Tsukuba

Introduction 47 HIV treatment centers existed in 2012 such as University/Regional/District Hospitals in Madagascar. However the key populations at higher risk such as Sex Worker(SW), Men who have sex with Men(MSM) and Injecting Drug User(IDU) have difficulties to reach to hospitals. This research aimed to examine how a public health center can improve the access of the key populations to HIV testing .
Outline JICA HIV Prevention Strengthening Project in collaboration with Ministry of Health supported the Municipal Hygiene Bureau(BMH) Isotry as a pilot HIV Comprehensive care center(CSI), in order to have the marginalized people access HIV testing and to offer socio-psychological supports and the treatment to people living with HIV(PLHIV), from May 2010 to December 2012. We analyzed the number of HIV testing for marginalized people from 2009, before the intervention, to 2012, after the intervention.
Discussion Seven hundred and seventy key populations such as SW, MSM and PLHIV participated in socio-psychological activities from April 2011 to December 2012. The HIV positive cases increased from 15 in 2009 to 36 in 2012 regardless of the national shortage of testing kits in 2012. One hundred and fifty seven SW were tested in 2011, 69 MSM in 2011 and 211 IDU in 2012. However the testing cases for key populations might be influenced by the survey. The difficulties for counsellors to identify the key populations were recognized.
Conclusion The CSI has improved the access to key populations, however the way to identify those people by counselors should be improved for better monitoring.

Predictors of loss to follow-up among HIV-infected patients at two large HIV clinics in Hanoi: the data from a prospective cohort study

SHOKO MATSUMOTO[1], JUNKO TANUMA[1], KEIKO SAITO[1], Cuong Duy Pham Do[2], Thuy Thanh Thi [2], Tuan Quang Nguyen[2], Dung Thi Hoai Nguyen[3], Lam Tien Nguyen [3], Kinh Van Nguyen [3], SHINICHI OKA[1]
[1]AIDS Clinical Center, National Center for Global Health and Medicine, Tokyo, Japan., [2]Infectious Disease Department, Bach Mai Hospital, Hanoi, Vietnam, [3]National Hospital of Tropical Disease, Hanoi, Vietnam

Background: The retention in care is one of the challenges for HIV treatment in the resource-limited settings. We investigated the risk factors related to the loss to follow up (LTFU) during 2007-2013 in two HIV clinics in Hanoi, Vietnam.Methods: The data was obtained from a prospective cohort of adult HIV-infected patients receiving antiretroviral therapy at two referral clinics in Hanoi. The subjects for analysis were those who enrolled from October 2007 to December 2012, and followed up every 6 months until December 2013. LTFU was defined as those who missed at least two consecutive visits and did not return by December 2013, excepting for those who were incarcerated at loss to follow. Factors associated with LTFU were analyzed using cox proportional hazards regression models.Results: forty-five (3.1%) out of 1431 patients in the cohort were lost to follow-up by December 2013, of whom 19 (42.2%) were lost within 6 months after enrollment. In multivariate cox proportional hazards model, age younger than 30 (HR=2.3: 95%CI, 1.2-4.5 vs. 30-39 years old) and baseline CD4 count less than 200/μl and 350/μl or more (HR=2.7: 95%CI, 1.1-6.5 and HR=3.1: 95%CI, 1.2-8.3, respectively vs. baseline CD4 count 200-349/μl) were independently associated with LTFU. Conclusion: High retention in care was noted in the cohort of patients receiving HIV treatment in Hanoi. Younger age group and less and greater CD4 count groups were considered as predictors for LTFU. These findings could be helpful for enhancing retention of patients in HIV treatment in Vietnam.

Genetic analysis of HIV-1 subtypes and drug resistance mutation in Savannakhet Province, Lao PDR

Tadashi Watanabe[1], Natsuki Nabara[1], Mayu Araki[2], Azumi Ishizaki[3], Panom Phongmany[4], Vatsana Sopraseutha[5], Hiroki Kagawa[1], Jun Kobayashi[6], Hisami Watanabe[7], Hiroshi Ichimura[3], Masahiro Fujimuro[1]
[1]Department of Cell Biology, Kyoto Pharmaceutical University, Kyoto, Japan, [2]Yokohama College of Pharmacy, Yokohama, Japan, [3]Department of Viral Infection and International Health, Graduate School of Medical Science, Kanazawa University, Kanazawa, Japan., [4]Savannakhet Provincial Health Department, Lao PDR., [5]Laboratory Unit, Savannakhet Provincial Hospital, Lao PDR, [6]Department of Global Health, School of health sciences, University of the Ryukus, Okinawa, Japan, [7]Centre of Molecular Biosciences, University of the Ryukyus, Okinawa, Japan

[Background] Epidemiological surveillance of HIV-1 subtypes is an important element for global antiviral interventions and prevention of virus diffusion. It's known that CRF01_AE is a major subtype, which is dominant in Southeast Asia, including Thailand, Vietnam and Cambodia. Subtype B is the major viral strain in Europe and U.S, and also found in China. On the other hand, the genetic information of HIV-1, especially local regions in Lao PDR, has not been researched. We evaluate subtype and drug resistance of HIV-1 viral genome circulating in Southern Lao PDR.
[Method] Plasma samples derived from 73 HIV-1 infected patients, informed consent was obtained, in Savannakhet provincial hospital were used for analysis. Viral RNA samples were subjected to RT-nested PCR, and analyzed sequences cording Protease (PR) and Reverse Transcriptase (RT).
[Results] Genotyping analysis found that 66/73 patients were infected with CRF01_AE. Subtype B was found from 5/73 patients. In 2 other patient samples, Recombinant strains having characteristics of CRF01_AE and Subtype B were found. It indicates that HIV-1 heterogeneous recombination of both subtypes has been occurred. Several major resistance mutations for RT inhibitors were found(NRTI: V75L, NNRTI: G190A), however major mutation for PR inhibitors was not found.
[Conclusion] Continuation of monitoring HIV-l genotype and drug resistance are essential for the appropriate antiretroviral therapy and better clinical interventions in southern regions of Lao PDR. Further study on genotype is needed in relation to socio-demographic study, to investigate virus diffusion route.

MSF's Medical Activities in The Republic of South Sudan

Hiroyuki KATO, Keiko OTANI, Yasuko SHINOZAKI, Nobuko KUROSAKI
Medecins Sans Frontieres Japan

(Background) The Republic of South Sudan was founded in July 2011, but civil war followed. According to the data of WHO in May 2014, 4,200,000 people were in need of humanitarian assistance and there were 959,000 IDPs. 278,600 people were driven away to neighboring countries. Villages were violently destroyed by deliberate attacks and violence against civilians. There was a shortage of food, and insufficient of primary health services. Medecins Sans Frontieres has been providing medical care and humanitarian aid before independence of South Sudan. (Method) Investigate the situation of malnutrition in the South Sudan projects of MSF, and treated. (Result) In 2014 first half year, over 13,270 children have admitted the MSF therapeutic feeding programs in South Sudan. There was hospitalized in 2,810 cases in May and June 2014 in Leer project. Bentiu project has been specializing in severe malnutrition; 239 malnutrished children accompanied by complications diarrhea, pneumonia and dehydration were hospitalized, and 42 cases died. (Discussion) Number of hospitalized malnutrition cases was 18,125 in 2013, which extends to 73% of the last year in the six months this year. It was 2,014 cases in Leer project in 2013, and the number of May and June in 2014 is already greater than 2013. The result shows how the situation of this year is inferior. Health care is restricted by violence, displacement, and poor infrastructure, however, it should be important to improve reinforcement of primary health, establishment of secondary health, and improvement of weakness of the system in future.

Consideration of Applying the MIRA to the Health Needs Assessment

YOSUKE TAKADA[1,2], TATSURO KAI[1,2], TAKASHI UKAI[2], TAICHI TAKEDA[2], SAKURAKO NARITA[2], SHINJI TAKAMURA[2]
[1]Disaster Reduction and human renovation Institution (DRI), Kobe, Japan.,
[2]Humanitarian Medical Assistance (HuMA), Tokyo, Japan

[Introduction] The Multi-Cluster/ Sector Initial Rapid Assessment (MIRA) is designed to identify strategic humanitarian priorities during the first few weeks following an emergency, and to send the most appropriate support to the victim/place more quickly.
[Outline] In November 2013, the Republic of the Philippines was stricken by the typhoon Haiyan, causing serious damage widely. I was dispatched there as a member of an advance team of HuMA (Humanitarian Medical Assistance) and carried out the health needs assessment by using MIRA. The MIRA framework consists of A) Impact of the crisis, B) Response capacity, C) Access and gaps analysis. The results of analysis helped us decide strategic humanitarian priorities. The MIRA approach of Secondary Data Analysis (SDA) and Community Level Assessment (CLA) were then adapted to each part of the MIRA framework.
[Result] HuMA had done the rapid assessment in the Cebu and Western Leyte based on SDA result. And HuMA asked to the medical officers working in the affected municipalities that number of patients a day and mobility, number of medical personal etc.
[Discussion] MIRA is a systematic and effective assessment tool because every steps are clear, and enables us to recognize the situations clearly.
[Conclusion] MIRA, a systematic assessment tool makes it easy to evaluate objective evaluation, and effective in encouraging strategic decision making for humanitarian assistance.

Next steps for application of the Ministerial Order Scope of Nursing Practice Guideline to actual situation in Lao P.D.R.

MAYUMI HASHIMOTO[1], YUKO OTOMO2, MASAKI AOKI[3], Phengdy INTHAPHANITH[4]
[1]1st Expert Service Division Bureau of International Medical Cooparetion National Center for Global Health and Medicine, Tokyo, Japan, [2]IC Net.Limited, Tokyo,Japan,
[3]JICA Sustainable Development of Human Resources for Health to improve MNCH Services Vientiane Lao PDR,
[4]Health Care Professional Office, Ministry of Health, Vientiane, Lao PDR.

Background Laotian Ministry of Health established Nursing and Midwifery Regulation, Scope of Nursing Practice Guideline and the other two guidelines. This study assess any gaps between the current nursing practices and the Scope of Nursing Practice Guideline, and explore a strategy of how to apply the Guideline to the actual situation for quality of nursing practices.
Methods Structured interview, questionnaire and direct observation were used. A Japanese expert mainly conducted the survey and Laotian national nursing administrative officers accompanied all processes. The data were collected from the biggest teaching hospital, eight provincial hospitals, six district hospitals and three health centers.
Results Many nurses asked about the meaning of terms in the Guideline during the survey. Personal Care of patients such as meal assistance were provided by patients' family members. Nursing records were used almost the same forms at all health facilities. Few nursing assessments were written. More than half of hospitals directors did not know the Guideline. The gaps between the Guideline and the current nursing practices are the widest in health centers, and the next could be district hospitals where there were no or a few medical doctors.
Conclusion The guideline is difficult to be applied at every health facilities, because the guideline is based on the assumption that there will be medical doctors. The results indicate needs to revise the Guideline in terms of actual situation surrounding nursing practices and intelligible terms, to enhance nurses'competencies, and to implement the Guideline collaborating with other professionals.

Enhancement of Nursing Competency in Indonesia

KUMIKO IGARASHI[1,2], Yuma Fujinami[2], Satomi Ogawa[3], Suryowinoto Tritarayati[4], Suhartati Yuti [4]
[1]National Center for Global Health and Medicine, [2]JICA Project for Enhancement of Nursing Competency through In-service Training, [3]Japanese Red Cross Kyushu International College of Nursing,
[4]Ministry of Health, Republic of Indonesia

Introduction: One of the prioritized strategies of the National Policy on Development of Human Resources for Health (HRH) in Indonesia is to improve quality of nursing care in line with to enhance Universal Health Coverage. Since 2012,JICA has been conducting a project which focus on the development of the national standard of a Ladder System as part of a continuing educational program for enhancement of nursing competency. Results: The project involves with the introduction of nursing competency assessment system with evaluating indicators based on the degree of nurses' mutuality. 8 pilot hospitals have adapted the system. The project has been promoting a better understanding of a comprehensive competency assessment system among nurses. A baseline survey has been conducted to 85 nurse managers and 1,483 nurses. A hospital with low understanding of nurse managers on ladder system shows low perception of nurses. Understanding of ladder system shows difference between one hospital which has already introduced ladder system and others without introducing the system. Discussion: It is expected to conduct evaluation of effectiveness of ladder system in the next step. It was realized that the understudying of nurse managers effects the understanding of individual nurses in ladder system. Conclusion: The project values the approach based on the degree of mutuality together with the development of curriculum for the CPD. A competency based ladder system can be applied not only to nurses but also to all HRH. Therefore, it is necessary to establish monitoring system to evaluate the effectiveness of Ladder System.

Implementation and implication of integrated maternal neonatal and child health services on outreach activities in Salavan province, Lao PDR

MASAMI MIYAKAWA[1,2], BOUNKONG SAYSOMBATH[2,3], TOMOMI KITAMURA[1,2], HIRONORI OKABAYASHI[1,2], KHOUTDARA VONGSARAVANH[2,3], SEIKI TATENO[1,2]
[1]Bereau of International Medical Cooperation, National Center for Global Health and Medicine, Japan,
[2]The Project for Strengthening Integrated Maternal, Neonatal and Child Health Services, JICA, Lao PDR,
[3]Salavan Provincial Public Health Department, Ministry of Health, Lao PDR

Introduction: Coverage of maternal neonatal and child health (MNCH) services are generally low in Lao PDR. Integrated MNCH services at health facilities or on outreach activities are counted as a key strategy in the Strategy for the Integrated Package of MNCH Services 2009-2015. We analyzed proportion of service coverage provided on outreach activities in the overall coverage in Lao fiscal year 2009-2013 in Salavan province, southern Lao PDR, then considered operational challenges. Outline: The provincial health department developed a practical handbook regarding implementing procedure on integrated services on outreach activities for service providers by translating national guideline into real context, then conducted training and on-site supervisory visit. Three and five rounds of outreach activities were implemented in 2012 and 2013 respectively. Coverage of most services have improved after integration, and one in three of the first antenatal care and three quarters of postnatal care for women were provided on outreach activities in 2013. Discussion: Health center staff demonstrated decent capacity of operational management and basic medical skills on MNCH in implementing integrated services on outreach activities, and the integration brought about increase in service coverage for women. Some challenges including high dependence on external financial sources, and negative effect of increase in duties outside health facilities under the limited human resource should be discussed from the perspective of sustainability. Conclusion: Further accumulation of practical experiences and close studies on contextually appropriate operational approaches regarding integrated MNCH services on outreach activities are expected.

The health system of Bolivia with a focus on Tocroca Health Post in Potosi Province

Megumi Matsumoto[1], Ryota Okamoto[2], Taiki Fukushima[3], Motoyuki Yuasa[4]
[1]Saitama Medical University, School of Medicine, Moroyama, Saitama, Japan,
[2]Yamanashi University, School of Medicine, Kofu, Yamanashi, Japan,
[3]Akita University, School of Medicine, Akita, Japan,
[4]Department of Public Health, Juntendo University Graduate School of Medicine, Tokyo, Japan

[Introduction]Bolivia has a large land area and various geographical features and climates. In the severe environment, Health Posts play an important role in primary health care. A Health Post is established in each small village to supply basic health services. To survey the strengths and weaknesses of Bolivian Health Posts' function, we visited and investigated the Tocroca Health Post located in Tupiza, Bolivia, in March 2014.[Methods]We visited the Tocroca health post to do a field-work and a home-visit with an assistant nurse working there. Then, we researched his daily work and the actual health activities of patients and community.[Results]The Health post brought great benefits. Villagers had opportunities to receive basic health services easily because of a convenient location and close relationship between the assistant nurse and them. Moreover, it cost less operating expenses. However, the study revealed a shortage of health staff and quality of medical skills and devices in the Health Post. The national and local governments should support basic training and educational programs for every assistant nurse, and also supply student nurses as additional human-power resources. Through such plans, all Health Post staff can enhance their health skill capabilities in order to better serve the community and promote preventive care activities via community participation.[Conclusion]Although Health Post has some problems and limitations in terms of physical and human resources, and especially emergency acre, we found that these health facilities are indispensable in the Bolivian health system.

Situation and Direction of Development Partners coordination in Kenya Health Sector

SATORU WATANABE
JICA Senior Policy and Technical Advisor on UHC/Health Financing,
Minitry of Health, the Government of Kenya, Nairobi, Kenya

[Introduction]The donor coordination in health sector in Kenya had been very fragmented in the past.In 2007, therefore, the Government of Kenya and all of Development Partners (DP) in health sector including Japan signed on the Kenya Health SWAPs Cord of Conduct under IHP+ flame work. Recently, under the Health Sector Coordination Committee (overall coordination mechanism between MoH and DPs), Inter-agencies Coordinating Committees (ICCs) for several key subsectors have been established.[Outline]There are 19 ICCs that have been established. However some of them are not functioning. Therefore, Ministry of Health and DPHK are planning to reorganize the structure to reduce numbers and to increase efficiency. At the same time, since the County Government has been responsible for service delivery under the devolution starting from the last year, the necessity of the donor coordination at the county level has been increasing. While there is the Intergovernmental Liaison Office in MOH especially in HRH, HF and Medical Products, its capacity is still too week to manage them with the County Governments.[Discussion/Conclusion]Author has been actively participating in DPHK, the Committee on UHC, ICC on HRH and HF. The trends of donor coordination from the past through the present will be overviewed. Then, the directions of donor coordination mechanisms in HRH and HF will be reported. Finally, the future mechanisms including Sector Direct Budget Support will be discussed as one of lessons on the donor coordination.

Analyses of JICA health projects contributing to health system strengthening

Yoshihisa Shirayama, Motoyuki Yuasa
Department of Public Health, School of Medicine, Juntendo University

[Objective] The importance of primary health care and health promotion has been reassessed in the light of post-MDGs or Universal Health Coverage policies. The study aimed to analyze the contribution of recently implemented health projects by Japan International Cooperation Agency (JICA) for strengthening the health system in the respective countries.
[Methods] We searched JICA's web-based database and identified 31 Project Design Matrices (PDM) implemented between April 2011 and March 2014. To assess the extent of the contributions of JICA projects to the health system strengthening, we developed an original matrix, organized in six rows and six columns, based on the six building blocks of WHO's health system framework. It has 36 cells as outputs and activities of PDM, respectively. Using the matrix, we categorized the projects by type, based on contributions to health system strengthening. Whether the PDM includes community as keyword, was also checked.
[Results] Thirty one projects were categorized into 36 cells (the table will be shown on the day). Three PDM included community as a keyword in their target and activities. Whereas JICA's position paper says that the community empowerment is an important aspect of their health projects, only three projects mentioned the community. More projects could be developed based on the health promotion concepts.

A Spatial model for Dengue Virus Infection in Urban Areas

STEPHAN KARL[1], Nilimesh Halder[2], Joel Kelso[2], Scott Ritchie[3], George Milne[2]
[1]Walter and Eliza Hall Institute of Medical Research, Melbourne, Australia,
[2]The University of Western Australia, Perth, Australia, [3]James Cook University, Cairns, Austrlalia

There are recent indications that the health burden caused by dengue virus infection has been vastly underestimated. While World Health Organization (WHO) estimates for the global number of infections usually range between 50-100 million cases per year, recent studies estimated approximately 3 times higher case numbers. The geographic range of dengue virus transmission is extending with the disease now more frequently occurring in areas such as southern Europe. Northern Australia is endemic for Aedes aegypti, one of the most prominent dengue vectors and the city of Cairns in northern Australia frequently experiences dengue epidemics.We have developed a spatially explicit individual based mathematical model to predict Dengue transmission in the city of Cairns. The model incorporates detailed census and mosquito trapping data and accounts for the spread of infections through mosquito dispersion and human movement. The model is capable of realistically reproducing past dengue epidemics in Cairns and can be used for planning interventions and scaling the effectiveness and cost effectiveness of potential vaccination programmes. Our current model setup incorporates conventional interventions such as in house residual spraying and removal of mosquito breeding sites.To our knowledge this is the most comprehensive, realistic and versatile model for urban dengue transmission which is currently available. The model can be transferred to other urban scenarios where census and mosquito trapping data are available.

Detection of dengue virus nonstructural protein 1 (NS1) in urine samples by using ELISA as a laboratory diagnostic method for dengue virus infection

YUKA SAITO[1,2], MENGLING MOI[1], AKIRA KOTAKI[1], MAKIKO IKEDA[1], SHIGERU TAJIMA[1], HAJIME SHIBA[2], KUNIAKI HOSONO[2], MASAYUKI SAIJYO[1], ICHIRO KURANE[3], TOMOHIKO TAKASAKI[1]
[1]Department of Virology 1, National Institute of Infectious Diseases, Tokyo, Japan, [2]College of Bioresource Science, Nihon University, Kanagawa, Japan, [3]National Institute of Infectious Diseases, Tokyo, Japan

[Background] Detection of NS1 antigen (Ag) in urine is relatively practical as compared to serum samples because blood collection would be a challenge in areas with limited resources, in patients with severe bleeding tendencies and in infants. In this study, the utility of urine samples in detecting NS1 Ag for DENV infection diagnosis by NS1 Ag ELISA was evaluated.
[Methods] The panel consists of 175 serum samples and 118 urine samples from 96 dengue patients. Patient samples were examined for the presence of viral genome by RT-PCR and, NS1 antigen and DENV IgM and IgG antibody by ELISA.
[Results] NS1 Ag was detected in urine samples of dengue patients obtained at days 2 to 17 after onset of disease and, positive detection rates of NS1 antigen ranged from 13-43%. Using real time RT-PCR, positive detection rates of viral genome in urine samples ranged from 20-33% at days 0 to more than 15. DENV IgM and IgG antibodies were at levels below detection for most urine samples.
[Conclusion] We found that NS1 antigen and viral genome were detected in urine samples, although at lower detection rates as compared to serum samples. IgM and IgG antibodies were at levels below detection for most urine samples. These characteristics indicate DENV laboratory diagnosis using urine samples by NS1 Ag ELISA is potentially useful in conditions where blood sampling is not feasible, and particularly in combination with RT-PCR.

Analysis of human neutralizing and enhancing antibodies using mouse monoclonal antibodies against dengue type 1 virus

ATSUSHI YAMANAKA, EIJI KONISHI
BIKEN Endowed Department of Dengue Vaccine Development,
Research Institute for Microbial Diseases, Osaka University, Osaka, Japan

Background: Dengue hemorrhagic fever is a severe form of dengue fever. Although the mechanism(s) has not been clarified yet, one of the most potential hypotheses is involvement of infection-enhancing antibodies. We have generated mouse monoclonal antibodies (MAbs) against dengue type 1 virus (DENV1) and analyzed their neutralizing and enhancing activities. Most MAbs showed neutralizing activity at high concentrations and enhancing activity at low concentrations (NEAb). However, some showed only neutralizing (NAb) or enhancing (EAb) activity. Here, we made cocktail antibodies using these MAbs and compared their antibody dose-dependent activities with those shown by sera from people in an endemic area.
Methods: DENV1 (Mochizuki strain) was used. Serum samples collected from Indonesian residents and MAbs (1 mg/ml: ascites) generated from DENV1-immunized mice were used. An assay to measure a balance of neutralizing and enhancing activities and a conventional neutralization test were performed. Three MAbs, EAb (IgG1; complement-independent) and complement-dependent NEAb (IgG2b) and NAb (IgG3), were mixed to prepare cocktail antibodies.
Results: Most sera from individuals in the endemic area showed complement-independent enhancing activities. In contrast, most sera that showed neutralizing activity depended on complement. A cocktail consisting of three MAbs (EAb:NEAb:NAb) at a ratio of 8000:2000:1 showed antibody dose-dependent activities similar to those obtained by sera from infected humans.
Conclusion: Infected individuals induce antibodies in a polyclonal status. Cocktail of MAbs would be a useful approach to analyses of human immune responses. The present study suggested that infected humans induce mainly complement-independent EAb.

Peripheral T cell dynamics correlate to disease severity in Dengue fever

KENJI HIRAYAMA[1], Nguyen Thi Phuong Lan[2], Van Nguyen Thuong[2], Thanh Le Chi[3], Thang Minh Cao[2], Nhon Cao Thi My[4], Quoc Kien Do[5], Tran Thi Ngoc Ha[1], Tien Nguyen Huy[6], Tran Ton[3], Quang Chan Luong[5], Van Tran An[4], Vu Thi Que Huong[2,6], Shusaku Mizukami[1], Dao Huy Manh[1]
[1]Immunogenetics, NEKKEN, Nagasaki University, Nagasaki, Japan,
[2]Immunology and Microbiology, Pasteur Institute Ho Chi Minh City (PIHCMC), Vietnam,
[3]HIV Laboratory, PIHCMC, [4]Nguyen Dinh Chieu Hospital, Ben Tre Province,
[5]National program for Dengue control, PIHCMC, [6]Clinical Product Development, NEKKEN, Nagasaki University

Background: The protective or pathogenic role of lymphocytes during the acute phase of Dengue virus (DENV) infection is poorly understood. This study aimed to monitor the dynamics of activated lymphocyte subsets, CD4 T helper 1/2 (Th1/ Th2) and CD8 T effectors, during the acute phase of infection in relation to genetic background, past history, and clinical severity.Methods: A prospective hospital-based study was conducted at Nguyen Dinh Chieu hospital, Ben Tre province, Vietnam. Sixty two patients were enrolled. Cases were classified based on 2009 World Health Organization (WHO) guidelines. Fever days were calculated based on the day of defervescence. Percentage of lymphocytes and absolute values of T cell subsets in blood samples were determined by flow-cytometry.Results: We observed the distinction in activated CD4 Th1 and CD8 T effector responses between the severe and mild group in the acute phase of DENV infection. CD4 Th1 were more activated in the mild group ($p=0.0409$) plus CD8 T effectors decreased more drastically in the severe group ($p=0.0333$) at Day -1. Plasma levels of cytokines could support the same tendency.Conclusion: This study reveals the importance of CD4 Th1 and CD8 T effectors in the course of protective immune response against DENV infection. They also might serve as early predictors for severe dengue manifestations.

The relation of depression with health behaviours and social conditions of dependent community-dwelling older persons in the Republic of Chile, from a nationwide survey

FELIPE SANDOVAL[1], Nanako TAMIYA[2], Peter Lloyd-Sherlock[3]
[1]University of Tsukuba, [2]University of Tsukuba, Faculty of Medicine, Health Services Research Department, Tsukuba, Japan,
[3]University of East Anglia Norwich, School of International Development, Norwich, United Kingdom

Objective: The aim of this study was to examine the relation between depression with health behaviours and social conditions among dependent community-dwelling older persons in the Republic of Chile.
Methods: This is a cross-sectional, inferential study using nationwide secondary data. We used logistic regression on 640 dependent community-dwelling older persons from all over Chile, with a valid depression assessment, excluding those taking antidepressants. The Geriatric Depression Scale (GDS-15) was used as outcome. As exposure, we used the health behaviours and social conditions of the older persons. Socio-demographic and physical conditions were used as covariates.
Results: 48% of the older persons presented depressive symptoms. Logistic regression shows significant harmful factors for being depressed are: visiting the doctor &gt4 times because of acute diseases (Odds Ratio-OR-:2.9, Confidence Interval-CI-:1.1~7.5), living alone (OR:2.1,CI:1.1~4.0), feeling uncomfortable with their living arrangement (OR:3.7,CI:1.1~12.0), and feeling discriminated (OR:2.2,CI:1.4~3.6).
Discussion: Among health behaviours, visiting the doctor because of acute diseases shows that, unlike chronic visits, subject may get used to health check-ups, but not to the stress of acute episodes. Among social conditions, living alone, burdens users, along with feeling uncomfortable with their living arrangement. Subjects that feel discriminated tend to be depressed.
Conclusion: Further analyses on ways to support older persons living alone to avoid depression are needed, along with a deeper understanding of the comfort with their living arrangement. Finally, ways to prevent discrimination among older persons should be encouraged.

Depressive symptoms and social engagement among older people at the community level

YURI SASAKI[1], YUIKO NAGAMINE[1], YASUHIRO MIYAGUNI[1], HIROYUKI HIKICHI1, TAMI SAITO[2], KAZUHIRO KAKIMOTO3, KATSUNORI KONDO[1]
[1]Chiba University, [2]National Center for Geriatirics and Gerontology, [3]Osaka Prefecture University

Objective: The world is focusing on how Japan, one of the fastest ageing countries, is dealing with the challenges. This study examined the correlation between depressive symptoms, and social participation and role of older people at the community level.
Method: We used the data from 29 municipalities in the Japan Gerontological Evaluation Study. A total of 62,723 who was restricted to those 65-74 years and not receiving public long-term care insurance benefit, were analyzed adjusting for sex. The 15 geriatric depression scale was used.
Result: Prevalence of depressive symptoms ranges from 19.3% to 32.3%. The prevalence was high in the cities where the prevalence of equivalent income less than two million yen (r=.61), living alone (.44), no spouse (.63), going out less than once a week (.47) or the decrease frequency (.48), were high. Participation of activities like volunteer (-.47) was negatively related, while involvements of community event administration were positively related especially among women. As for the relationships with others, spouse were negatively related to depressive symptoms at the community level, but some other relatives were not.
Conclusion: Factors associated with depressive symptoms at individual level may also be associated at the community level. It might be important to create community environment where older people were not only able to receive care, but also give care, or listen to spouse's complaints etc, while the high burden of administrative activities, taking care of some others may lead to depressive symptoms at the community level.

Association between household water insecurity, quality of life, and postnatal depression in urban Nepal

YOKO AIHARA[1], Thilottama Pun[2], Jyoti Sharma[3]
[1]Educational and Research Center for Community Collaboration, Kobe-city College of Nursing, Kobe, Japan.,
[2]Netaji College of Nursing Science, Kathmandu, Nepal.,
[3]Institute of Medicine, Tribhuvan University, Kathmandu, Nepal.

[Background] Due to rapid urban population growth, water scarcity is one of the serious social development issues in Nepal. The estimation of demand of daily water supply in this region is 195 million litres, whereas the supply level is 100 million litres in dry season. The present study aimed to examine household water insecurity scale (HWIS) and its' association with psychological health and quality of life among postnatal women living in urban Nepal.[Methods] Study subjects were postnatal women who were admitted to X hospital in Kathmandu, Nepal. Interview survey was carried while women admit to hospital and one-month after being discharged. In total, 267 women were followed-up and their data were analysed. Sociodemographic and parturition characteristics, postnatal depression, physical health-related QOL (HRQOL), and HWIS (0-24 points, and higher score means lower security). Multivariate regression models were applied to examine the association between HWIS, depression, and HRQOL. [Results] The mean score of HWIS was 8 points, and Chronbach's alpha was 0.73. A month after, 17 women (6.4%) had probable postnatal depression. Women who had higher HWIS more likely to have postnatal depression (Odds ratio = 1.43, 95% confidence interval = 1.01-2.02). In addition, higher HWIS associated with lower physical HRQOL (be-ta coefficients = -0.11, p-value = 0.01). [Conclusion] The study suggests that the water scarcity issues may be associated with not only diarrhoeal illness, but also psychological illness and QOL among women. The validity and reliability of HWIS should be tested in further study.

High prevalence of risk factors for non-communicable diseases in Palau

CHIFA CHIANG[1], TRAVIS S SINGEO JR[2], EDOLEM IKERDEU[2], HIROSHI YATSUYA[1,3], KAORI HONJO[4], TAKASHI MITA[5], YAN ZHANG[1], ESAYAS H HILAWE[1], CHAOCHEN WANG[1], NOBUO KAWAZOE[1], MAYU UEMURA[1], RENZHE CUI[6], HIROYASU ISO[6], BERRY M WATSON[2], ATSUKO AOYAMA[1]
[1]Department of Public Health and Health Systems, Nagoya University School of Medicine, Nagoya, Japan , [2]Bureau of Public Health, Ministry of Health, Republic of Palau, Koror, Palau, [3]Department of Public Health, Fujita Health University School of Medicine, Toyoake, Aichi, Japan , [4]Global Collaboration Center, Osaka University, Suita, Osaka, Japan, [5]Institute for Academic Initiatives, Osaka University, Suita, Osaka, Japan, [6]Department of Social and Environmental Medicine, Osaka University Graduate School of Medicine, Suita, Osaka, Japan

Background: Although NCDs have become the predominant health problems of Palauan society, there was no comprehensive data for developing effective control strategies. The first Palauan national STEPS was, therefore, completed in mid-2013 to provide information for its adult population aged 25-64. This study aims to grasp the magnitude of major NCD risk factors in Palau by analyzing the given data. Methods: We conducted descriptive statistical analysis for the population-based data. A total of 2,184 valid participants were selected, excluding 2 without answers of sex, 28 pregnant women and 12 outside the targeted age range. Participants were stratified into two groups, 25-44 and 45-64 years of age, for the subsequent analyses. Results: Approximately 75% of the total subjects were Palauans, and 19% were Filipinos. Overweight/obesity was observed in 70% of 25-44 group and 80% of 45-64 group. Hypertension was more frequent in males than females, and the old group had a higher prevalence than the young group (38% vs 63%). Diabetes, raised triglycerides and raised total cholesterol were also more common in the old age group, 13% vs 25%, 39% vs 49% and 15% vs 30%, respectively. Although the percentage of cigarette-smoking was 17%, about 60% of the adults were current tobacco users in terms of all kinds of tobacco products. Participants who almost do not eat vegetables and fresh fruit were 12% and 38%. Conclusions: This study revealed a very high prevalence of NCD risk factors in Palau. It indicates that urgent actions for NCD prevention and control are required.

Collaboration in Doctor Training System with Neighboring Country: Exchange Training Project for Anesthesiologist-Trainees between Japan's General Hospital and Philippines' Children's Hospital

TATSURO SASAKI[1], RICHARD LUCERO[2], JOSE ALCANTARA[2],
TERESITA BATANES[2], MARICHU BATTAD[2], TAISHIN ITO[1], YASUHIRO MAEHARA[1]
[1]Department of Anesthesiology, National Center for Global Health and Medicine, Tokyo, Japan,
[2]Department of Anesthesiology, Philippine Children's Medical Center, Manila, Philippines

Japan's new clinical training system for doctors that started in 2004 has been modified multiple times as it created regional imbalance of the distributions of doctors. In recent years, the regionally balanced distribution of doctors' training system has been required, and improving the program in order for each hospital to create unique training program has been encouraged. There has been shortage of children patients in the clinical training grounds due to the dwindling birth rate, shortage of pediatricians or the imbalanced distribution of them, but in order to supplement the shortage of opportunities of clinical trainings with children, collaborating with medical institutions in those countries with the higher infant rates in the populations is one of the possible solutions. We look into the prospects and possibilities of the exchange training system for anesthesiologist-trainees between National Center for Global Health and Medicine and Philippine Children's Medical Center.

Development of ecohealth education curriculum in Teacher Training College toward the realizing ESD, Lao PDR - lesson learnt from demonstration lesson about Garbage and Health -

KOJI NAGATA[1], SACHI TOMOKAWA[1], NATSUMI KOBAYASHI [1], ASAKURA TAKASHI[2],
RYUICHI WATANABE[1], SHOHEI KOKUDO[3], KHAMHEANG UTTHA[4], KANHAVONG BOUNSENG[4],
LUDETMOUNSONE BOUAPHANH[4], DOUANGCHANH PHOUMY[4], KAZUHIKO MOJI[5]
[1]Graduate School of Education, Shinshu University, Nagano, Japan, [2]Faculty of Education, Tokyo Gakugei University, Tokyo, Japan, [3]Graduate School of Human Development and Environment, Kobe University, Japan,
[4]Faculty of Education, National University of Laos, Vientiane, Lao PDR,
[5] Graduate School of International Health Development, Nagasaki University, Japan

Introduction Recently, environmental problems become tangible in Lao PDR. At the same time, health problems which are caused by environmental destruction are become seriously. In this back ground ecohealth education was developed in Lao PDR. The purpose of ecohealth education is to acquire good behavior about environment and health. However, there are not sufficient teaching method and lack of teaching materials for ecohealth education. In addition lesson about garbage-disposal problem and health was not well developed.
Outline Lesson plan and teaching materials about garbage-disposal problem and health were developed and demonstration lesson was conducted to 60 students of Teacher Training College in Savannakhet on March 2014. The lesson aimed to make understand 1) influence of littering garbage on environment and human's health, 2) the necessity of using garbage box. Moreover teacher used quiz about littering plastic bag and a can in the developed lesson. After finish the lesson, the impression and evaluation of lesson collected from students by self-writing questionnaires, from Laotian's researchers who observed the lesson by oral interview.
Discussion The students got interest influence of lettering on human's health and animal. Laotian's researchers evaluated effectiveness of utilizing quiz in the lesson. In addition they pointed out that necessity of teaching about separating trash and demonstrate of familiar example for students.
Conclusion In the lesson about garbage-disposal problem and human's health, it was suggested that demonstrating familiar example for students, introducing necessity of separating trash and using the quiz in the lesson were quite effective.

Development of ecohealth education curriculum in Teacher Training College, Lao PDR - lesson learn from demonstration lesson about linkage - between environment and health -

SACHI TOMOKAWA[1], TAKASHI ASAKURA[2], RYUICHI WATANABE[1], SYOHEI KOKUDO[3], KEOSADA NGOUAY[4], KHAMHEANG UTTHA[4], KANHAVONG BOUNSENG[4], LUDETMOUNSONE BOUAPHNH[4], DOUANGCHANH PHOUMY[4], KAZUHIKO MOJI[5]
[1]Faculty of Education, Shinshu University, Nagano, Japan, [2]Faculty of Education,Tokyo Gakugei University,Tokyo,Japn, [3]Graduate School of Human Development and Environment,Kobe University,Japan, [4]Faculty of Education,National University of Laos,Vientiane,Lao PDR, [5]Graduate School of International Health Development,Nagasaki University,Japan

Introduction Recently, in Lao PDR, environmental issues have become social problems as health problems caused by environmental destructions have become serious. In this background, ecohealth education has been developed. The purpose of the ecohealth education is to acquire good behavior about environment and health. However, in the current curriculum of teacher training, there are problems of the lack of time to teaching methods and teaching materials on ecohealth education.
Outline In this study, lesson plan and teaching materials were developed in Teacher Training College (TTC) in Savannakhet Province, Lao PDR on March 2014. We conducted a demonstration lesson for 60 students and some Laotian researchers to evaluate them. The lesson consisted of following three parts, 1) write down the important or essential items in student's life, school, country and world, 2) mark the items related to health and environment, respectively, and 3) discuss relationships between marked items. After the lesson, we corrected the impressions and evaluations of lesson from students by self-writing questionnaires and from Laotian researchers by oral interview.
Discussion The students found the importance of environment, health and linkage between environment and health through the lesson. Laotian researchers appreciated the effectiveness of dialog and discussion in the lesson.
Conclusion We designed the lesson to promote an understanding about linkage between environment and health, and through the lesson, the students found the importance of environment and health in their life. Therefore, it was indicated that the developed lesson in this study was applicable and adequate in TTC, Lao PDR.

Challenge of Decentralization of the training management for the Provincial level in Cambodia

YUKO TAKAHASHI[1], IZUMI MURAKAMI[1], YURIKO EGAMI[1,2]
[1]JICA Project for Improving Maternal and Newborn Care through Midwifery Capacity Development, Phnom Penh, Cambodia, [2]Japan Bureau of International Medical Cooperation, National Center for Grobal Health and Medicine

Introduction: Human resource is an important property for provision of good health services. In order to assure the qualified human resources, Ministry of Health (MoH) has a strategy on their production, deployment and retention. MoH has revised CPA guideline with special focuses on trainings by setting up the instruction of establishing a training unit (TU) at CPA3 hospitals.Outline: Kampong Cham Provincial Hospital(KCM PH) has established a TU in collaboration with National Maternal and Child Health Center (NMCHC) and JICA project. KCM PH is the first CPA3 hospital to have its own TU and training center in the country. In order to share its experiences, guideline for TU in KCM PH and a document which shows the process of establishment have been prepared. Organogram of TU in revised CPA guideline was quoted from KCM guideline. KCM TU was officially recognized by MoH to conduct trainings and monitor trainees within the province and its region. This decentralizing system would be applicable to Basic Emergency Obstetric and Newborn Care (BEmONC) training in the near future.Discussion: Working in accordance with government policy has contributed to establish the foundation of training management system for decentralization. Commitment of MoH and NMCHC is an important factor to strengthen this training system.Conclusion: It is important to pile up good practices and challenges in KCM. KCM TU is expected to play a leading role as a model government structure for all the CPA3 hospitals in nationwide.

The results for 2014 of endoscopy training in Asia endoscopy center (AEC): a collaboration between Nagoya University and Hue University of medicine and Pharmacy (HUP), Central Vietnam

KENTA YAMAMOTO[1], Ryoji Miyahara[2], Hiroki Kawashima[2], Kazuhiko Hayashi[2], Yoshiki Hirooka[3], Huy Tran Van [1], Hidemi Goto[2]
[1]Department of Endoscopy, Hue University of Medicine and Pharmacy,
[2]Department of Gastroenterology and Hepatology, Nagoya University Graduate School of Medicine,
[3]Department of Endoscopy, Nagoya University Hospital

Introduction: There is a high demand on endoscopic skills from Japan. The training is necessary as limited number of endoscopist in Vietnam(0.5/100.000 population) compared to Japan(13.2/100.000).
Outline: We established the AEC at HUP since September 2013. We send 5 medical teams with 31 doctors and 18 nurses for 1 week advanced training course and one doctor stayed 1 year for basic training.
Discussion: The number of endoscopy is increasing; Esophagogastroduodenoscopy:EDG 5.953cases (188.7% year on year), Colonoscopy:CF 428(372.2%), Endoscopic retrograde cholangiopancreatography:ERCP 20(250%), Polypectomy:PP 66(507.7%), endoscopic ultrasonography:EUS 119, and double balloon endoscopy:DBE 4 until May 2014. The number of endoscopist is also increasing: CF: 7 doctors (6 doctors in last year), ERCP: 3 doctors(2), PP: 5 doctors(3), EUS 2 doctors.Common diseases are gastric cancer: 39 cases (41 cases at last year), Esophageal cancer: 14(9), Colon cancer: 5(6), Gastric ulcer: 24(53), Duodenal ulcer: 51(83). Forty doctors visited to study. Our center was reported in 6 newspaper and television coverage. The center becomes popular as we have been active in involving media reports. Several patients visited from all of Vietnam. The increasing cases provides the chance to improve skills. But the technology transfer of DBE and ERCP was relatively difficult because the number of patients is still limited.
Conclusion: Intensive and repeated training by Japanese team has been reported by many media. The reputation increase the number of patients and provide chance to study for doctors. It thus makes technology transfer easier. However, we must try to accelerate transfer of DBE and ERCP.

Contribution to the society through healthcare overseas training

ARISA YAMASHITA
TERUMO

[Introduction] I joined the 'Ryushoku' program which was coordinated by Human Resource Development Department in TERUMO Corporation in order to experience "contributing to the society through healthcare" of our company policy which we have never changed since our company was established in 1921. In this program, I worked at local NGO for two months and solved the problems which NGO had. [Outline] I solved two problems with NGO staff. First is to make the procedure of syringe quality control. Indonesian government has distributed the syringes for contraception. But these syringes have the quality problems. So NGO had made the draft to analyze problems. Second is to conduct the educational activity for residents. The number of diabetes patients is increasing in Indonesia. And most patients are not aware of their illness at early stage. So we conducted the educational event for them. [Results] Two problems were solved by cooperation with NGO staffs. (1) The procedure was accepted by government and I trained the operators to conduct this test. NGO could conduct from this year. (2) NGO staffs and I conducted the diabetes educational activity for people who live near NGO's clinic. At first they didn't know about the diabetes in detail. After that activity, they got the knowledge of diabetes and wanted to check their blood sugar level. We could also find pre-diabetes parson after test. [Conclusion] Through this program, I could learn how to manage the project and consider how to contribute the society in the world through the healthcare.

Global Health Security under Challenge of Equity: Sustainability of Global Pathogen Surveillance

Tomohiko MAKINO
Infectious Diseases Surveillance Center, National Institute of Infectious Diseases

Objective: After SARS in 2003, global health community has committed the transparent sharing of any emerging outbreaks. This momentum crystalized the revision of International Health Regulations (IHR) in 2005, which now binds member states to notify any potential public health emergencies within 24 hours to World Health Organization (WHO). In the extension of this global disease surveillance mechanism, pathogen surveillance and genome information sharing play key roles for both risk assessment and development of medical countermeasures. However, question still lies in whether benefits of surveillance are equitably shared by the member states. This research aims to analyze the intersection of different values and norms regarding the pathogen sharing and benefit sharing. Methods: WHO's Influenza A(H5N1) sample sharing dispute was chosen for the stakeholder analysis. The contrast was made clear between developed and developing nations on the viewpoints of various ethical concepts, including security and equity.Results: Different ethical values justified each side. Developed nations were for sample sharing because of security concern. They recognized the benefit of sample sharing as the preparedness and prompt response to be beneficial to the global community. Developing nations, in contrast, were hesitant to share pathogens unless equitable attribution of benefit to the originating nations was guaranteed. They challenged the sample sharing as the biopiracy, claiming the revenues of diagnostics and vaccines were monopolized by pharmaceutical industries mostly located in developed nations. Conclusion: Equitable sharing of benefit should be ensured for global health security by practical operation of disease surveillance.

Novel viruses hide among bats in Vietnam

FUTOSHI HASEBE[1,2,4], Thi Thu Thuy Nguyen[3], Hoai Linh Ly Pham[4], Thi Hang Pham [4], Co Thach Nguyen[3], Thi Dinh Dang[3], Takeshi Nabeshima[2], Fuxun Yu[2], Thi Quynh Mai Le[3], Kouichi Morita[2]
[1]Research in Tropical / Infectious Diseeases, Center of International Collaborative Research, Nagasaki University, Nagasaki, Japan, [2]Department of Virology, Institute of Tropical Medicine, Nagasaki University, Nagasaki, Japan, [3]Department of Virology, National Institute of Hygiene and Epidemiology, Hanoi, Vietnam, [4]Vietnam Research Station, Institute of Tropical Medicine, Nagasaki University, Hanoi, Vietnam

Since 2007, we have been conducting bat surveillance in Vietnam to (a) determine the prevalence of zoonotic viruses in bats, (b) investigate the status of bat hunting, cooking and dietary habits among the residents, and (c) assess risk of infection from bats to humans. Serum samples from bats were collected in Vietnam. Through indirect IgG ELISA, antibodies cross-reactive to Nipah virus of the genus Henipavirus were detected in 31 (41.9%) of 74 Rousettus leschenaultii and 3 (2.8%) of 109 Cynopterus sphinx. By neutralization test, 2 of R. leschenaultia were found positive. By RT-PCR, henipavirus gene was amplified from spleen specimens of 3 (2.0%) of 150 Cynopterus sphinx and 11(17.5%) of 63 Chaerephon plicata (insectivorous bat). Phylogenetic analysis of the sequenced PCR amplicons showed that the viruses detected in the present study belong to the different cluster of the paramyxovirus group. Our study suggests that henipaviruses and other paramyxoviruses are maintained among nonpteropid bats in Central and Northern Vietnam. In 2012, our study site was moved to Soc Trang province where Pteropus spp (natural reservoir host of Nipah virus) lives. Serum, fecal and urine specimens were collected from Pteropus spp. New alphaherpesvirus were isolated from 3 urine samples. Identification of these viruses is currently being analyzed. Thus, a long-term systematic surveillance of bats is needed to determine the ecological relationship between bats, humans, other animals and the environment.

Impact of food safety campaign on the Streptococcus suis infection in humans in Thailand

DAN TAKEUCHI[1], YUKIHIRO AKEDA[1], SHIGEYUKI HAMADA[2], KAZUNORI OISHI[1,3]
[1]Laboratory for Clinical Research on Infectious Diseases, RIMD, Osaka Univ.,
[2]Thailand-Japan Research Collaboration Center on Emerging and Reemerging Infections, Nonthaburi, Thailand, [3]National Institute of Infectious Diseases, Infectious Diseases Surveillance Center, Tokyo, Japan

Streptococcus suis is a gram-positive facultative anaerobic coccus and known to cause zoonosis in swine and human. S. suis infects humans through the consumption of raw pork products or the contact with pigs, or both. The previous population-based study at Phayao province in northern Thailand in 2010 revealed that fatality rate reached as high as 16% and more than 60% of cases developed meningitis, causing severe neurological sequelae including hearing loss. The calculated incidence rate of 6.2/100,000 suggested that more than 700 cases occurred in northern Thailand every year. Because the consumption of raw pork products was the major route of infection there, public intervention to change the eating behavior of local residents seemed to be effective to control the infection. Based on these facts, we have implemented the food safety campaign at Phayao province between January and December 2011. In the present study, we report the marked decrease of the incidence rate in 2011 and 2012 compared with that in 2010. Moreover, we estimated the infectivity of the pathogen from epidemiological data. (Collaborators: Anusak Kerdsin, Wantana Paveenkittinporn (Thai NIH))

Development of a new rapid diagnostic system which can differentiate one flavivirus infection from others

EIJI IDO[1,2], Steve Ahuka[3,4], KENTARO IBUKI[5], James Brandful[2], Jean-Jacques Muyembe[3,4]
[1]Graduate School of Medical and Dental Sciences, Tokyo Medical and Dental University, Tokyo, Japan.,
[2]Noguchi Memorial Institute for Medical Research, University of Ghana, Accra, Ghana.,
[3]Faculty of Medicine, University of Kinshasa, Kinshasa, Democratic Republic of Congo.,
[4]National Institute of Biomedical Research, Kinshasa, Democratic Republic of Congo.,
[5]Graduate School of Medical Sciences, Kyoto University, Kyoto, Japan.

Background: The flaviviridae is a large family consisting of various mosquito-borne viral pathogens such as dengue, West Nile, yellow fever viruses, etc. The importance of this family as emerging and reemerging infectious diseases is obvious because not only imported cases but occasional outbreaks are reported in subtropical and temperate zones probably in relation to the global warming and frequent human migrations. Their genomic structures are quite alike each other, and their clinical symptoms are also similar. Thus we needed to develop a new rapid diagnostic system that can differentiate one flavivirus from others.
Methods: Based on the sequence data on NCBI, we aligned all the sequences of each virus group to maximize the homology among all the members. We specifically paid attention to the NS5 genes encoding a methyl transferase and an RNA polymerase, both of which are essential enzymes for the viral replication.
Results: We have found that all the NS5 genes possess several highly conserved regions consisting of eight or more amino acids with the almost identical sequences. By setting the universal primers there, we have clarified that the phylogenetic topology obtained from relatively short fragments (ca. 600 bp) are almost equal to those from the whole NS5 genes and the full-genomes.
Conclusion: The present results suggest that this new PCR system can be used for a rapid differentiation purpose. In order to verify the usefulness of this new system, it would be necessary to further test various degenerative primers using clinical samples to achieve the optimal performance.

Prospective survey on Maternal near-miss in Burundi

KAYO KANEKO[1], Juma Dr.NDEREYE[2], NSHIMIRIIMANA Caritas[2],
Fabrice Dr.KAKUNZE[2], Keiko OSAKI[3], Kae Furukawa[1]
[1]JICA Expert, MOH Burundi, Burundi, [2]National Reproductive Health Programme, Ministry of Public Health and Fight Against HIV/AIDS, Burundi, [3]Senior adviser, Japanese International Cooperation Agency

<Introduction> In Burundi, free maternal and child care service started in 2006 and coverage of services has improved, although ever since MMR (500: DHS 2010) is still higher than in other Sub-Saharan countries. MMR in health facilities (84.6: MSPLS 2012) is still of particular concern. However not enough relevant information has been collected. Thus, JICA and MOH conducted a Prospective survey on maternal near-misses in 9 hospitals.
<Outline> 48 cases of maternal near-misses were found in one month. The breakdown: 15 cases of severe postpartum hemorrhage, 7 of ruptured uterus confirmed, 6 of ruptured uterus suspected, 6 of use of blood products, 4 of severe systemic infection, 4 of Eclampsia and others. 5 out of the 7 cases of ruptured uterus confirmed had a scarred uterus. 12 cases started delivering within 3 hrs from arrival and 5 stillbirths were observed.
<Discussion> Severe postpartum hemorrhage is considered a major cause of maternal death globally. The introduction of the free health care services coincided with the performance of cesarean section in all district hospitals. The risk of ruptured uterus can therefore be linked to the presence of scared uterus.
<Conclusion> Training to nurses for improving detection/response to severe postpartum hemorrhage and strengthening the sensitization skills on the risk of spontaneous vaginal delivery by women with the scared uteruses could reduce maternal mortality.

Survey on current situation of ENOC provided by general nurses in Burundi

KAYO KANEKO[1], Juma Dr.NDEREYE[2], Bonose Illuminée NAHABAGANWA[2],
Fabrice Dr.KAKUNZE[2], Kae FURUKAWA[1]
[1]JICA EXPERT, MOH BURUNDI, BURUNDI, [2]National Reproductive Health Programme, Ministry of Public Health and Fight Against HIV/AIDS, Burundi

<Introduction> In Burundi, due to 12 years of sociopolitical crisis, there have been delays in setting health regulations. Reproductive Health Care norms and related Protocols were approved in 2012. To develop a training implementing the Protocol on "Essential Obstetrical and Neonatal Care (ENOC) ", JICA and MOH conducted a survey targeting nurses (n = 100) using a self-administered questionnaire with questions assessing knowledge, practice and in-service training received.
<Outline> 72% nurses were working for different services in rotations. Proportion of nurses with knowledge on the following interventions; Prenatal Counseling 8%, ANC 15%, Active Management of the Third Stage of Labor 9%, Kangaroo Care 18%, PNC 18%, PMTCT 70%. Percentage of attended in-service training was; Prenatal Counseling 0%, ANC 6%, Use of Partogramme 42%, Active Management of the Third Stage of Labor 9%, Kangaroo Care 3%, PNC 3%, PMTCT 36%.
<Discussion> The nursing curriculum was only recently revised and the in-service training opportunities have been insufficient, which could explain the lack of knowledge. Knowledge, practice and in-service training received were expectedly high as many development partners supported PMTCT training. Practice rate use of Partogramme, PMTCT was higher than in-service training received. This could be due to skills learned on the job.
<Conclusion> A training package based on the concept of continuing care and the integration of services could be effective, due to the fact that nurses rotate in different services with lack of knowledge.

Observation point and Assessment for midwifery care at tertiary and referral hospital in Cambodia

KANAKO FUKUSHIMA[1], Hiroko Ooishi[2], Chea Ath Chhay Sveng[3], Yuriko Egami[1,2]
[1]Bureau of International Medical Cooperation, National Center for Global Health and Medicine, Tokyo, Japan., [2]JICA Project for Improving Maternal and Newborn Care through Midwifery Capacity Development, Cambodia, [3]Natinal Maternal and Child Health Center, Cambodia

Introduction: Project for Improving Maternal and Newborn Care through Midwifery Capacity Development has been carried out from 2010 until 2015 at National Maternal and Child Health Center (NMCHC), Kampong Cham Provincial Referral Hospital, and in Kampong Cham region in Cambodia. Midwifery trainers in target health facilities experienced evidence based midwifery care workshop. This study is to figure out the observation point and assessment of midwifery care in both hospitals.Outline: Direct observation of midwives' activity and semi-structured interview were conducted to midwives in both hospitals.Discussion: 10 cases were observed and 13 midwives are interviewed in NMCHC, 7 cases and 16 midwives in Kampong Cham, in June2014. 51 observation points were regarded from the time of entering the delivery room until the start of second stage of labour at NMCHC and 37 in Kampong Cham. 28 observation points were regarded from second stage of labour to third stage of labour and 13 in Kampong Cham. 20 observation points were regarded from third stage of labour until transfer to maternity ward and 33 in Kampong Cham. Delivery progress was assessed through power, passenger and passage and general condition.Conclusion: Delivery progress was assessed mainly through physical condition of woman. Challenges are remained on and after delivery.

Role of Village Midwives in Communities-Findings from the Survey in Sinnar State, the Republic of Sudan

AKIKO SHIMIZU[1], KUMIKO NAKANO[1], AKIKO HAYASHI[1], MICHIRU SUDA[2], YUKO SAITOU[1], JUNKO NAGANO[1], HIROMI INOUE[1], AYA YAGI[1], YASUHIDE NAKAMURA[1], Gaafaer Dr. Nada[3]
[1]Non-profit Organization HANDS (Health and Development Service), [2]TAC International Co., Ltd., [3]Director of Mother and Child Health Directorate, General Directorate of Primary Health Care, Federal Ministry of Health

[Background] Improving MCH is challenging but urgently needed in Sudan. Because of high proportion of home delivery in rural areas, it is indispensable to strengthen capacity of village midwives (VMW). Mother Nile Project was carried out since 2008, aiming for empowerment of VMW with in-service trainings in Sinnar, followed by phase II (MNP2) which was launched in 2011. MNP2 conducted the Comprehensive Approach in Sinnar including support of follow-up trainings, strengthening linkage between health facilities and VMW, and community activities.[Methods] Baseline Survey (BLS) and Endline Survey (ELS) were conducted before/after MNP2 interventions to measure the effectiveness of the Comprehensive Approach. Semi-structured questionnaires and in-depth interviews were conducted to examine background and service delivery information of VMW. 57 VMW were valid respondents in both surveys. [Results] 1) 30% of VMW had no formal education and 37% could not read/write Arabic. 2) Monthly average number of home delivery assisted by a VMW increased from 1.4 to 3.3. 3) Annual average number of emergency cases referred by a VMW rose from 1.6 to 3.5. 4) The number of VMW utilizing official monthly reports mounted from 33% to 80%. 5) VMW involved in community awareness activities increased from 74% to 88%. Qualitative analysis indicated that VMW referred mothers at risk by effectively employing community support and saved their lives.[Conclusion] Increased number of delivery and referral assisted by VMW might be associated with enhanced trust towards VMW in the communities, strengthened linkage between health facilities and VMW, and development of supervision system.

Why Potosi province in Bolivia has higher mortality rate of mother, infant, and neonate

MAKO KIMURA[1], Naoki Muroya[2], Ryo Yoshimaru[3], Motoyuki Yuasa4
[1]University of Toyama, Faculty of Medicine, Toyama, Japan, [2]Asahikawa Medical University, Faculty of Medicine, Asahikawa, Hokkaido, Japan, [3]Miyazaki Medical University, Faculty of Medicine, Miyazaki, Japan, [4]Department of Public Health, Juntendo University Graduate School of Medicine, Tokyo, Japan

[Introduction]Compared with the entire country of Bolivia, Potosi Province had higher mortality rates of mothers (352 per 100,000 live births), infants (101 per 1,000 live births) and neonates (52 per 1,000 live births) in 2002. The study was conducted to examine the reason why these mortality rates were higher in Potosi Province.[Methods]We had an opportunity to visit Bolivia using a student field matching program coordinated by the Japan Association for International Health Student Section. We interviewed common mothers, professionals working in the field of maternal health and pediatrics in Potosi such as medical workers and the Japan International Cooperation Agency (JICA) staff, regarding the possible reasons.[Results]We identified the following four reasons. First, the information system to monitor and evaluate health issues for mothers and infants was fragile. Second, health insurance to mothers was not completely accessible. Third, some mothers didn't recognize the importance of medical checkups and vaccinations. Mothers seemed to have low interest in their own health. In addition, health workers were likely to decline to use advanced or specialized medical remedies. Fourth, most childbirths were performed at home without Skilled Birth Attendance.[Conclusion]To decrease maternal and infant mortality rates in Potosi Province, it is necessary for healthcare workers and mothers to change their mindsets through training and health education.

The association between Mother and Child Health Handbook (MCHHB) and uptake of antenatal care (ANC) services and mother's health behavior in Palestine

HARUMI KITABAYASHI[1,2,3], KEIKO OSAKI[2], ATSUKO AOYAMA[3]
[1]Global Collaboration Center, Ochanomizu University, Tokyo, Japan,
[2]Japan International Cooperation Agency, [3]Nagoya University School of Medicine

Background: Introduced in 2008, the mother and child health handbook (MCHHB) covered 80% of newborns in Palestine in 2010. However, only half of mothers received MCHHB during pregnancy.Objective: This study aimed to identify the association between the receipt of MCHHB during pregnancy with mother's uptake of ANC services and health behavior.Method: Cross-sectional study was conducted using data set of Palestinian Family Health Survey 2010. Univariate logistic regression analysis was conducted on the data set of births in less than 12 months prior to the survey (n=2250) to examine the association.Result: Receipt of MCHHB during pregnancy was strongly associated with receiving information on diet (OR:1.51), high pregnancy risk (OR:2.22), breastfeeding (OR:1.99), family planning (OR: 2.14), postnatal care (OR:1.89), AIDS (OR: 1.59) and folic acid and iron tablet (OR:1.66). While receiving urine test (OR: 1.44), blood test (OR: 1.28) and breast cancer screening (OR: 1.88) was positively associated, ultrasound diagnosis (OR: 0.538) was negatively associated with MCHHB. Reception of MCHHB during pregnancy was also associated with the type of health facilities where ANC services were provided. When adjusted by the type of facilities, the association between MCHHB and service uptake was no longer statistically significant.Conclusion: Receipt of MCHHB during pregnancy increased the uptake of ANC services and information. Seeking care at the health centers of the ministry of health and UNRWA, which provide ANC services stipulated in the guidelines, may concurrently influence receipt of MCHHB and higher uptake of ANC services.

Development of a Data-Based Semi-Quantitative Food Frequency Questionnaire for the Eastern Rural Area of Rwanda

YUKIKO AMITANI[1], NORIKO SUDO[1], AYUMI YANAGISAWA[1], MAKIKO SEKIYAMA[2], YUKO CABALLERO[1], AOUN NAEL[2], Mukamugema Christine[3], Runesha Simeon[3], TAKUYA MATSUOKA[4], HIROAKI IMANISHI[4], TAKAYO SASAKI[4], HIROTAKA MATSUDA[2]
[1]The Graduate School of Humanities and Sciences, Ochanomizu University, Tokyo, Japan, [2]Graduate School of Frontier Sciences Integrated Research System for Sustainability Science, The University of Tokyo, Tokyo, Japan, [3]World Vision Rwanda, Kigali, Rwanda, 4World Vision Japan, Tokyo, Japan

<Background> Dietary assessments have seldom been carried out in Rwanda. This study aimed to develop a data-based semi-quantitative food frequency questionnaire (FFQ) that can be used for large-scale assessment with limited time and manpower. <Methods> A dietary survey was conducted using one-day weighed food records (FR) for 205 Rwandans aged 1 to 69 in March and August 2013 and March 2014 in the eastern rural area. The average portion size of food (consumed per meal) was determined by sex and age groups. Out of the food observed during the FR, we selected food items to be listed in our FFQ. The inclusion criteria were (a) the commonly consumed foods in this area, (b) expensive and irregularly consumed animal foods, (c) major contributors to energy intake, (d) purchased foods, and (e) foods with different consumption frequency by gender due to social and cultural reasons. Some cooked foods were further divided into subgroups according to the inclusion/exclusion of certain ingredients, which altered the nutritional value. <Results> Eighteen food items were selected and classified based on each criteria; (a) 11 items, (b) 2 items, (c) 2 items, (d) 2 items, and (e) 1 item. Porridge, soup, and agatogo, simmered dish with banana or potato, were divided into subgroups according to their ingredients. Nine categories were defined for the frequency of food intake, ranging from "never" to "3 times a day." <Conclusion> We developed an 18-item FFQ for the eastern rural area of Rwanda. Its validity and reproducibility will be examined an upcoming survey.

Impact of Nutrition Improvement Project in the Eastern Rural Area of Rwanda

AYUMI YANAGISAWA[1], NORIKO SUDO[1], YUKIKO AMITANI[1], MAKIKO SEKIYAMA[2], YUKO CABALLERO[1], AOUN NAEL[2], MUKAMUGEMA CHRISTINE[3], RUNESHA SIMEON[3], TAKUYA MATSUOKA[4], HIROAKI IMANISHI[4], TAKAYO SASAKI[4], HIROTAKA MATSUDA[2]
[1]The Graduate School of Humanities and Sciences, Ochanomizu University, Tokyo, Japan., [2]Graduate School of Frontier Sciences Integrated Research System for Sustainability Science, the University of Tokyo, Tokyo, Japan, [3]World Vision Rwanda, [4]World Vision Japan

<Background>In Rwanda, various projects are carried out by the government and NGOs. Based on a dietary assessment by a newly developed food frequency questionnaire (FFQ), effects of nutrition project by World Vision (WV), an international NGO were examined. <Methods>In March 2014, FFQs were administered to 71 adults in Rukara and Mwiri sectors. In Rukara, WV provides nutrition education for malnourished children and their parents. Mwiri served as a control sector. We asked the study participants to report their habitual consumption frequency of 18 food items listed in our FFQ by choosing from nine categories ("never" to "3 times a day"). Energy and nutrient intake was calculated by multiplying consumption frequency per day by energy and nutrient contents in mean portion size (average amount of intake per meal for men and women). The estimated energy and nutrient intake was compared with the reference values established by WHO and FAO (energy and protein requirements and recommended intake for calcium, iron, and vitamin A). <Results>Intake of energy and all nutrients was deficient in both areas. Calcium intake in Rukara and Mwiri met 41% and 25% of the reference values, respectively. Mode of consumption frequency of " Milk " , a major contributor to calcium intake, was "2-4 times a week" (36%) in Rukara and "never" (35%) in Mwiri. <Conclusion>Increased intake of milk and calcium observed in Rukara could be partly due to the intervention by WV. Other influential factors should be further examined.

Intra-Household Food allocation and Nutritional Status of rural residents in Eastern Province, Rwanda

MAKIKO SEKIYAMA[1], Nael Aoun[1], Noriko Suto[2], Yukiko Amitani[2], Yuko Caballero[2], Ayumi Yanagisawa[2], Mukamugema Christine[3], Runesha Simeon[3], Takuya Matsuoka[4], Hiroaki Imanishi[4], Takayo Sasaki[4], Hirotaka Matsuda[1]
[1]Graduate School of Frontier Sciences, the University of Tokyo, Tokyo, Japan, [2]Ochanomizu University, The Graduate School of Humanities and Sciences, [3]World Vision Rwanda, [4]World Vision Japan

[Background]Eradication of child malnutrition remains a social challenge in Rwanda, where 44% of under-five children are stunted (RDHS, 2010). As Eastern Rwanda is prone to droughts and famines, improvement of nutritional status among children through ensuring local food security is of significance. This study examines the relationship between intra-household food allocation under the current regional food security, and nutritional status among rural residents.[Methods] The sectors of Rukara and Mwiri in Eastern Rwanda constitute the study sites. Rukara is the target area of a child nutrition project implemented by World Vision, while Mwiri is the control one. Ninety percent of local people engage in subsistence agriculture. Low agricultural productivity causes serious poverty (WVR, 2009). Anthropometric measurements and a questionnaire survey were conducted in March 2013. One-day weighed food records were collected in March/August 2013 and March 2014.[Results]The prevalence of underweight among adults was not high (10.2%), while 32% of under-five children were stunted. Mean adult BMI in Rukara and Mwiri were 22.4 and 22.8, while HAZ of under-five children were -1.00 and -1.21, suggesting that child nutritional status has been improved due to the intervention. Children who frequently consumed animal products and milk had higher HAZ. Furthermore, HAZ of children were positively associated with the mother's access to kitchen garden and her knowledge on food/nutrition, indicating that these factors improved intra-household food allocation.[Conclusion]Ensuring mother's access to food resources is a sustainable strategy for improving child nutrition in eastern Rwanda.

The factor of nutritional condition of children living in rural villages in Republic of Zambia

JUNKO HASEGAWA[1,2], TARO YAMAUCHI[3]
[1]School of Rehabilitation Sciences, Health Sciences University of Hokkaido, [2]Graduate Schoold of Health Sciences, Hokkaido University, [3]Faculty of Health Sciences, Hokkaido University

[Background] In developing countries nutritional evaluation is mainly based on anthropometric measurement, consequently we can recognize only already malnourished children. Malnutrition in young children will remain into adolescence and adulthood owing to brain underdevelopment hence there is strong need of predictive method to assess malnutrition high-risk children. We have done preliminary research in southern province, Republic of Zambia in August 2013. We found maize-depended diet and shortage of both fat and animal protein intake in the area. The aim of this study is to detect major contributors to child malnutrition and verify relevancy between child nutrition and living environment.
[Method] We organize field research in the same area in August 2014. The field study contents following; 1) Child anthropometric data (length, weight, upper arm circumference). 2) Mother's anthropometric data (height, weight, skinfold thickness). 3) Semi-structured interview about household's living environment (family structure, hygiene, parental education, etc). We will validate related factors to children's nutritional status by calculating odds ratio between child's anthropometric data and their living environment.
[E xpected outcome] Comprehensive child nutrition screening method in consideration of living environment will be expected to develop in object area by specifying related factor of child nutritional status.

The efficacy of Japanese style school health program, in a lower educated country

KHALED RESHAD
NPO Karez Health and Educational services, Shizuoka, Japan

The health and education in Afghanistan is in a very lower level, because of the long term un-stability. The NPO of Karez Health and Educational Services performed clinical and educational activities in a rural area of Kandahar, Afghanistan. In past 12 years about 370,000 patients were treated and almost 100,000 children and women were vaccinated in the area, as a well health education was performed in 243,338 school children and to their mothers.

While, the rate of 5 year old children mortality and mother's mortality on labor is still very high in this country. For improving these factors, we are planning a Japanese style school health program to be tried in a model area in Kandahar. Allocation of a health worker in the school, for training the school children in their health promotion, for drinking healthy water and the necessity of vaccination and so on is expected to be of great value.

Obstacles due to the gape of traditional and educational restrictions are to be solved for the better implementation of the program.

Development and implementation of teacher training program for introduce school based health check up to primary and secondary school in Lao PDR

KAZUMA TORISAWA[1], SACHI TOMOKAWA[1], TAKASHI ASAKURA[2], RYUICHI WATANABE[1], SHOHEI KOKUDO[3], TOHRU KOISO[4], KEOSADA NGOUAY[5], KHAMHEANG UTTHA[5], KANHAVONG BOUNSENG[5], DOUANGCHANH PHOUMY[5], KAZUHIKO MOJI[6]
[1]Faculty of Education, Shinshu university, Ngano, Japan , [2]Faculty of Education, Tokyo Gakugei University, Tokyo, Japan , [3]Graduate School of Human Development and Environment, Kobe University, Japan, [4]School of Health and Sport Sciences, Chukyo University, Chiba, Japan, [5]Faculty of Education, National University of Laos, Vientiane, Lao PDR, [6]Graduate School of International Health Development, Nagasaki University, Japan

Introduction We have conducted an operational research for introducing health check-up (HC) to primary and secondary schools in collaboration with the Faculty of education of National University of Laos. In this research, we aim to clarify the situation of children's health and their nutrition status by conducting HC (height and weight measurement, hearing and eye test), and utilized the HC data in health education.

Outline We conducted a teacher training about school-based HC on September 2014 in Luang Prabang Teacher Training College (TTC), Lao PDR. Teachers from eleven primary and three secondary schools which are located near the TTC participated in this training. One week after the training, four trainers of the training visited the concerned schools and observed the good practices, challenges and points to be improved in each school. The data was analyzed by qualitative methods, and we conducted a descriptive study to clarify the challenges of teacher training.

Discussion As good practices, trained teachers conducted HC in an appropriate manner, shared well the contents of training and worked cooperatively with other non-trained teachers. On the contrary, as challenges and points to be improved, teachers set measuring instrument at inappropriate places. Moreover, the instructions given to the students, especially those about eye test, were not sufficient.

Conclusion For conducting effective HC, it needs to emphasize the importance and necessity about " how to set measurement instruments correctively", " instructions to school children" and "sharing trained contents with colleagues and cooperative work" in the further teacher training.

Effect of School Health Program during Primary School Days towards Present Parasite Infection among Secondary School Students - Research in Vientiane, Lao PDR

MICHIKO CHOSA[1,2], ITSUKO YOSHIDA[1,3], KONGSAP AKKHAVONG[1,4], SOMCHIT AKKHAVONG[1], TORU CHOSA[1,5]
[1]NPO JADDO, Kagoshima, Japan, [2]Wakamatsu Memorial Hospital, Kagoshima, Japan, [3]Tenri Health Care University, Nara, Japan, [4]Institute of Public Health, Lao PDR, [5]St. Mary's Hospital, Fukuoka, Japan

[Background] A school-based parasite control program becomes widespread as a part of school health curriculum in Vientiane, Lao PDR. This study aimed to investigate whether the parasite infection among secondary students today is associated with the degree of school health implementation during their primary school days. [Methods] In this cross-sectional study, 196 secondary school students aged 10-18y.o. are recruited. Stool examinations with Kato-Katz method and questionnaire survey were conducted. [Results] Parasite infection rate was 61.2% as a whole, Liver fluke 39.9% and Hookworm 37.1% respectively. Students with high scores of school health program at primary school showed significantly low infection rate in Hookworm, but not in Liver fluke. Hookworm infection was related to student's age, the father's low income and regular de-worming. Liver fluke infection was related to parents' low income. [Conclusion] Parasite control requires long-term effort. Even in Vientiane capital, continuity of the project is essential. For Hookworm control, strengthening the school health program and implementing control program at secondary school are suggested. For Liver fluke control, current school health program may not be effective, because it is related to poverty.

Basic Study for Parasite Control of School Children in Kammouane, Lao PDR

TORU CHOSA[1], KONGSAP AKKHAVONG[2,3], SOMCHIT AKKHAVONG[2], DAISAKU URABE[1,4], HIROAKI YAMAZAKI[1,4]
[1]St. Mary's Hospital, Fukuoka, Japan, [2]NPO JADDO, Kagoshima, Japan,
[3]Institute of Public Health, Lao PDR, [4]NPO ISAPH, Tokyo, Japan

[Background]NPO "JADDO" have been supporting school health in Lao PDR. This basic study is done to evaluate the severity of parasite infection among school children in Sebangfai district, Kammouane province.
[Methods]Health and Education local offices are interviewed. 3 primary schools are chosen near NPO "ISAPH" nutrition project site. 29 school children of Dong Makba are interviewed. 138 School children of Tung are tested stool specimen with Kato- Karz method.
[Results]Many of 29 children have correct knowledge on parasite transmission. They put shoes, wash hands, drink safe water and never take raw fish in school, however, NOT in their own houses. Stool exam shows that 45% of children are parasite positive, 43% liver fluke, 19.5% hook worm and 5.8% double infected.
[Conclusion]It's really important not only promoting people's correct knowledge, but also making action plan to trigger their behavior change.

Molecular characterization of human metapneumovirus circulating in Japan and the Philippines

Yujiao Yu[1], Michiko Okamoto[1], Kanako Otani[1,2], Makiko Kishi[1], Akira Suziki[2], Mariko Saito[3], Raita Tamaki[3], Segubre-Mercado Edelwisa[4], Lupisan Socorro[4], Olveda Remigio[4], Hitoshi Oshitani[1]
[1]Department of Virology, Tohoku University Graduate School of Medicine, Sendai, Japan., [2]Virus Research Center, Sendai Medical Center, Sendai, Japan., 3Tohoku-RITM Collaborating Research Center on Emerging and Reemerging Diseases, Muntinlupa City, the Philippines., [4]Research Institute for Tropical Medicine, Muntinlupa City, the Philippines.

[Background] Information regarding molecular characterization of human metapneumovirus (hMPV) around the world is still limited. We therefore investigated detection rate and genotypes of hMPV circulating in Japan and the Philippines.
[Methods] Nasopharyngeal swabs were collected from outpatients with acute respiratory infection at two pediatric clinics in Sendai, Japan (January 2010 to December 2013) and from outpatients with influenza-like illness including both adults and children in Leyte island and hospitalized children with severe pneumonia in Leyte and Palawan islands, the Philippines (January 2009 to December 2013). Real-time RT-PCR was performed to detect partial N gene of hMPV. As for hMPV positive samples, nucleotide sequence of partial F gene was analyzed to define hMPV genotypes.
[Results] The detection rates were 3.6% (115 out of 3161), 4.1% (93 out of 2270) and 4.5% (127 out of 2841) among outpatients in Sendai, among outpatients in Leyte Island, and among inpatients in both Leyte and Palawan islands, respectively. We could not determine dominant genotype before 2012 because hMPV positive samples are few. In Sendai, Japan, dominant genotype switched from A2b (in 2012) to B2 (in 2013). In the Philippines, dominant genotype was A2b in 2012 through 2013. Notably, we observed increasing detection of A2b genotype hMPV both in Japan and the Philippines in 2012.
[Conclusion] We found common and different characteristics in molecular epidemiology of hMPV in Japan and the Philippines. Continuous and global monitoring is needed to understand molecular epidemiology of hMPV.

Emerging GII.21 norovirus in Bhutanese children with diarrhea

Takashi Matsumoto[1], Takaaki Yahiro[1], Takeshi Wada[1], Chimmi Dorji[1], Sonam Wangchuk[1], Kunzang Pem Tshering[2]
[1]Department of Microbiology, Faculty of Medicine, Oita University, Yufu, Oita, Japan., [2]Public Health Laboratory, Deaprtment of Public Health, Ministry of Health, Royal Government of Bhutan, Thimphu, Bhutan., [3]Department of Pediatrics, Jigme Dorji Wangchuk National Referral Hospital, Thimphu, Bhutan.

Diarrhea is a leading cause of childhood morbidity and mortality in developing countries such as Bhutan, where diarrhea is widely prevalent among children, is no exception. However, information regarding etiology of viral diarrhea is scarce. Development of a comprehensive diagnostic, treatment and control strategy understanding the proportion of different viruses causing diarrhea is important. Norovirus (NoV) is one of the major viral agents of diarrhea, and accounts for 5-36% of sporadic cases. In an attempt to understand the diversity of viral agents, the present study sought to determine the proportion of diarrhea caused by NoV and their genotype distribution in Bhutanese children. From February 2010 through December 2012 stool samples were collected from children under 5 years of age who presented with watery diarrhea at Jigme Dorji Wangchuk National Referral Hospital in Thimphu. RNA was extracted from rotavirus negative stool samples and NoV was determined by RT-PCR. Genogroups and genotypes were determined by submitting nucleotide sequences of the amplicons to Norovirus Genotyping Tool. NoV was positive in 23.7% (64/270) cases. Among them 60 and 4 belonged to genogroup GII and GI, respectively. Yearly genotype distribution showed that in 2010, GII.3 was dominant followed by GII.4 and GII.7. In 2011, GII.21 was dominant followed by GII.6. In 2012, GII.21 continued to be the dominant strain. We conclude from the present study that NoV is a major cause of diarrhea in Bhutanese children with GII.21 as the dominant type which differs from the dominant GII.4 strain in other countries.

Travel-Related Co-Infection Case of *Salmonella enterica* serovar Typhi producing CMY-4 AmpC β-lactamase and Enteroaggregative *E. coli* Producing ESBL isolated from India

NOBUYUKI MISHIMA[1,2], Issaku Nakatani[1,2], Hideo Nakaya[3], Masumi Taguchi[4], Hiroyuki Amano[2], Toshimasa Nishiyama[1,2]
[1]Department of Public Health, Kansai Medical University, Osaka, Japan., [2]Center for Travel Medicine, Kansai Medical University Takii Hospital, [3]Division of Clinical Laboratory, Division of Infection Control and Prevention, Kansai Medical University Takii Hospital, [4]Division of Bacteriology, Department of Infectious Diseases, Osaka Prefectural Institute of Public Health

A 30-year-old Japanese male patient was admitted to our hospital with the complaints of sustained high-grade fever and watery diarrhea. He traveled and stayed at New Delhi and Varanasi for two weeks. Before the consultation, the former physician administered tosufloxacin tosilate hydrate and cefteram pivoxil orally for four days under the suspected diagnosis of bacterial enteritis. However, his condition did not get better with the administrations. And then *Salmonella enterica* serovar Typhi was detected from both blood culture and stool culture using samples collected at the first day of admission. Each bacterium showed different drug susceptibility, respectively. *S.* Typhi grown in stool culture produced CMY-4 AmpC β-lactamase. Simultaneously, ESBL-producing enteroaggregative *Escherichia coli* was detected from same diarrheal samples. Therefore, this patient was diagnosed with *Salmonella* Typhi and enteroaggregative *E. coli* co-infection. It was considered that the estimated infected area was in India, so that oral azithromycin was selected as a therapeutic agent. Although we confirmed negative results in blood culture, watery diarrhea lasted and the value of serum CRP revealed re-escalation. Consequently we changed to use ceftriaxone (1 x 2 g/day, 7 days) administered by intravenous root. After that, watery diarrhea was improved and his general status recuperated within several days. Laboratory findings also turned to normal values. Three times stool examination after discharge from hospital revealed no *S.* Typhi in the end. No case report of AmpC-producing *S.* Typhi infection in Japan at present. Accordingly we report this case with bibliographical consideration.

Evaluation for Maternal Deaths Audits in Benin with the routine health information system after 1 year of the institutionalization

TOSHIHARU OKAYASU[1,3], Clement Glele Kakai [2], MASAHIDE KONDO[4]
[1]JICA Benin, Cotonou, Benin, [2]Service of Epidemiology, Health Surveillance for Borders, Ports and Airports , National Public Health Department, Ministry of Health, Cotonou,Benin, [3]Department of Healthcare Policy and Management, Graduate School of Comprehensive Human Sciences, University of Tsukuba, [4]Department of Health Care Policy and Health Economics, Faculty of Medicine, University of Tsukuba

Introduction The maternal mortality rate is still high (397/100000, Demographic Health Survey 2006) in Benin. The reduction of the rate is one of the highest priorities for Ministry of Health(MOH). The Donors such as UNFPA, WHO, JICA, USAID have supported the training for Maternal Mortality Death audits since 2010. The MOH has institutionalized the obligation for organizing each maternal death audit with the ministerial ordinance since May 2013.
Outline In order to monitor the impact of the ministerial ordinance after 1 year of application, we would like to evaluate the actual achievement with the monthly routine database systems(SNIGS) and with the surveillance system for maternal death Audit and Riposte (SDMR).The data of SDMR was available from January to 15th week(mid-April) in 2014 at the moment of the study.
Discussion In 2012, 702 maternal death cases have been reported to the Direction of Mother and Child Health(DSME) of MOH, on the other hand 629 cases have been recorded in SNIGS. 471 cases have been registered in SNIGS in 2013. After introducing weekly reporting system, SDMR from January 2014, 114 maternal death cases have been reported with 32 cases audited (28. 1%) up to 15th week. The realization rate per department varied from 0% to 100%. Among 12 departments, one department was not yet reported. The low rate of the realization might be related with paramedical staff strike in 2014.
Conclusion The realization rate for maternal death audits is still low in 2014. SDRM should be still reinforced.The early solution for the strike is desired.

The status and factors influencing of adolescent pregnancy in rural Bangladesh: Evidence from Bangladesh Demographic Health Surveys.

Md. Harunor Rashid[1], Enbo Ma[2], Yukiko Wagatsuma[2]
[1]Graduate School of Comprehensive Human Sciences, University of Tsukuba,
[2]Faculty of Medicine, University of Tsukuba

Introduction: Pregnancy in adolescence has an increased risk of developing both maternal and fetal complications. Currently the World Health Organization recommended for reviewing and revising the adolescent health policies. This study aimed to determine the status and factors influencing the adolescent pregnancies in rural Bangladesh. Methods: This study used the Bangladesh Demographic Health Survey (BDHS) database from 1994-2011 and extracted 9510 adolescent cases and were analyzed. The BDHS had a large sample and used internationally accepted method. Results: The pregnancy rate was 50.5% in 1994 and 51.5% in 2011 (p&lt0.001). The rate was 56.1% of married women &lt18 years and 10.9% &lt18 years. The pregnancy rate was higher (61%) among non-educated women than higher educated women (25.2%). The rate was 60% for illiterate husbands and 38.9% for higher educated husband. The contraceptive use rate was lower (29.5%) among non-pregnant and 70.5% of pregnant women (p&lt0.0001). Adjusted odd ratio (95% Confidential Interval) was 11.6 (6.9-19.5) for women married &lt18 years, 4.2 (2.4-7.4) for non-educated women, 1.74 (1.2-2.5) for husband no education and 1.1 (0.8-1.4) for poor women. The odd ratio of contraceptive use was .281 among non-pregnant women than pregnant women. Conclusion: The marriage before 18 years is the major factor for the adolescent pregnancies. This information would be useful in reviewing the adolescent health policies for reducing the adolescent pregnancies in rural Bangladesh.

The expected role of traditional birth attendants in rural area in Sierra Leone

CHIEMI FUJII, YASUHIDE NAKAMURA
Graduate School of Human Sciences, Osaka University, Osaka, Japan

<Background>Recently in Sierra Leone, the role of Traditional Birth Attendants(TBAs) was confined to conduct delivery with Skilled Birth Attendants (SBAs) and to promote community people to deliver in the health facility. However, regardless of policy, 70% of pregnant women deliver at home by TBAs in rural area. Is it really possible to conduct all deliveries in the health facility? The objective is to clarify the role of birth attendants in rural area from the viewpoint of both birth attendants and pregnant women.<Methods>The field survey was conducted at Kambia District in Sierra Leone in March 2013.The semi-structured interview was made for 21 SBAs, 60 TBAs, and 280 mothers. <Results>60.4% of mothers delivered at home by TBAs and 53.8% among them chose TBAs because of trust. In the case of delivering in the health facility, 75.3% of mothers stayed within 1Km from the health facility.23.6% of mothers delivered in the health facility was referred by mainly TBAs. <Conclusion>In rural area, SBAs meet needs of community people who live near the health facility and TBAs respond to trust from community people who live far away from the health facility. Community people use both birth attendants as their situation demands and SBAs and TBAs coexist in rural area. Under this circumstance, the policy which delivery at home by TBAs is prohibited does not suit the present situation in rural area. Consequently, training and provision of medical equipment should be conducted for TBAs who play important role in maternal and child health in rural area.

Is postpartum hemorrhage underestimated in El Salvador?

EMI SASAGAWA[1], Lizeth Rosa María Elías de Buendía[2], Guillermo Antonio Ortiz Avendaño[2], Alba Marina Díaz de Navarro[2], Hector Anibal Barrera Erazo[2], Dalia Xochitl Sandoval López[2], Kiyoshi Kita[1], Chizuru Misago[3]
[1]School of International Health, Graduate School of Medicine, the University of Tokyo, Tokyo, Japan, [2]National Maternity Hospital "Raul Arguello Escolan", San Salvador, El Salvador,
[3]Department of International and Cultural Studies, Tsuda College, Tokyo, Japan

[Introduction] Postpartum hemorrhage (PPH), losing 500ml of blood or more after vaginal delivery, is a leading cause of maternal mortality, accounting for 25% of death. The current worldwide standard practice of postpartum blood loss assessment is visual estimation. However, according to the previous studies, visual estimates were inaccurate, 30-90% less than actual values for blood loss.
[Outline] To determine the prevalence of PPH, a cross-sectional study was conducted in a tertiary national maternity hospital in El Salvador, where the official data of PPH prevalence was 1.7% in 2013. Gestational age of 35 weeks or more were eligible for the study. Collection of blood loss was initiated immediately after delivery until 4 hours of delivery, and blood loss was measured using by scale and measuring cup.
[Discussion] From 1 April to 31 May,2014, 521 women were participated in the study, and 173 (33.2%) and 31 (6.0%) women were identified as PPH and severe PPH (losing 1,000ml blood or more), respectively. The result suggested the prevalence of PPH was 20 times higher than the official prevalence of PPH of this hospital.
[Conclusion] It was found that PPH had been underestimated in El Salvador. Early detection of PPH leads to the early and appropriate treatment; therefore, accurate measurement of blood loss should be strengthened.

The Continuing Professional Education (CPE) program for transformation-GORIN(Field Training Program)

SATOKO HORII, SHINJI TAKEMURA, YUICHI ANDO, TOMOFUMI SONE
Department of International Health and Collaboration, National Institute of Public Health, Saitama, Japan

Introduction:National Institute of Public Health (NIPH) has been engaged in the CPE since 1938 to strengthen the capacity of public health professional belonging to local governments for ensuring the quality of public health services in Japan. We introduce one of the NIPH's CPE program, GORIN (Field Training Program) which corresponds to Japan's health issues transition.Outline:GORIN started in 1961 as a NIPH's original training program. This program is carried out in the real field by the interprofessional team. Steps of GORIN are following; 1.setting up joint teams composed by different disciplines professions (GOdou) for dealing with the problems of a real public health situation. 2. going to the field (RINchi) and conducting research to identify and consolidate issues with local staffs. 3. planning, implementation, and evaluation of the activities identified by the research.4.feedback the results to the field and NIPH. The participants are required to accomplish all these steps during three months. Discussion:Since this program enables to simulate the real public health issue, the participants experience different situation from the routine activities and it enables to re-learn and rethink about public health. Through the process of problem solving, they can realize importance of team building and they can acquire leadership. It enables them to transform their organization to response their public health issues after return to their place.Conclusion:By extracting the core essence of GORIN, this program could be a model of CPE program in developing countries.

A new educational approach using information and communications technology for International Health classrooms

ISAMU KONISHI[1], Cunanan C Carmelo[2]
[1]Department of Physical Therapy, Niigata University of Health and Welfare, Niigata, Japan.,
[2]Angeles University Foundation, Angeles City, the Philippines.

Introduction People's concerns for global health have been contributing to the community of international health professionals. International Health classes in universities for health and welfare personnel are one of the key opportunities for encouraging young people's concerns. Globalization and the progress of information and communications technology (ICT) demand alternative educational approaches for 21st-century students.
Outline A new educational approach using ICT was applied to a class session of International Health for 160 students from 10 courses for health and welfare professionals. The students were in their third year of study at Niigata University of Health and Welfare (NUHW), and the study took place in June 2014. The students were given opportunities to participate in overseas interactions through Skype video. Their partners were university students studying Physical Therapy, who were the same age, from Angeles University Foundation, Pampanga Province, the Philippines. The Filipino students were field interns for Community-based Rehabilitation in the province. The NUHW students were divided into small groups, and 4 groups were simultaneously assigned to manage approximately 15-minute interactions with their Filipino partners using English in the classroom.
Discussion The new educational approach was successful in that it increased the NUHW students' self-awareness of their English-speaking potentiality. The topics for their interactive communication were general topics. In-depth discussion of more specific topics, such as poverty and disability, may be possible in the future.
Conclusion The new educational approach using ICT for International Health classrooms is feasible. Further study is required to evaluate the course outcomes.

A local community hospital in Hokkaido provides job opportunities for young medical staffs to support their work in developing countries

HIROTAKE MORI[1,2,3]
[1]Center for International Cooperation and Medical Training, Yoichi Hospital, Hokkaido, Japan,
[2]Carrier development for doctors, Non-governmental organization, GLOW,
[3]Faculty of Tropical Medicine, Mahidol University

A plenty of young medical staffs in Japan dream of working in developing countries. The opportunity of job after coming back from abroad, however, is not well prepared. Many community hospitals in rural areas in Japan, in contrast, are seeking for medical staffs.
A local community hospital in Hokkaido provides job opportunity to medical doctors who work/study abroad in developing countries. Food, shelter, and a car are provided so that they can come with empty hands. In addition, we provide scholarship for all hospital medical staffs up to two years who hope to work/study abroad. We also collaborate with a non-governmental organization, GLOW, working on the issue of gap on health and medical service in Japan and developing countries.
Increasing the job opportunity in Japan for young medical staffs who dream of working in developing countries is necessary for supporting their carrier. The multidisciplinary programs will be required.

Mental health research for Japanese who are living in Thailand

TAKESHI YODA[1], KATSUNORI YOKOYAMA[2], MARIE YORIKI[2], HIROMI SUZUKI[1], TOMOHIRO HIRAO[1]
[1]Department of Public Health, Faculty of Medicine, Kagawa University, Miki cho, Japan,
[2]Graduate school of Medicine, Kagawa University, Miki cho, Japan

It is officially reported that approximately 55,000 Japanese are living in Thailand. Generally speaking, foreign residents tend to have many stresses and sometimes they have mental disorders. We used self-administered questionnaires for an internet survey to investigate mental health condition for Japanese residents in Thailand. Total 48 residents were answered, of which male residents were 35 (72.9%) and mean age is 35.8 (SD 9.25) years old. We used Insomnia score (IS) from the Structured interview guide for the Hamilton depression rating scale (SIGH-D) and K6 score for the evaluation of the depression. We defined IS higher than 3 as serious insomnia and K6 higher than 5 as possible depression from former studies. Number of people with IS higher than 3 and K6 higher than 5 was 4 (8.3%) and 13 (27.1%), respectively. The characteristics of both serious insomnia and possible depression people were as follows: Poor Thai language skills in writing and reading, living less than 1 year or more than 5 years, having chronic diseases and smoker. There were significant associations with male, living alone and living less than 1 year from multiple logistic regression analyses on the association of possible depression after adjustment for age, gender, education, employment status, annual income, living period, language skills, and IS. Although the sample size is small, our research results suggest that relatively high number of people may suffer from mental illness compare with the Japanese living in Japan. To confirm our results, we will proceed to further research.

Factors associated with depression of caregivers involved in nursing care for community dwelling older people: Using National Data in Indonesia

SHIHO KASHIWAGI, Felipe Sandoval, Nanako Tamiya
Medical science, University of Tsukuba, Ibaraki, Japan

Background: Asian countries, where the speed of aging is quite rapid, might be necessary to discuss whether they will depend on the family based-care or adopt the Long-Term Care Insurance system like Japan and Korea in the near future. Researches regarding caregiver's burden in East Asian countries have been developed, however, those related to caregivers who nurse old peoples in Southeast Asia has not been fully documented yet.
Objective: We aim to clarify the factors associated with the depression of caregivers involved in nursing care for community dwelling older people in Indonesia where the problem of aging is one of the most important challenges for social policy.
Methods: We used the data of Indonesia Family Life Survey 4, which was investigated by Research and Development (RAND, USA) in 2007 and 2008 in Indonesia. We set lower than 10 points as "Low risk depression group" and higher than 11 points as "High risk depression group". Dependent variable was the depression, and independent variables were factors pointed out by reviewed literature. We selected variable with the significance level of ($p<0.2$). The logistic regression models were used for this analysis.
Results: The factors related positively to the high depression for caregivers in Indonesia were marriage status, while those related negatively to the depression were age and self-rated health condition.
Conclusion: From a secondary analysis, we found out that caregivers who tended to have a high risk of depression were older people, no marriage status, and low self-rated health condition in Indonesia.

Change during 6years of oral health status of children in Cambodia

REIKO NUMAGUCHI[1], MISATO FUJUYAMA[1], MASAKO SASAKI[1], CHIYOKO HAKUTA[2]
[1]NPO Kham Kham Khmer , Tokyo, Japan, [2]Tokyo Medical and Dental University, Tokyo, Japan

Introduction: We investigated the oral health status in the rural area in 2008, which showed the children living there had less decayed teeth than those in the suburban area. In 2014, we made the same investigation. This poster presentation reports the differences of the results of the two. Based on the results, we will continue the activity to prevent oral disease.
Outline: This survey was conducted in the elementary school of the rural area in January, 2008 and 2014. In 2008, the subjects were 37boys and 32 girls, age 5-15. In 2014, the subjects were 51 boys and 51 girls, age 4-14. In 2014, P0(no plaque))was 2.9%, P(plaque)was 97.1%, G0(no gingivitis)was 7.3%, G (gingivitis) was 62.8% and percentage of decayed milk teeth(t) was 35.8%.Compared to 2008, P increased 1.7 times (<0.01), G increased 1.9 times (<0.05)and percentage of t increased 1.7 times(<0.01).
Discussion: The results of this survey showed that the oral health status of the children in rural area had worsened. Those suggested the need of eating habits instruction, as well as oral and systemic health education to prevent oral diseases.
Conclusion: We should review the health education programs adapted to suit to the needs of the region by strengthening collaborations with teachers, parents and the primary school support groups.

Development of Nepalese growth curve and factors associated with nutritional status of aged 5-19 years school children, adolescents in Nepal

KAYAKO SAKISAKA[1,2], Kyo Hanada[3], MR Banjara [4], RK Pokharel[5], Mitsukuni Sugimoto[3]
[1]Chuo University, Tokyo, Japan, [2]Depatment of Community and Global Health, Graduate School of Medicine, the University of Tokyo, [3]Japan International Co-operation Agency (JICA), Tokyo, Japan,
[4]Public Health and Infectious Disease Research Center (PHIDReC), New Baneshwor, Kathmandu, Nepal,
[5]Child Health Division, Department of Health Services, Nepal, Teku, Kathmandu, Nepal

Background: A nationwide cross-sectional study was conducted to develop first national growth chart(weight for age, height for age, weight for height, and body mass index) for Nepalese school children. Methods: Anthropometric measurements were performed by asking the age and measuring and recording of height and weight. Multistage random sampling was used for the selection of schools, and students to include different ethnicity, age (5-19 years) and sex. The survey was conducted in 15 districts representing different ecological and administrative regions. Results: In total 6887 (Boys 3435, Girls 3452) students participated. Nepal's child growth chart was totally lower than WHO Child Growth Chart in height, weight, and BMI. In particular, weight and BMI curve once deteriorated from 5 and 8 years old. Determinants of lower BMI (below median) were, living in Terai, Hill ecological belt, far western development regions, religious minority group, and attending in community school.Conclusions: This study determined the growth standard for Nepalese children which can be used as reference values to determine nutrition status of Nepalese population.

Prevalence of Malnutrition and Associated Factors Among Children Aged 5-24 Months in the Eastern Province of the Republic of Cameroon

CHIKAKO NAGAHORI[1], Jean Paul TCHUANI[3], TARO YAMAUCHI[4]
[1]Faculty of Nursing, Okinawa Prefectural College of Nursing,
[2]Graduate School of Health Sciences, Hokkaido University,
[3]Batouri Health Office, Ministry of Public Health, Cameroon, [4]Faculty of Health Sciences, Hokkaido University

Background: The Republic of Cameroon has one of the highest child malnutrition rates, which has serious effects on children's lives and health. The present study was conducted to clarify nutritional status and associated factors in 5-24-month-old children in the district of Batouri, Republic of Cameroon.Methods:Participants were mothers with children aged 5-24 months who visited one of five health centers in February and March of 2013. Mothers were interviewed using a semi-structured questionnaire, and the child's weight, length, head circumference (HC), and mid-upper arm circumference (MUAC) was collected. The data were compared with child growth standards proposed by the WHO using Z-scores. The participants were divided into a malnutrition group (Z-score&lt-2) and a normal/good group (Z-score≥-2) based on these factors. Results:A total of 100 mother-child pairs participated in this study; valid data from 100 pairs were subjected to analysis. The rates of wasting, stunting, underweight, low HC, and low MUAC in the participating children were 6%, 31%, 14%, 0%, and 16%, respectively. Mothers' age, child's age, mothers' educational level, mothers who had family planning information, and type of water source were significantly associated with malnutrition. Conclusion:There was not significantly associated with children's malnutrition and complementary feeding. The high percentage of stunting was positively related to high chronic malnutrition. The result suggests that insufficient nutrition was more likely to begin after weaning.

Discussion regarding the cause of chronic malnutrition of the children aged 6 months to 5 years attending the Nepal's Lamjung District Community Hospital OPD during a one- month period in September 2013

MASAHIKO SAKAMOTO, KANAKO SAKAMOTO
Department of Pediatrics,Saku Central Hospital Advanced Care Center

[Introduction]Malnutrition among children has been a long-standing problem in Nepal. Though Lamjung is regarded as a relatively wealthy district in Nepal, a number of children remain malnourished. In 2013, we assessed the nutritional status of children attending the Lamjung District Hospital Outpatient Dept. (OPD). It showed 22% of the children were moderately to severely malnourished (reported on April, 2014). Here, we exploratively analyzed the causes of chronic malnutrition in these children.[Outline]A total of 206 children (6 months to 5 years old) attending the Lamjung District Hospital OPD on September 2013 were interviewed regarding their family background, perinatal period and food habit etc. Anthropometric measurements were used to assess the WHO Child Growth Standard. These data were analyzed using the Mann-Whitney U test.[Results]There were no significant differences in sex, religion, ethnicity, parents' educational level and family income, between 46 chronically malnourished children and 160 normally nourished children. Birth weight was lower in chronically malnourished children (p=0.03). Children with low birth weight (n=33) had a higher percentage of chronic malnutrition than children with normal birth weight (n=161)[Conclusion] There were no significant relationship between chronic malnutrition and social background of the children in our research. The differences in patients' socio-economic background may be smaller at the Lamjung District Hospital. However, birth weight seemed to be associated with chronic malnutrition. In order to improve chronic malnutrition, it is important to record birth weights and to follow-up on low birth weight children.

The cause-specific risk factors associated with neonatal mortality in Bangladesh

FARZANA FERDOUS[1], Sumon Kumar Das[2], Shahnawaz Ahmed[3],
Fahmida Dil Farzana[4], Yukiko Wagatsuma[5], Abu Syed Golam Faruque[6]
[1]Department of Clinical trial and Clinical Epidemiology, Graduate school of Comprehensive Human Sciences, University of Tsukuba, Japan, [2]International Centre for Diarrhoeal Disease Research,
Bangladesh (icddr,b), Dhaka, Bangladesh, [3]International Centre for Diarrhoeal Disease Research,
Bangladesh (icddr,b), Dhaka, Bangladesh, [4]International Centre for Diarrhoeal Disease Research,
Bangladesh (icddr,b), Dhaka, Bangladesh, [5]Department of Clinical trial and Clinical Epidemiology,
Graduate school of Comprehensive Human Sciences, University of Tsukuba, Japan,
[6]International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b), Dhaka, Bangladesh

Background: Inadequate perinatal care has adverse ramification on neonates including neonatal mortality. Objective: The aim was to identify the cause-specific factors associated with neonatal mortality in rural Bangladesh. Methodology: A prospective study was carried out in demographic surveillance system area of Mirzapur from 2008 to 2012. A total of 568 neonates died. WHO's verbal autopsy tools and ICD-10-code were used for diagnosing the causes of death.Results: A significant increased trend in neonatal death was observed from 15% in 2008 to 26% in 2012 (X2 for trend p&lt0.001). In multivariate analysis, associations were observed between early neonatal deaths (&lt=7-day) and ever able to suckle or bottle fed [0.15(0.06, 0.38)], fever [0.30(0.08, 1.06)], fast breathing [0.29(0.12, 0.70)], grunting [3.36(1.13, 9.99)], abdominal distention [0.08(0.02, 0.42)], postural skin rash [0.05(0.002, 0.92)] and yellow palm or sole [0.15(0.03, 0.92)]. Perinatal asphyxia (43%) was identified as primary cause of death followed by preterm delivery (24%), sepsis (14%) and pneumonia (9%). Newborns died within 1-day (3 times more), was associated with puffy face of mothers during pregnancy (4 times more), complications during delivery (2 times more), and other parts except head came first (3 times more) were associated with perinatal asphyxia. Moreover, preterm gestation [0.50(0.26, 0.95)], water broke before labor [0.32(0.15, 0.69)], those newborns who died within 0-1 day [0.25(0.13, 0.50)] and fever in neonates [3.48(1.43, 8.48)] were associated with sepsis deaths.Conclusion: Improved maternal health, special attention during pregnancy and neonatal health related risks, and perinatal care would reduce the neonatal death.

Loop-mediated isothermal amplification for rapid visual detection of *Naegleria fowleri*

Aongart Mahittikorn[1], Hirotake Mori[1], Supaluk Popruk[1],
Yaowalark Sukthana[1], Duangporn Nacapunchai[2]
[1]Department of Protozoology, Faculty of Tropical Medicine, Mahidol University, Bangkok, Thailand,
[2]Department of Parasitology, Faculty of Medical Technology, Mahidol University, Bangkok, Thailand

Loop-mediated isothermal amplification (LAMP) was designed for detection of *Naegleria fowleri*, which is a causative agent of the fatal disease called primary amebic meningoencephalitis. The primers set for the virulence-related gene consist of five primers targeting on specific gene of the parasite.
The LAMP assay could be performed within 60 min at 64° C in a water bath. Amplification products were visualized by hydroxy naphthol blue and agarose gel electrophoresis.
Sensitivity of the LAMP assay for detection of *N. fowleri* in pure cultures was 1 trophozoite per reaction. No cross reactivity with other tested relevant microorganisms such as *Acanthamoeba*, *Entamoeba*, *Giardia*, *Cryptosporidium*, *Escherichia coli* was found. After testing with spiked clinical samples and water collected from natural sources in Thailand, LAMP might be a promising tool for easy, rapid detection of *N. fowleri*.

Comparison of Toxoplasmosis diagnosis by indirect fluorescent antibody test and modified agglutination test in free-range chicken in Thailand

Udonsom Ruenruetai [1], Aongart Mahittikorn[1], Patcharee Chaichan[2], Yaowalark Sukthana[1]
[1]Department of Protozoology, Faculty of Tropical Medicine, Mahidol University, Bangkok, Thailand,
[2]Department of Parasitology, University of Limoges / Inserm UMR 1094 Tropical Neuroepidemiology, France

Toxoplasma gondii infection in free-range chicken is a good indicator for oocyst contamination in the environment because they feed on the ground. Moreover, if undercooked chicken meat is ingested raw, it will be an important source of infection for humans and other animals. The objective of this study was to determine the specific IgG antibody and compare the practical diagnostic tools for toxoplasmosis diagnosis in chicken using indirect fluorescent antibody test (IFAT) and modified agglutination test (MAT). In the preliminary study, we validated the two serological diagnostic tools for *T. gondii* infection with the positive chicken control sera. Our results showed a high agreement for *T. gondii* IgG antibody detection by two methods. Then the tools were evaluated with 150 free-range chicken. The seroprevalence of *T. gondii* by IFAT was 27.3% (41/150) with the titer varies from 1:16 to >256 and 7.3% (11/150) by MAT with a low titer from 1:10 to 1:40. In the next step, the seropositive results will be compared with the detection of *T. gondii* in the organ of the positive chicken by using animal inoculation and PCR.

Occurrence of Amphistomes in Lao-PDR Livestock

SURAPOL SANGUANKIAT[1], Marcello Otake Sato[2], Megumi Sato[3],
Tiengkham Pongvongsa[4], Moji Kazuhiko[5], Jitra Waikagul[1]
[1]Faculty of Tropical Medicine, Mahidol University, Bangkok, THAILAND, [2]School of Medicine, Universidade Federal do Tocantins, Palmas, Tocantins, Brasil , [3]School of Health Sciences, Faculty of Medicine, Niigata University, Niigata, Japan, [4]Station of Malariology, Parasitology & Entomology, Savannakhet Province, Lao PDR, [5]Graduate school of International Health Development, Nagasaki university, Nagasaki, Japan

Paramphistomes, the rumen flukes, are parasites that infect ruminants including cattle, goats, sheep, and water buffaloes. Paramphistomosis is a major cause of productivity and economic losses in different countries, the infection with these flukes cause an acute parasitic gastroenteritis with high morbidity and mortality rates, mostly among young animals. Eighteen adult parasites were randomly collected from different bovine stomachs selling at the market. This is the first report of the occurrence of the Paramphistomum in Laos. Morphological analysis revealed that they are Paramphistomum cervi (Family Paramphistomatidae). The PCR assays amplified the amplicons of 446 and 374 base pairs for CO1 of mtDNA and ITS2 of rDNA respectively. Pairwise alignment of those 2 genes sequences were performed and confirmed the morphological results. In an eco-health aspect, a livestock suffering from paramphistomosis can give direct impact to the human health. Since, meat products are important source of protein for local people, lower nutrition conversion of livestock may negatively influence to the nutritional status of the people. Once Paramphistomum have been described in southeastern Asian countries as Thailand with high prevalence, further studies on the prevalence of this amphistome in Laos and the ecology of the snails (intermediate hosts) is necessary. Further studies are needed to confirm the presence of this parasite in goats, other possible host and clear the natural history of this parasite infection of ruminants in the region and measure the indirect impact of this parasite infection in the health of the people.

Nematode infection among ruminants in monsoon climate (Ban-Lahanam, Lao PDR) and its role as food-borne zoonosis

TIPPAYARAT YOONUAN[1], Marcello Otake Sato[2], Megumi Sato[3],
Tiengkham Pongvongsa[4], Moji Kazuhiko[5], Jitra Waikagul1
[1] Department of Helminthology, Faculty of Tropical Medicine, Mahidol University, Bangkok THAILAND 10400,
[2]School of Medicine, Universidade Federal do Tocantins, Palmas, Tocantins, Brasil ,
[3]School of Health Sciences, Faculty of Medicine, Niigata University, Niigata, Japan,
[4]Station of Malariology, Parasitology & Entomology, Savannakhet Province, Lao PDR,
[5] Graduate school of International Health Development, Nagasaki university, Nagasaki, Japan

Trichostrongylids infection has gained significant public health importance since those infections have been reported in humans in Lao PDR. In this study, gastrointestinal nematodes were identified and the intensity of infections was determined in goats and cattle in Lahanam Village, Lao PDR. Fecal samples were collected from 14 goats and 11 bovines. Ninety three percent (13/14) of goats and 36% (3/11) of cattle were infected, with an average of 1,728 and 86 eggs per gram of faeces (EPG), respectively. Coproculture showed Trichostrongylus spp. (goats 16%; bovines 48%), Haemonchus spp. (goats 69%; bovines 37%), Cooperia spp. (bovines 8%) and Oesophagostomum spp. (goats 15%; bovines 6%). The necropsy of an adult goat confirmed the coproculture result and showed also infection by Trichuris spp. Morphologic diagnosis of Oesophagostomum spp., H. contortus and T. colubriformis were confirmed by sequencing of the ITS region. DNA sequences of T. colubriformis found in goats and in human previously reported in the same area were identical. Due to interactions between humans and goats in Lahanam Village and EPG results showing a high intensity of gastrointestinal nematode infection in these animals makes Trichostrongylus spp. infection in livestock an important public-health issue. The great quantities of eggs of Trichostrongylus in animal feaces contaminate the environment and plants used for human consumption and causing human gastroenteritis. Other ruminant parasites, such as Oesophagostomum and Haemonchus, found in caprines and bovines, are reported to be causes of zoonosis and their presence in humans should be investigated in future field surveys in this area.

Clinical and Molecular Diagnosis of Emerging Viral Infections in Champasak province, Laos, July to August 2013

VIENGVALY PHOMMANIVONG[1], Kanda Seiji[1], Andrew Waleluma Darcy[1], Takaki Shimono[1],
Lamaningaoa Pheophet [1], Bounthanh Phaytanavanh[2], Toshimasa Nishiyama[1]
[1]Department of Public Health, Kansai Medical University,
[2] Champasak hospital, Department of health Champasak province, Ministry of Health, Lao PDR

Background: Dengue fever (DF) is caused by an arboviral infection affecting many parts of the tropical and the subtropical regions worldwide mostly endemic in the Southeast-Asia countries. DF as an infectious diseases are still a major challenge for public health in Laos. By July 2012 the Ministry of Health (MOH) declared that the nationwide data on the dengue outbreak is presented with more severe cases hospitalization and more deaths than before; DENV-3 was dominant strain circulating in Champasak province that year. This study therefore was designed to identify the Dengue virus (DENV) types and molecular diagnosis of infectious diseases during outbreaks.
Methods: A study was conducted in Champasak province where 40 blood samples were collected from DF suspected patients during outbreaks, July to August 2013. Viral RNA was extracted from plasma samples using TRIzol reagents and analysed by RT-PCR using specific primers for DENV and Chikungunya virus (CHIKV). The amplified fragments from these samples were sequenced and phylogenetic analysis was performed using these sequence data.
Results: Among these patients, fever and headache was the most common symptoms in both disease, followed by myalgia, and arthralgia. The identification of the Dengue RNA in 13 samples, and also CHIKV has been seen in 10 cases. Only one sample had the co-infection with DENV-2 and DENV-3. A co-infection with DENV-3 and CHIKV was found in 2 samples.
Conclusion: However doctor diagnosed only DENV infections during the outbreak, we detected DENV-2 and DENV-3, and also detected CHIKV within these 40 patients' samples in Champasak province, Laos.

The role of endoscopy in the diagnosis of parasitic diseases at Asia Endoscopy center in central Vietnam (Hue)

KENTA YAMAMOTO[1], Ryoji Miyahara[2], Hiroki Kawashima[2],
Kazuhiko Hayashi[2], Yoshiki Hirooka[3], HUY TRAN VAN [1], Hidemi Goto[2]
[1]Department of Endoscopy, Hue University of Medicine and Pharmacy,
[2]Department of Gastroenterology and Hepatology, Nagoya University Graduate School of Medicine,
[3]Department of Endoscopy, Nagoya University Hospital

Introduction: Endoscopy is useful to diagnose and catch parasites, but many textbooks and lectures often use unclear pictures because parasites were observed by old machines in developing countries. Clear endoscopic pictures are required for better diagnosis
Outline: We investigated 14 cases of parasitic diseases (3 *Ascaris lumbricoides*, 5 *Necator americanus*, 5 *Trichuris trichiura*, 1 *Clonorchis sinensis*) which were diagnosed during from September 2013 to May 2014 in 5953 esophagogastroduodenoscopy, 428 colonoscopy and 20 endoscopic retrograde cholangiopancreatography.
Discussion: The median age was 54(34-77) years old. Male-female ratio was 2:3. The most common symptom was abdominal discomfort or pain (11 cases). The symptomatic duration was from 3 days to 1 year. About *N. americanus*, all cases were found at second part of duodenum. Connected head on the duodenal wall became clear when we took it away with clamp. The discrimination of male or female was clear because most cases showed the copulatory posture. *T. trichiura* were often found at ileocecal colon. All were found under the bad preparation for colonoscopy, it was suspected the good preparation might have washed parasite away. *A. lumbricoides* were found at duodenum, bileduct and colon. We recommend to catch it like a portfolio-style with polypectomy snare to avoid damage. Even the old endoscope can diagnose, but the new endoscope would be better for observation of parasites. We thought the number of parasitic patients may be less than real because about 70% patients came to hospital after taking anti-parasite-drug.
Conclusion: New endoscope is useful to take clear endoscopic pictures.

The Imported Case of Chikungunya Fever From Commonwealth of Dominica

KAZUO IMAI[1], TAKUYA MAEDA[1], YASUKO HONMA[2], SATORU MIYAKE[2], AOI MITARAI[2], YUKIKO KOBAYASHI[2], TOMOHIKO TAKASAKI[3], KOUKI KAKU[4], YASUSHI HIYAHIRA[5], AKIHIKO KAWANA[1]
[1]Department of Internal Medicine, Infectious and Pulmonary Disease, National Defense Medical College, Saitama, Japan., [2]Narita Airport Quarantine, Chiba, Japan., 3National Institute of Infectious Diseases, Tokyo, Japan., [4]Infectious Disease Epidemiology Analysis & Response, National Defense Medical College, Saitama, Japan., [5]Global Infectious Diseases and Tropical Medicine, National Defense Medical College, Saitama, Japan.

[Introduction] Chikungunya (CHIK) fever is an acute febrile illness, caused by mosquito-borne Chikungunya virus (CHIKV) infection. Previously restricted only to the countries in Asia and Africa, the first CHIK fever cases were reported from the Latin Caribbean regions in December 2013. The accumulation of patients was eventually recognized as an outbreak which was alerted by the CDC and ECDC in 2014. We here report the first imported case of CHIK fever from Commonwealth of Dominica in Japan.
[Outline] This present case is a 53-year-old Japanese woman who had stayed in the Commonwealth of Dominica from June X to X+15, 2014. On June X+15, she had developed high fever and malaise accompanied with generalized expressive joint pain. When arriving at Japan on X+18, she was immediately subjected to medical inspection in Narita airport, and then diagnosed as CHIK fever based on the RT-PCR. On June X+20, spotted rashes had appeared on whole body except her face. The laboratory analyses revealed leukopenia and occult uric blood on June X+21. CHIKV genome was not detected by RT-PCR in serum, saliva and urine samples on the same day. CHIKV specific IgG and IgM were detected.
[Discussion] The present report is the first evidence of travel-associated CHIK fever in Japan imported from the Latin Caribbean regions.
[Conclusion] Travelers who visit the Latin Caribbean regions should be appropriately alerted to the risk of CHIK fever. The epidemiological surveillance and laboratory diagnosis for the disease should be properly performed to prevent CHIKV-related unexpected events in Japan.

Intervention and Impact on Infection at the National Maternal and Child Health Center in Cambodia

SITHA MEAN[1], Tomoko Mori[3], Chie Iitake[3], Shinichi Hosokawa[3,] Azusa Iwamoto[2], Yuriko Egami[4], Sody Seang[1]
[1]Department of Neonatology, National Maternal and Child Health Center, Phnom penh, Cambodia,
[2]Bureau of International Medical Cooperation, National Center for Global Health and Medicine, Tokyo, Japan,
[3]Department of Pediatrics&Neonatology,National Center for Global Health and Medicine, Tokyo, Japan,
[4]Project for Improving Maternal and Newborn Care, Japan International Coopertion Agency

<Background>The National Maternal and Child Health Center (NMCHC) in Cambodia has an important role as a national top referral hospital for maternal and neonatal care. In 2012, the Neonatal Care Unit (NCU) experienced an outbreak of nosocomial infection. Since then, we did strengthen nosocomial infectious control in cooperation with experts (WHO/JICA). This report aims to assess the intervention after three years passed.
<Method>The intervention on nosocomial infection in the NCU consisted of standard precaution and cleaning of equipment. We compared both morbidity and mortality rate due to infection determined clinically in between 2011 and in 2013 using the statistics of NMCHC retrospectively.
<Results>The staff in NCU strongly strengthened the standard precaution such as hand wash and cleaning the machine as instructed by the manual. They avoided using same equipment for different patients strictly. The consumption amount of antiseptics has increased five times. The prevalence of clinical infection has decreased from 29.4% in 2011 to 24% in 2013(p=0.028). Mortality rate caused of clinical infection also decreased from 4.1% to 1.3 %(p=0.003).
<Conclusion>The interventions on nosocomial infection in the NCU are effective with limited resources. The strong efforts of staff for infectious control because of lessons learned from the past outbreak seem to be the key to success. In the future, not only NCU staff but also the infection control team should be involved as supervisor to maintain this ideal condition.

ASSESSMENT OF THE QUALITY OF GLIBENCLAMIDE AND METFORMIN TABLETS IN THE PHILIPPINES AND CAMBODIA

AYANO KOSUGI, YUMI SANAMI, ANDREA OROZCO,
NAOKO YOSHIDA, HIROHITO TSUBOI, KAZUKO KIMURA
Department of Drug Management and Policy, Institute of Medical,
Pharmaceutical and Health Sciences, Kanazawa University, Ishikawa, Japan

[Background]Spurious/falsely labeled/falsified/counterfeit (SFFC) and substandard medicines may induce the therapeutic failures. This study investigated the quality of diabetes therapeutic medicines, glibenclamdie (GLB) and metformin (MTF) tablets distributed in the Philippines and Cambodia.[Methods]The samples were collected in randomly selected outlets in Batangas province in Philippines (February, 2012), and the capital and surrounding areas in Cambodia (August, 2013). Quality test (quantity, content uniformity, and dissolution test), and authenticity investigation were performed.[Results]GLB (Philippines/Cambodia, 53/52 samples) and MTF (Philippines/Cambodia, 67/60 samples) were collected from 143 outlets. Any SFFC medicines were not found. In Cambodia, withdrawal GLB product was sold. In GLB samples, 13 samples (25%) collected in Philippines and 11 samples (21%) collected in Cambodia were unaccepted in one or more quality tests. All MTF samples collected in Philippines were accepted in all quality test; 2 samples (3%) collected in Cambodia were unaccepted in dissolution test, and one of them did not show the formulation property as sustained release tablet. Collected samples were kept beyond the stated temperature in the outlets without pharmacist in Philippines and all outlets in Cambodia.[Conclusion]The SFFC medicines were not found in this study, but some might be substandard. In some outlets, the medicines were kept in inappropriate storage condition. The quality management in the manufacturing and the distribution processes should be improved.[Acknowledgements]- JPMA/FDA in the Philippines 2011- WPRO/MOH in Cambodia 2013

Analysis of participant observation data to identify multi-angled viewpoints using text-mining approaches

MAYA IWASA[1], YUKO OHNO[2]
[1]Faculty of Nursing, Senri Kinran University, Osaka, Japan.,
[2]Dept. of Mathematical Health Science, Osaka University Graduate School of Medicine, Osaka, Japan

Objectives This study uses text mining to analyze participant observation data and identify the multi-angled viewpoints from which the observers perceive a topic.Method The participant observation data cover the cultural values of and the meaning and practice of " care " for the Serer ethnic group in the Republic of Senegal. The study divided the data into 18 descriptive classes and analyzed them by text mining to extract component words. It then examined the relationship between words that occur frequently and characteristic words that occur both frequently and infrequently in the data overall and in the 18 classes.Results Text mining produced 609 component words overall, of which 162 words occurred at least ten times, with the word "children" occurring most frequently, at 144 times. Of the 18 classes, that with the largest number of component words was "Values, beliefs, and taboos regarding diet and nutrition," with 73 instances. "Children," which occurred most frequently overall, occurred zero times in the classes "5. Worldview," "12. Political or legal influences," and "18. Work." Thus, words that occurred frequently overall were not necessarily frequently occurring characteristic words in every class, but rather appeared as infrequently occurring characteristic words. Conclusion Characteristic words that occurred infrequently may serve to caution researchers against oversight when collecting data by class. Exercising this caution is also thought to nurture new perspectives in future data collecting activities and help to interpret the data not only by class but also as a whole.

Layered Relationships in Caring for the Elderly in Southwestern Ethiopia

MARIKO NOGUCHI
Graduate School of Asian and African Area Studies, Kyoto University, Kyoto, Japan

In Ethiopia, care for the elderly living in rural areas has become an issue of growing concern. The changes brought about by urbanization are eroding the culture of intergenerational solidarity and mutual support that has long been prevalent. Recent studies have shown that in rural Ethiopia, 7.49% of the population aged 65 years or older lives alone. Many of them receive no support from family or relatives. However, in rural areas where much of the population relies on subsistence farming, how can the elderly who live alone or who live without their kinships maintain their livelihood?
In this presentation, I will describe the sort of care these elderly people need and who provides it to them, focusing in particular on activities of daily living (ADLs) and livelihood support. I have conducted field research since 2008 in Z area, which has a population of 292 in 75 households (as of 2009) and is located in M village, southwestern Ethiopia.
In Z area, there are 13 elderly households. Of these, 5 households consist of spouses and unmarried children, 5 households consist of only spouses, and 3 households consist of widows who live alone. Two of the widows have no relatives. All the elderly individuals in Z area are independent in basic ADLs. Many of them participate in livelihood activities insofar as they can handle labor and obtain necessary daily support. Each elderly household has a different living arrangement based on reciprocity within layered relationships such as kinship, neighbors, and mutual associations.

Consciousness Transformation of Healthcare Staff about Health Management Capacity at South Sulawesi Province in Indonesia

SAEKO HATTA, YASUKO KINJOU
Department of Nursing, Facluty of Human Health Sciences, Meio University, Okinawa, Japan

[Introduction] Project for Improvement of District Health Management Capacity was conducted in Indonesia in an attempt to reduce the east-west disparities within the nation, and to establish an effective regional healthcare administration system for decentralization. Interviews, consisting a part of Project, with the healthcare staffs who took part in the project revealed transitions of consciousness among them. [Overview] A healthcare staffs consisting of doctors and nurses at health center and health office, were interviewed for several times, and the results were put to analysis. In the initial year of the project, the staffs were observed with the sense of "doubt and isolation" in their project activities and an awareness that the project was their "task". In the second years, however, they received "positive stimulation from the local residents" and felt the "joy of engaging in their professional works", and were "awake to the importance of health prevention activities". While they agreed with the concept of the project, some were observed with a perception that they are "professional healthcare staff who help and guide local people".[Discussion] In Indonesia, regional healthcare lacked sufficient budget, and in the current instability resulting from the decentralization trend, the healthcare staffs found it difficult to maintain their motivation as the professionals responsible for healthcare. Through the project, they were reassured of the importance of their work and regained confidence in the notion, "prevention is the best cure". [Conclusion] We have clarified consciousness transformation in healthcare staffs who participated in the community health project.

Progress and future perspective to strengthen national health research system in Lao PDR

MIKI MIYOSHI[1], Sengchanh Kounnavong[2], Boupha Thongmalayvong[3], JUN KOBAYASHI[4], CHIAKI MIYOSHI[5]
[1]Center for International Collaboration and Partnership, National Institute of Health and Nutrition, Tokyo, Japan, [2]Health Research Division, National Institute of Public Health, Ministry of Health, Lao PDR, [3]Ethic Research Committee, National Institute of Public Health, Ministry of Health, Lao PDR, [4]Departmenr of Global Health, School of Health Sciences, University of the Ryukyus, Japan, [5]Bureau of International Medical Cooperation, National Center for Global Health and Medicine, Japan

Objective: In Lao PDR there has been no database on the existing health researches. Under the limited uniformity of the information obtained by hearings at the concerned organizations, several problems have been recognized including duplication of the collected data. Our research project therefore aims to strengthen national health research system in collaboration with the Ministry of Health (MoH), Lao PDR. Of which, this presentation focuses on "National Health Research Forum (NHRF)" and "Lao PDR Health Research Portal (Lao HRP)".
Method: The 7th NHRF was held on October 15-16 2013 at the National Institute of Public Health (NIOPH). Information on Lao HRP was obtained through the lecture at the NHRF and by interviews at the NIOPH.
Results and Discussion: At the 7th NHRF held in the NIOPH as a national collaboration institute, almost all of the major donor agencies supporting health researches participated, thereby having 134 participants in total. Notably, the Japanese Consortium for Lao Health Research mediated between the NIOPH and the University of Health Sciences of Lao PDR to organize the 7th NHRF jointly. It is expected that the country's ownership of research has been strengthened under the collaboration of these two institutions.
Lao HRP is a web-based system that has been developed on the NIOPH server. The main purposes were 1) Establishment of database for searching health research in Lao PDR and 2) Introduction of the online application system for ethical clearance. Further efforts are required to enhance the database and increase the utilization of online application system.

Distributions of severe occupational injuries by type of industry and type of event or exposure: a comparison between Thailand and Japan

Michiyo Yamakawa[1], Pornchai Sithisarankul[2], Takashi Yorifuji[3],
Sarunya Hengpraprom[2], Toshihide Tsuda[3]
[1]Okayama University, Center for Regional Research, Okayama, Japan,
[2]Department of Preventive and Social Medicine, Chulalongkorn University, Thailand,
[3]Department of Human Ecology, Graduate School of Environmental and Life Science,
Okayama University, Okayama, Japan

Background: To examine the distributions of severe occupational injuries by type of industry and type of event or exposure in Thailand, comparing with those in the same year and in the past in Japan. Methods: Data of severe injuries (death and disability with 4 or more working days loss) by the two types were obtained from the Thai Workmen's Compensation Fund records for 2009 and the Japanese national occupational accident statistics for 1999 and 2009. We calculated a proportion of severe injuries due to an event or exposure among the total in each industry. We then calculated proportionate ratios (PRs) for the proportion, comparing with trade (reference), and their 95% confidence intervals (CIs). Results: In 2009, the proportion of severe injuries due to contact with objects was 76.4% in Thailand. The proportion was 38.5% in Japan, followed by falls (38.0%). These types of event or exposure accounted for 50.2% and 33.3% of the total severe injuries in 1999, respectively. Specifically, in 2009, the proportion of severe injuries due to contact with objects was 85.4% in manufacturing with 1.3 of PR (95% CI: 1.3 to 1.3) relative to trade in Thailand. In Japan, the proportion was 59.2% with 2.0 of PR (1.9 to 2.0). The proportion was higher in 1999 (69.9%) but the PR was similar. Conclusion: This study indicated needs to address severe injuries due to contact with objects in manufacturing in Thailand. The occupational safety and health experiences of Japan could be informative for reducing these injuries in Thailand.

Consultation behavior of travelers whom the Narita Airport Quarantine Station referred to domestic medical settings in Japan

YUKIKO OGAWA, KEIKO WADA, YUKO SUZUKI, SACHIKO MIYOSHI, YASUKO HONMA, SATORU MIYAKE
Narita Airport Quarantine Station, Narita, Japan

The number of travelers who had clinical symptoms and visited to the health consultation rooms of Narita Airport Quarantine station was 5,254 between Jan 1st to Dec 31st. Some of them were referred to domestic medical settings to seek further medical examination and medical treatment with the referral forms from the Quarantine station. Two-hundred and twenty two cases of them were confirmed to have visited clinics after getting into Japan using by the consulting reports from the clinics. Out of them, 121 were men and 101 were female. And 20s age group visited clinics more than other age groups. More than half of them visited clinics on the day or the next day of arriving Japan. About 28% of them (63 case)were the cases of animal bites while fever (105 cases) and diarrhea (86 cases) were the most frequent symptoms. It was affirmed that the referral for rabies vaccination is one of the most important role of the health consultation at the Quarantine.

Comparison of end-of-life perspectives and issues between Kazakhstan and Japan

YURI AOCHI[1], SAHO OANA[2], ATSUSHI SAITO[3], ZHANAR AMANTAY[3],
SAMAL KADIKOVA[3], TAKAHIRO IWAGAKI[1], ATSUSHI OGIHARA[4]
[1]Graduate School of Human Sciences, Waseda University, Tokyo, Japan,
[2]School of Nursing, Luke's International University,
[3]Al-Farabi Kazakh National University, 4Faculty of Human Sciences, Waseda University

<Objective>The purpose of this study was to compare end-of-life perspectives and issues between Kazakhstan and Japan to reveal ethnic differences in end-of-life awareness and the view of the family.<Methods>1) A questionnaire survey was administered to 140 Kazakh students and 185 Japanese students. The questionnaire included questions from 13 topics about "view of the family","neighborhood relations", "family events"and "end-of-life care"in both Japanese and Russian. We performed a chi-square test for each question and calculated the kappa coefficient. We used the HALWIN for the analyses. 2) We performed semi-structured interviews with three Kazakh doctors regarding nursing care, end-of-life care, and problems with nursing and end-of-life care. We categorized the interview data using the Steps for Coding and Theorization (SCAT) method for qualitative data analysis. <Results/Discussion>1) Kazakh students, to a greater extent than Japanese students, showed high awareness of family relationships, good neighborhood relationships, strong valuation of tradition, and family-first thoughts during end-of-life situations2) Opinions surfaced that considerable burden was placed on the family. In addition, it was revealed that family relations were diluted, and there were several cases in which family care was consigned to a nursing facility.It is necessary to further investigate the current situation and elucidate family burden in caring for the elderly among Kazakh individuals. Further, this calls for an evaluation of social welfare policy.

The reconstruction of the smallpox outbreaks of at Amakusa region in early 19 century: a case study of the historical epidemiology

TOMO ICHIKAWA[1], Bonoru Higashi[2], Satoshi Murayama[3], Kanako Taira[4], Taro Yamamoto[1]
[1]Department of International Health, Institute of tropical medicine, Nagasaki university, Nagasaki, Japan.,
[2]Faculty of Letters, Kyoto Prefectural University, Kyoto, Japan., [3]Faculty of Education, Kagawa University, Takamatsu, Japan.,[4]School of Medicine, Nagasaki University, Nagasaki, Japan.

[Background]In the late Edo period, a large-scale smallpox epidemic broke out at Takahama village of Amakusa region. Because this disease is not a carrier disease and has no asymptomatic carrier state, it is suitable to reconstruct the epidemic situation and cause of that by using historical documents.[Methods]The historical documents of the village headman (Shoya) are available for the epidemiological analysis. Through the calculating the records of the population and isolation, it is possible to compare infection rate, fatality rate and verify the gap of each age and sex.[Results]At Takahama with a population of 2960, 182 people infected and 79 of them died. The epidemic period was 101 days from July 1807 to April 1808. Among the male the infection rate under 10 years old was 9.2%, teens was 5.5% and twenties was 4.8%. Among the female this rate of teens was 8.8% and of twenties was 7.9%. The fatality rate was over 20% at most generation.[Conclusion]Firstly we can reconstruct the smallpox epidemic through the historical records. Secondly this disease diffused with the opportunity of the people massed such as a funeral. After that by infection within the families patients much increased. Therefore Takahama village had the customary isolation to avoid the infection with the smallpox. Thirdly the smallpox often occurred among families which numbers were from 5 to 11. The smallpox epidemic became large-scale in this year because this disease was infected mainly in the households.

Sero-epidemiological survey of border malaria in Quang Tri, Vietnam

Le Thanh[1], Hoang Ha[1], Lam Chi Duc[2], Tran Van Thanh[3], Mai Nam[3],
Yuba Inamine[4], Changchun Li[4], Shusuke Nakazawa[5], Hisami Watanabe[4]
[1]Quang Tri Preventive Medicine Center, Quang Tri, Vietnam,
[2]Huong Hoa District Health Center, Quang Tri, Vietnam, [3]Quang Tri Health Service, Quang Tri, Vietnam,
[4]Center of Molecular Biosciences, University of the Ryukyus, Japan,
[5]Institute of Tropical Medicine, Nagasaki University, Japan

BACKGROUND: The prevalence of falciparum malaria was generally decreased under an effective malaria control in Indochina, while Pv malaria comes to front. This has been observed in the border areas between Vietnam and Laos. To promote Pv malaria control, information on the incidence of G6PD deficiency is essential. METHODS: We performed malaria survey to elucidate the border malaria in Vietnam, in Xy and Thanh commune, Houng Ha District by sero-epidemiological methods throughout the year. RESULTS and CONCLUSION: The border malaria in Vietnam was predominantly Pf malaria (156 patients/commune/year). The peak of malaria incidence was from August to October in these areas, except in 2012, in which the peak was observed between January and July. In these areas, malaria incidence was not decreasing. On the contrary, in Sepon District, Laos, malaria incidence was decreasing year by year. One of the reasons seems to be the movement of people between Vietnam and Laos in Xy commune. G6PD assay was employed by the paper filter samples. The lower G6PD activity cases of patients were about 4% in both communes. On the other hands, the anemia cases of younger villagers, under 15 y.o., tended to decrease as compared with that of Laos. This study demonstrates that Xy commune in the border area may be one of the last hotspots of malaria in Indochina.

Analysis of the mechanism of glucose metabolism inhibition, as a cause of hypoglycemia, in falciparum malaria using a systems biology approach

Koichi Shirakawa [1], Mitsuaki Nishibuchi[2]
[1]Department of Pathogenic Microbiology, Graduate School of Medicine, Kyoto, Japan,
[2]Division of Human-Nature Dynamics, Center for Southeast Asian Studies, Kyoto University, Kyoto, Japan

Background: It is well known that hypoglycemia occurs in falciparum malaria. This is caused by inhibition of glucose metabolism in the cell of the infected patients. Molecular genetic technique is generally used to elucidate the pathophysiology of the symptoms. However, it is hard to overview the interaction of the host factors and the factors of malaria parasites implicated in the disease symptoms. The metabolic pathway of glycolysis has been well studied in systems biology, but its relationship with a factor(s) derived from the malaria parasite is not well known. Methods: In our study, we analyzed pathway in the glycolysis of the host cells to see whether or not they are possibly implicated in the disease process and/or the symptoms by Cell Illustrator. This is one of the methods employed in systems biology, which is a new study field attempting to understand the system of the life as a whole using bioinformatics approaches. Results: Our result indicated that inhibition of glucose metabolism by the falciparum factors is not simple; not only a factor, the enzyme known to be responsible for the disease process, but also some other enzyme(s) are also implicated in the inhibition of the glucose metabolism in the host cells under the influence of the parasite factors. Conclusion: Although elucidation of exact identity of other enzymes awaits further pathophysiological and systems biological analyses, our results suggest that other enzymes than a factor and related host factors can be the targets of anti-malarial drugs.

Determining the associate factors of self-implemented preventive measures against malaria by ex-patients of microscopists in the Province of Palawan, the Philippines

EMILIE L. MATSUMOTO-TAKAHASHI[1,2], Pilarita Tongol-Rivera[3], Elena A Villacorte[3], Ray U Angluben[4], Shigeyuki Kano[1], Masamine Jimba[2]
[1]Department of Tropical Medicine and Malaria, Research Institute, National Center for Global Health and Medicine, [2]Department of Community and Global Health, Graduate School of Medicine, The University of Tokyo, [3]Department of Parasitology, College of Public Health, University of the Philippines Manila, the Philippines, [4]KLM/Pilipinas Shell Foundation, Inc, Palawan, the Philippines

Background: To take a significant step toward the elimination of malaria in Palawan island, the present study was conducted determining the associate factors of the self-implemented preventive measures against malaria by the ex-patients of the microscopists.
Methods: Structured interviews were carried out from January to February in 2012, in 20 remote malaria-endemic villages throughout Palawan. In total, 141 ex-patients who had previously been diagnosed malaria-positive by the microscopists, volunteered to participate in the present study. Structural equation modeling was conducted to determine factors associated with self-implemented preventive measures against malaria, which included: (1) place of residence; (2) socio-demographic characteristics; (3) knowledge on malaria; (4) participation in community awareness-raising activities for malaria prevention; and (5) satisfaction with microscopists.
Results: Structural equation modeling identified six significant factors independently associated with self-implemented preventive measures against malaria; ethnicity, knowledge on malaria transmission, knowledge on vector species, knowledge on vector's most active time, participation in awareness-raising activities for malaria prevention by microscopists, and satisfaction with microscopists. Tagalog ethnicity (the predominant ethnic group) was positively related to better self-implemented preventive measures.
Conclusion: In conclusion, aside from providing early diagnosis and treatment, the strengthening of awareness-raising activities by microscopists was suggested to be an effective strategy for reducing malaria re-infection in Palawan. These activities should be particularly strengthened to improve preventive measures implemented by ex-patients traveling to mountain areas and to enhance the knowledge on malaria transmission particularly among indigenous residents.

Nucleoside transporter 1 plays an essential role for transporting nucleosides and nucleobases in *Plasmodium berghei* in the asexual phase

MAMORU NIIKURA, Shinichi Inoue, Megumi Inoue, Fumie Kobayashi
Department of Infectious Diseases, Kyorin University School of Medicine, Tokyo, Japan

Background: Protozoan parasites rely on the host for the supply of purine nucleosides because they are unable to synthesize purine rings de novo. Nucleosides and nucleobases are transported across the parasite plasma membrane by nucleoside transporters (NTs) of *Plasmodium spp*. The *Plasmodium falciparum* (*Pf*) genome sequencing project revealed four nucleoside transporters, PfNT1, PfNT2, PfNT3 and PfNT4. In the genome of *P. berghei* (*Pb*) ANKA which is a rodent malaria parasite, the orthologues of PfNT1, PfNT2 and PfNT4 were identified. However, the roles of PbNT1, PbNT2 and PbNT4 in *Pb* ANKA remain unclear.
Methods: To investigate the roles of PbNT1, PbNT2 and PbNT4, we generated *pbnt1*-disrupted *Pb* ANKA (*pbnt1*-KO parasites), *pbnt2*-disrupted *Pb* ANKA (*pbnt2*-KO parasites) and *pbnt4*-disrupted *Pb* ANKA (*pbnt4*-KO parasites), and evaluated the effect of *pbnt1*, *pbnt2* or *pbnt4* disruption on the outcome of infection with *Pb* ANKA.
Results: We showed that the rapid increase of wild-type *Pb* ANKA (WT parasites) in mice early in infection was significantly inhibited by disruption of *pbnt1*. Moreover, *pbnt1*-KO parasite-infected mice showed neither cerebral paralysis nor cerebral haemorrhage, and all mice spontaneously recovered from infection. By contrast, mice infected with *pbnt2*-KO or *pbnt4*-KO parasites showed features similar to those of mice infected with WT parasites.
Conclusion: In the present study, we demonstrated that the high virulence of *Pb* ANKA in the asexual phase is suppressed by disruption of *pbnt1* but not *pbnt2* or *pbnt4*.

Antimalarial tropones: possibility of *Plasmodium falciparum* glyoxalase I (*pf*GLOI) inhibitor

AKI ISHIYAMA[1,2], MASATO IWATSUKI[1,2], TSUYOSHI YAMAMOTO[3],
HIROMI MIURA[2,3], KAZUHIKO OTOGURO[1], SATOSHI OMURA[3]
[1]Research Center for Tropical Diseases, Kitasato Institute for Life Sciences, Kitasato University, Tokyo, Japan,
[2]Graduate School of Infection Control Sciences, Kitasato University, Tokyo, Japan,
[3]Kitasato Institute for Life Sciences, Kitasato University, Tokyo, Japan

Background: During our screening program to discover new antimalarial compounds, we isolated several tropone-related compounds from the culture broth of *Penicillium viticola* FKI-4410 (1). Among them, puberulic acid showed potent antimalarial activity against *Plasmodium falciparum* K1 strain, with an IC_{50} of 0.051 μM. It has been reported that some tropone derivatives (hinokitiol etc.) show yeast glyoxalase I inhibitory activity and also have *in vitro* antimalarial activity (2). Therefore, we tested the *P. falciparum* glyoxalase I (*pf*GLOI) inhibitory activity of puberulic acid and its related compounds to determine if the mode of action of them is *pf*GLOI inhibitory activity. **Method**: Puberulic acid, stipitatic acid and viticolins A, B and C were obtained from a culture broth of FKI-4410. *Pf*GLOI inhibitory activity was evaluated using recombinant *Pf*GLOI with 96-well microtiter plate format. Formation of *S*-D-lactoylglutathione was monitored by the increase in absorbance at 240 nm in kinetic mode and percentage inhibition was calculated. **Results**: Puberulic acid showed moderate activity with the IC_{50} values of 60.43-84.13 μM. The IC_{50} values of stipitatic acid and viticolins A, B and C were more than 100 μM. Hinokitiol showed potent inhibitory activity against *Pf*GLOI, displaying IC_{50} values of 3.42-16.25 μM. **Conclusion**: These results indicate that the antimalarial mode of action of puberulic acid and its related compounds is not *via pf*GLOI inhibition in contrast to hinokitiol. 1) *J Antibiot*. **64**, 183-188 (2011), 2) *Biochim Biophys Acta*. **1208**, 127-135 (1994)

The efficacy of safer sex programme for men who have sex with men (MSM) in Malindi, Kenya

AKIKO FUKUDA[1], SHUTARO SUNAI[2]
[1]Graduate School of International Health Development, Nagasaki University, Nagasaki, Japan,
[2]Japan Overseas Cooperation Volunteers, Nagasaki, Japan

[Background]MSM is one of the most at risk population of HIV infection since 15.2% of the new HIV infections occurred among MSM and prisoners(NACC2009). Consequently, a study showed that 37% of MSM in Kenya have been engaged to anal intercourse without condoms. The programme aiming to promote condom use among MSM for their safety was developed by the authors, based on a health belief model and considering of MSM's unique sexual behavior through a focus group discussion. Community-based organization members were trained as peer educators. This study assessed the efficacy of the programme.[Methods]Interventions were held in Malindi, Kenya from November 2013 to February 2014. There were 170 MSM participants who were recruited by CBO members and pre and post surveys were conducted. The data analysis was conducted by "R". Wilcoxon Signed-rank Test was performed to determine the significant changes.[Results]Among 140 sample analyzed, the mean age was 26.8(median 26.0; SD 6.9; range 18-54), and 26% homosexual and 74% bisexual. Significant positive changes were seen among the score of self-esteem($p<0.05$), safer sex intention($p<0.001$)and knowledge($p<0.01$).[Conclusion]The efficacy of the programme was identified. However, we have to consider of selective bias on intervention site and recruitment of participants. Further study is needed to determine participants' behavior change towards condom use.

Focus Group Discussion (FGD) on Double Burden of Tuberculosis and non-Communicable Diseases (NCD) in Kathmandu, Nepal

YUKO TSUDA[1], Takanori Hirayama[2], Shinsaku Sakurada[3]
[1]Department of Infectious Disease Control, Osaka city Public Health Office, Osaka, Japan,
[2]Department of Program Support, Research Institute of Tuberculosis,
Japan Anti-Tuberculosis Association, Tokyo, Japan, [3]Bureau of International Medical Cooperation,
National Center for Global Health and Medicine, Tokyo, Japan

Introduction The research aims at understanding the present situation of TB and NCDs double burden in NepalOutline FGD was conducted on January 14, 2014 in Kathmandu, Nepal. Participants were 4 Doctors, 1 Government officer, 1WHO TB Expert, and 6 Public Health Nurses (PHNs)/Dots providers. Three themes were discussed as below: 1) the present situation of TB and NCDs double burden in Nepal and its associated problems. 2) measures that may help to reduce delay in case finding 3) The current situation of TB infection spread inside Nepalese hospitals and measures to control this situation.Discussion Both the medical doctors and the PHNs mentioned that they sometimes receive patients who suffer from the double burden of TB and NCD. However, the problem of double burden is yet unrecognized at the national level. Doctors mentioned that registration and follow-up system is not efficient in Nepal and no systematic study has yet been conducted. PHNs mentioned that it is necessary to provide routine examination/ screening test. Furthermore, the participants were of the opinion that providing training and education for health care workers and patients increases the chance of early case finding. Conclusion Although it was long suspected that Nepal suffers from the double burden of TB and NCDs, no quantitative data are yet available to support this claim. Further systematic and quantitative studies should be performed in order to adopt effective measures to reduce the double burden problem.

Monitoring of *Vibrio cholerae* in environmental water in Nam Dinh province, Northern Vietnam

TAICHIRO TAKEMURA[1], Luong Thi Tran[1], Tu Dong Nguyen[2],
Cuong Tuan Ngo[2], Asako Tokizawa[1], Minh Binh Nguyen[2], Tetsu Yamashiro[1]
[1]1-12-4 Sakamoto, Nagasaki-shi, Nagasaki, Japan,
[2]Department of Bacteriology, National Insitute of Hygine and Epidemiology, Vietnam

Vibrio cholerae serogroup O1 and O139 are the causative agents of severe diarrheal disease, cholera. Since the cholera has been considered as water-borne diseases, monitoring of environmental water for *V. cholerae* strains is important to identify the source of epidemic of cholera. We carried out monitoring of *V. cholerae* and vibrio phages in environmental water in Nam Dinh province, northern Vietnam. Blackish water and shrimp swab samples were collected at 10 locations, in every two months from Jul 2013 to Feb 2014. *V. cholerae* strains were isolated by culture using selective medium. The presence of *V. cholerae* was detected by PCR targeting *O1, O139, toxR*, and toxigenic *ctxA* genes. Vibrio phages (fs1 and fs2) were also detected by PCR. We have succeeded to isolate serogroup O1 strains from three sites, but the *ctxA* genes were not possessed by the strains. O139 gene was detected from one sample, but *V. cholerae* O139 strains were not isolated. Throughout the monitoring period, *toxR* was detected in the 97% of the samples, but the *ctxA* was not detected from all the samples. Vibrio phage fs1 and fs2 were detected 26% and 57% respectively. Clear correlation to *V. cholerae* detection was not observed. In Vietnam, the reiterating cholera outbreak had been reported every 5 to 7 years until 2009, but new cholera patients have not identified after that. Continuous monitoring of *V. cholerae* in environmental water should contribute to detection of cholera outbreak, also clarifying the toxigenic conversion and genome dynamics of *V. cholerae*.

Hospital based study of childhood diarrhea in Vietnam

Asako Tokizawa[1], Binh Minh Nguyen[2], Van Trang Nguyen[2], Thi Thu Huong Vu[2], Dinh Thiem Vu[2], Thi Thanh Tao Nguyen[3], Van Chuyen Vu[4], Duc Luu Do[5], Taichiro Takemura[1], Thi Phuong Lan Nguyen[3], Tetsu Yamashiro[1]
[1]Institute of Tropical Medicine, Nagasaki University, [2]National Institute of Hygiene and Epidemiology, Vietnam, [3]Pasteur Institute Ho Chi Minh City, Vietnam, [4]Nam Dinh Children's Hospital, Vietnam, [5]Nam Dinh Preventive Medical Center, Vietnam

A hospital based study for childhood diarrhea in Nam Dinh province, northern Vietnam and in Dong Nai province, southern Vietnam has been conducted since December 2012. The objective of the study is to determine a broad range of etiologies of acute childhood diarrhea including enteropathogenic bacteria, gastroenteritis viruses, and diarrhoeagenic protozoa in two regions and compared. A pair of diarrheal stool and questionnaire was collected from approximately 1,800 patients who hospitalized for the treatment of diarrhoea in total, in the period between December 2012 and May 2014.
Rotavirus, norovirus, and any pathotypes of diarrheagenic Escherichia coli (DEC) were the three most frequently isolated enteropathogens both in northern and southern Vietnam. Enteropathogenic bacteria were more frequently isolated in south than in north. Total isolation frequency of Salmonella spp., Shigella spp., and Campylobacter spp. was less than 5% in two areas, indicating their less significance as etiologies in acute childhood diarrhea in Vietnam. No Vibrio spp. were isolated in the study. Detection rate of each of adenovirus, sapovirus, and astrovirus was in the range of approximately 1.4-6.2% of the samples tested both in northern and southern Vietnam. Diarrhoeal protozoa were relatively frequently detected in two areas in the study.

Preliminary survey of intermediate host snails of *Schistosoma mekongi* in Cambodia

MASASHI KIRINOKI[1], Naoko Hayashi[1], Sinuon Muth[2], Socheat Duong[2], Chuor Char Meng[2], Chennan Wang[3], Hiroshi Ohmae[4], Takeshi Agatsuma[3], Yuichi Chigusa[1]
[1]Laboratory of Tropical Medicine and Parasitology, Dokkyo Medical University, Tochigi, Japan, [2]National Center for Malaria Control, Parasitology and Entomology, Ministry of Health, Phnom Penh, Cambodia, [3]Department of Environmental Health Sciences, Kochi Medical School, Kochi University, Kochi, Japan, [4]Department of Parasitology, National Institute of Infectious Diseases, Tokyo, Japan

(Introduction) The infectious larvae of *Schistosoma mekongi* swim out of their intermediate host snails, *Neotricula aperta*, into river water and infect humans by skin penetration and cause human schistosomiasis. In Cambodia, a provincial mass drug administration campaign was successfully executed; however, some villages still have a high risk of infection, suggesting that a control strategy targeting only humans may be insufficient for eliminating schistosomiasis mekongi. Therefore, we conducted a field survey of intermediate host snails as the first step in the development of a control strategy. (Outline) Snails were semi-quantitatively collected from 17 survey sites in four villages in Kratie Province, Cambodia from 2011 to 2014. The data collected from these sites showed that the number of snails fluctuated from year to year. There were sites where a large number of snails were collected compared with other sites every year and sites where no snails were collected. (Discussion and Conclusion) The results showed that the population dynamics of the snails are fluid and complex. In our recent study, DNA analysis showed that in addition to *N. aperta* γ -race, other *N. aperta*-like snails, which are suggested as a new lineage belonging to the genus *Neotricula*, are included in the samples collected during this survey (Wang, et al., 2014). Susceptibility of the *N. aperta*-like snails to *S. mekongi* is unknown. Identifying high-risk areas such as the stable habitats of *N. aperta* might enable elimination measures. Further studies are required to establish countermeasures against intermediate host snails for the elimination of schistosomiasis mekongi.

Fragment Evolution identified novel inhibitors of Trypanosoma cruzi Spermidine synthase

YASUSHI AMANO[1], Ichiji Namatame[1], Yukihiro Tateishi[1], Kazuya Honbou[1], Eiki Tanabe[1], Yohsuke Hagiwara[1], Shinji Ito[1], Masashi Imanishi[1], Tatsuya Niimi[1], Hitoshi Sakashita[1], Kazuhiko Yamasaki[2], Tomomi Kubota[3], Osamu Tani[2], Koji Furukawa[2]
[1]Medicinal Chemistry Research Labs, Drug Discovery Research, Astellas Pharma Inc, Ibaraki, Japan, [2]Biomedical Research Institute, National Institute of Advanced Industrial Science and Technology, Ibaraki, Japan, [3]Glycomedicine Technology Research Center, National Institute of Advanced Industrial Science and Technology, Ibaraki, Japan

[Background] We founded a collaborative research consortium for Neglected Tropical Diseases (NTDs) with Tokyo University, Tokyo Institute of Technology, and Nagasaki University. Especially in NTDs, drugs for diseases caused by protozoan parasites, such as Leishmaniasis, Chagas' disease, or sleeping sickness, are required to be developed because there are no effective drugs at present. The consortium developed iNTRODB, an integrated system for searching drug-target proteins from parasitic protozoa genomes, and selected spermidine synthase (SRM) as one of the drug targets. To discover selective SRM inhibitors, we applied the Fragment Evolution (FE) method.
[Methods] Recombinant Trypanosoma cruzi SRM enzyme was prepared. An enzymatic assay, a surface plasmon resonance (SPR) assay, and a thermal shift assay (TSA) were developed. We conducted screening campaign against fragment library containing 8,000 compounds using these three assays. Cocrystal structures of fragment hits and SRM were solved and structural information, such as binding sites or interactions, was obtained. Based on structural information, we explored highly potent inhibitors of SRM.
[Results] 57 and 95 fragment hits were identified by SPR assay and TSA, respectively. Approximately one third of fragment hits had inhibitory activity. Structural analysis revealed that fragment hits bound either the substrate binding pocket or the dimer interface. Finally, we identified several inhibitors which show IC50 values below 10 μM.
[Conclusion] FE methods discovered novel potent inhibitors of Trypanosoma cruzi SRM. Anti-trypanosoma activities of these inhibitors are being investigated. SRM inhibitors are expected to be developed as future drug candidates.

Pattern recognition approach to high-speed counting of live tripomastigotes

HIDENORI SAKANASHI[1], HIROKAZU NOSATO[1], EIICHI TAKAHASHI[1], MASAHIRO MURAKAWA[1], KOJI FURUKAWA[2], KAZUHIKO YAMASAKI[2], TOMOMI KUBOTA[3], TETSUO YANAGI[4], KENJI HIRAYAMA[4]
[1]Information Technology Research Institute, National Institute of Advanced Industrial Science and Technology, Tsukuba, Japan, [2]Biomedical Research Institute, National Institute of Advanced Industrial Science and Technology, Tsukuba, Japan, [3]Glycomedicine Technology Research Center, National Institute of Advanced Industrial Science and Technology, Tsukuba, Japan, [4]Institute of Tropical Medicine, Nagasaki University, Nagasaki, Japan

[Background] Trypanosoma cruzi is the parasitic protozoan causing Chagas disease. The reproducing process of T. cruzi in animal hosts takes place inside the various cells through binary fission of amastigotes (AMs). The AMs then transform into trypomastigotes (TMs), an infectious form of the protozoan. Burst of the infected cells allows infection of other cells by the disseminated TMs. For the discovery of new of anti-protozoan drugs, we so far consumed a lot of time and labor; i.e., in drug-screening assay the numbers of live wriggling TMs in culture suspension or stained AMs in fixed cells were counted under microscopes following a period of exposure with drugs. To improve the throughput of this screening, we developed a novel image analysis technology.
[Methods] The numbers of wriggling TMs in the microscope video images were examined by a pattern recognition approach, using a model representing the correlation between the moving images and the number of TMs. The correlation model was generated by analysis of the videos, which ware prepared for training in advance, by means of machine learning and pattern recognition based on the cubic higher-order local autocorrelation (CHLAC) features. CHLAC is advantageous in extracting and quantifying the characteristics appearing in the shape and motion of TMs.
[Results] We applied the devised method to several sets of T. cruzi cultivation using the fibroblasts derived from adult mouse brain as the host cells, and have succeeded in counting the number of live TMs in the culture with high efficiency and accuracy.

Drug discovery for anti-trypanosoma and target protein-inhibitor complex structure analysis using protein-ligand docking simulation

RYUNOSUKE YOSHINO[1], NOBUAKI YASUO[2], YOHSUKE HAGIWARA[3],
KAZUKI OHNO[3], ICHIJI NAMATAME[3], MASAYA ORITA[4], MASAKAZU SEKIJIMA[1,2]
[1]Global Scientific Information and Computing Center, Tokyo Institute of Technology, Tokyo, Japan,
[2]Department of Computer Science, Graduate School of Information Science and Engineering, Tokyo Institute of Technology, Tokyo, Japan, [3]Drug Discovery Research, Astellas Pharma Inc, Ibaraki, Japan,
[4]Drug Discovery Research, Astellas Pharma Inc, Ibaraki, Japan

Chagas disease is an infectious disease caused by the parasitic protozoan *Trypanosoma cruzi* (*T.cruzi*). This disease affects people from approximately 20 countries, particularly those living in the southern United States and Latin America, with 15 million people estimated to be infected. Nifurtimox and Benznidazole are currently available for treatment of Chagas disease. However, there are serious problems associated with their use, including adverse effects and limited effectiveness during the chronic phase of this disease. Thus, developing new therapeutic agents against *T. cruzi* infection is warranted. In order to develop a novel anti-trypanosoma drug, protein-ligand docking simulation was carried out to discover drug candidate. Furthermore, in vitro assay was conducted to determine IC_{50} of the selected compounds against a target protein. First, we focused on spermidine synthase (SRM) as target protein. SRM is an enzyme that takes part in the biosynthesis of spermine. This enzyme catalyzes polyamine group transposition to synthesize spermidine. Polyamine compounds contribute polymerase activity and nucleic acid stability. Therefore, SRM is promising as anti-trypanosoma target protein.We performed protein-ligand docking simulation for SRM active site using medicine-like compounds approximately 4,800,000 and evaluated medicine-like compounds with high docking score by *in vitro* assay. As a result, we obtained active compounds which inhibit SRM with values for IC_{50} of 10uM order. The optimization of these compounds would be accelerated by the model structure of SRM-active compound complex we obtained.

Discovery of dengue virus protease inhibitors and their inhibition mechanism through docking simulation

SHUNTARO CHIBA[1], YOHSUKE HAGIWARA[2], KAZUKI OHNO[2],
KAZUYA HOMBOH[2], MASAYA ORITA[3], MASAKAZU SEKIJIMA[1,4]
[1]ACLS, Tokyo Tech, Kanagawa, Japan, [2]Drug Discovery Research, Astellas Pharma Inc, Ibaraki, Japan,
[3]Drug Discovery Research, Astellas Pharma Inc, Ibaraki, Japan, [4]GSIC, Tokyo Tech, Tokyo, Japan

Dengue virus (DENV) is prevalent in tropical and sub-tropical regions and causes severe diseases such as dengue fever. Despite the urgent need for new prophylaxis and therapeutic methods for DENV-related diseases, vaccines and therapeutic agents have not been approved. To address this issue, we focused on DENV protease, which is indispensable for replication of the virus, as a target protein to develop an initial anti-DENV drug candidate. First, we applied docking simulation for the protease to search for potent compounds with potential inhibition activity from the millions of compounds currently available on the market. We then conducted inhibition activity assays of the candidate compounds with suspected inhibitory activity. We determined the IC50 of the candidate compounds, followed by mass and NMR spectra experiments, which indicated that no false-positive results were obtained in the assays. The peak value was 10 μM. We compared the structures of the candidate compounds with those of known inhibitors of DENV protease in a database and found no similarity. This finding is particularly relevant since a novel compound structure is necessary to develop a better drug candidate. To determine the inhibition mechanism of the candidate compounds, we conducted a more elaborate docking simulation for our compounds; the results were compared to those for compounds that were assumed to have inhibition activity but actually did not. The results showed that our candidate inhibitory compounds occupy a region deeper in the binding pocket than non-inhibitory compounds, which would allow them to acquire higher potency in inhibiting DENV protease.

Health-seeking Behavior for Their Children by Foreign Mothers in Japan

KYOKO SUDO, Yoko Hamamoto
National College of Nursing, Japan, Tokyo, Japan

Background: Children's health is influenced by the physical, mental, and social health of their family members as well as the social environment in which they are reared. The mothers' perception and/or attitude regarding healthcare for their children is considered vital to the child's health. When accessing healthcare, children in Japan with foreign mothers are considered to be vulnerable because their mothers may not be familiar with the Japanese culture and environment and may struggle with the language. The purpose of this study is to explore the health-seeking behavior of foreign mothers for their children in Japan.Methods: A Focus Group Interview (FGI) was conducted with foreign mothers living in Japan with children who were less than six years of age. The data was analyzed using content analysis. This research was approved by ethical committee of National Center for Global health and Medicine.Results: Six FGIs were conducted with four mothers from different countries. Health-seeking activities for their children by these foreign mothers included: "gathering information about child health management," "selection of adequate healthcare facilities," "preventive healthcare behaviors," "home management when the child is sick or ill," "decision leading to take the child to a healthcare facility," and "acceptance of the Japanese healthcare system." The results indicated that the foreign mothers were influenced by the Japanese and their native environment, culture, and traditions. Conclusion: Foreign mothers need to accept and integrate the procedures of the prevailing culture with their traditional attitudes when seeking healthcare for their children.

The current state of and challenges faced by disseminating information on maternal and child healthcare among expatriate community

MARIKO SAKAMOTO[1], IZUMI ASANO[1], SEIKO MIZUTANI[1], YUKO OHASHI[2], HIDEMI HASHIMOTO[3]
[1]College of Nursing, Aichi Medical University of Nagakute city, Japan,
[2]College of Life and Health Sciences, Chubu University of Kasugai city, Japan,
[3]Mie Prefectural College of Nursing of Tsu city, Japan

[Background] With a remarkable surge in the number of foreigners opting to live in Japan for a long term, it has become an urgent task to disseminate information on maternal and child health to foreign residents in the child-rearing age range. However, the actual state of information dissemination has not been sufficiently grasped. This study aimed to clarify the current state of and challenges faced by disseminating information on maternal and child healthcare to the foreign residents in City A.
[Methods] We conducted a semi-structured interview of 11 collaborators who are involved in maternal and child health support for foreigners in City A from July to November in 2013.
[Results and Discussion] In City A, interpreters were staffed at related organizations near areas having a large concentration of foreign residents, and "access to information via interpreters" and "access to information through day-care centers for children run by foreign managers" emerged as important sources of information. Though an information network existed among expatriate community, it was of limited scope as the parents are busy. City A promoted the dissemination of information by obtaining "cooperation from foreign residents." Some key challenges included "lack of understanding of the Japanese health care system," "cultural and habitual differences," and "excessive burden on specific interpreters."
[Conclusions] This study revealed the need to consider how information dissemination could be facilitated by cooperating with interpreters who play the role of a liaison between foreign residents and related organizations, without imposing an excessive burden on the interpreters.

Determinants and outcomes of Health Literacy (HL) among Brazilian mothers living in Japan

HIDEMI HASHIMOTO[1], SATOKO YANAGISAWA[2]
1Mie Prefectural College of Nursing, Tsu, Japan, [2]Aichi Prefectural University, Nagoya, Japan

[Background] Health Literacy (HL) is the capacity to obtain, process and understand basic health information and services needed to make appropriate health decisions. We developed Health Literacy Scale for Brazilian Mothers living in Japan (HLSBM) which consisted of Basic Literacy (BL) and Critical Literacy (CL). Using the HLSBM, this study aimed to examine determinants and outcomes of HL for Brazilian mothers in Japan.[Methods]Of 1474 questionnaires distributed, 448 were analyzed. Before conducting multi-regression analysis, t-test or Mann-Whitney U-test was conducted to examine possible determinants of HL. For outcome analysis, three health behavior - dental consultation, medical consultation, and diet, were measured by three levels in practice: good, medium, and bad. Kruskal-Wallis test and Jonckeer-Terpstra's Trend test were conducted to examine the influence of HL on each behavior. [Results]Determinants of HLSBM were Japanese language ability, social support, receiving health information, history of health service use, and health knowledge ($F=56.65$, $p<.001$, $R2=.47$, Adjusted $R2=.45$).Total HL score significantly increased as health seeking actions improved in dental consultation ($p=.017$) and diet ($p=.039$). CL increased in all outcomes (dental consultation: $p=.018$, medical consultation: $p=.034$, diet: $p=.002$) as the maternal behavior improved.[Conclusions]Japanese language ability was the most influential determinant on HL of Brazilian mothers. Providing health information in their language also helps improving their HL. The outcome analysis revealed that CL is more influential on health behavior of the mothers than BL.

The Health Issues of the Brazilian school children

SATOMI SUZUKI, KUNIE MATSUMOTO, KAORI HIBI
Gifu College of Nursing Glfu, Japan

Objectives: In the investigation of the Brazilians adults in Japan, the tendency to suffer from lifestyle related diseases or mental disorder has become clear. It is therefore apprehensive that there is the same influence on the school children who have similar lifestyle.The purpose of this study is to clarify actual conditions which influence the health of the Brazilian school children. Data are collected through interviews for children and their guardians. Methods: The data was individually collected through interviews in the public elementary school. Children's investigation items are sleep, meal, exercise, school life, and physical symptom. Guardian's investigation items are children's health condition, consideration of sleep, a meal, exercise, and working situation. Results: One of five children had intermediate degree of obesity. Two children's favorite foods were meat. Two children's unfavorite foods were vegetables. Three children lack concentration. Two had headache.Guardians working in the early morning were parents of two children eating breakfast alone and one mother of two children not eating breakfast.Five guardians were careful about oil, fat, and vegetable consumption. However, nobody worried about not eating breakfast and eating breakfast alone. Conclusion: Physical symptoms were considered as a result of lack of sleep, hunger due to not eating breakfast, maladaptation of school life, and influence of guardian's working situation.This study continues data collection through interviews, in order to clarify relationship between the health issues and lifestyle, and to grasp the background which influences the health of the Brazilian school children.

An investigation into the realities of the difficulties in information gathering and doctor consultations at hospitals faced by foreign patients. Studied foreigners in Japan and Japanese in Chiang Mai, Thailand

YUKO UESUGI[1], KANA NOJIRINO[2], KEIKO SEKIDO[1], HIROYA MATSUO[1]
[1]Department of International health, Kobe University Graduate School of Health Sciences, Kobe, Japan,
[2]Kobe University Hospital, Kobe, Japan

Purpose: The internationalization of recent years has been accompanied by an increase of foreigners in Japan. Also, amongst the Japanese living overseas, there are over 50,000 residing in Thailand in Asia. This study was to describe the information gathering process for foreigners in Japan and Japanese in Chiang Mai. Method: The subjects of the study were foreign patients who can understand English who received consultations at a hospital with international internal medicine outpatient care, as well as Japanese patients at a hospital in Chiang Mai. A self-reporting questionnaire was used. Results: Subjects were 50 foreign patients in Japan and 10 Japanese in Chiang Mai. Regarding their course of action when they get sick, amongst the foreign nationals in Japan, 16 people (32%) reported that they "search on the internet". Meanwhile, 8 (80%) of the Japanese residents in Thailand said they" consult a friend". As for the reason for choosing the hospital, 19 (38%) of the foreigners in Japan said it was because "there are English-speaking staff" while 5 (50%) of the Japanese patients said it was because "there are Japanese-speaking staff". Discussion: As many foreign residents in Japan used the internet to gather information, more sufficient information should be made available online. We are thinking of a system to provide information to Japanese patients in Chiang Mai. The fact that in both Japan and Thailand, language-related issues were the main source of unease suggests that the sufficient provision of medical interpreters is an important issue for consideration

Foreign residents' barriers to health: findings from consultations at the International Town Healthcare Room in Hyogo

RYOKO NAKATA[1], JUNKO NASU[2], KIYOMI INOUE[1], RISAKO KURONO[1], KUMSUN LEE[3], AKIKO KUROTAKI[4], FUMIKO SATO[5], SAKIKO KANBARA[6], AIKO YAMAMOTO[4], XIAOYU WU[4]
[1]Faculty of Health Sciences, Kobe Tokiwa University,Kobe,Japan, [2]Graduate School of Nursing, Toho University,Tokyo,Japan, [3] College of Nursing Art and Science, University of Hyogo,Akashi,Japan, [4] Research Institute of Nursing Care for People and Community, University of Hyogo,Akashi,Japan, [5]Faculty of Nursing, Senri Kinran University,Osaka,Japan, 6Faculty of Nursing, University of Kochi,Kochi,Japan

The International Town Healthcare Room has been held to promote health mainly at the Research Institute of Nursing Care for People and Community, University of Hyogo. In accordance with the increase in number of the foreign residents with various cultures and international marriages, there are emerging needs related to health and daily life. The International Town Healthcare Room was established for those foreigners in September 2012 and held 11 times in 2013. The largest national group of visitors were Chinese, the rest were the Hong Kongers, Thai, and American et al. The number of the consultation cases was 1137. Their chief concerns were that they could not make themselves understood in Japanese when explaining their symptoms at clinic; they were ignorant of the effects of the prescribed medications; they could not afford to seek medical consultation due to lack of health insurance; they were seeking mental support as they are under stress due to alienation from 'Japanese culture or sense of value' especially in rearing families. Thus we have determined four barriers that affect the health of the foreigners: 1) the language barrier, 2) the inadequate knowledge about health insurance and medical system in Japan, 3) the barriers related to unfamiliar culture and sense of values, 4) difficulty to locate help to resolve the suffering resulting from poor adaptation to such unfamiliar cultures. To eliminate such barriers, it is important that the various professionals form the alliance, provide social resources such as linguistic services and family consultation.

Current situation of ARV accessibility in countries of origin of migrants in Japan

TAKASHI SAWADA[1], Yuko Yamamoto[2]
[1]Minatomachi Medical Center, Yokohama, Japan, [2]Services for Health in Asian & African Regions

Background: Majority of HIV positive migrants in Japan are from developing countries, where accessibility to AIDS treatment is limited. A research on HIV treatment and situation of health services in migrant's countries of origin was conducted to ensure correct information.
Method: WHO treatment guidelines were reviewed, and information was collected on AIDS treatment and situation of health services in countries of origin of HIV positive migrants planning to return home by contacting the local health care providers, NGOs and international agencies.
Result: Developing countries have mainly provided generic drugs, which was developed mostly before 1996. However these drugs have many side effects and limitations. Recent WHO treatment guidelines call for stronger monitoring and promote the use of safer and less toxic treatment options. The guideline, coupled with the decrease in global prices of safer treatment options, increase accessibility and availability to these treatment options are expected. In corresponding with local counterparts in migrant's country of origin, we received confirmation in most of countries, that there is improvement in availability of WHO recommended treatment. However, there is still limitation in regimen options and facilities, especially for those who have complicated situations.
Conclusion: International aid, reduction in drug prices and the work of civil society, has in general, led to improvement in treatment access in developing countries. However, there is still much limitations in treatment options and disparity among and within countries and remain large. Information on health care situation should be carefully collected before referring HIV positive migrants back home.

Health consultation for immigrants and its Oral Health Characteristics in Japan

KOICHI NAKAKUKI[1,2], TAKASHI SAWADA[2], Katsuya Ohwaki[2], TORU HONDA[2]
[1]Maxillofacial Surgey, Graduate School of Medical and Dental Sciences, Tokyo Medical and Dental University,
[2]SHARE, Services for the Health in Asian and African Regions

[Introduction] It may be difficult to visit medical institutions for immigrants in Japan, because of difficulty in language, health insurance system and economic issues. SHARE, Services for the Health in Asian and African Regions, regularly holds free dental consultation for immigrants from 1998. We studied their oral health characteristics.
[Outline] We performed 92 dental consultations, at free health consultations for immigrants in Tokyo area, between 1998 and 2013. 4587 out of 5767 received dental consultation, 79.6% of all attendants. Average attendants were about 70 until 2003, but it was decreased to 40 in 2004, because of the tighter control against undocumented foreign workers by Metropolitan government in Tokyo. We expanded the areas from 2009, and the attendants were increased to about 55. However it decreased to 40, after the great east Japan earthquake in 2011.
[Discussion] Many of the attendants were the Asians. Filipino's oral health conditions were the worst, many decayed or missing teeth, and many wear denture prostheses. Quick and cheap treatments were selected, even though many of them had experience of dental treatment and oral health education, and used interdental brush. While, Bangladeshis had many calculus, but few decayed teeth. Many of them had never had dental treatment or oral health education, and rarely recognize the problems. Those differences may suggest that dental health is greatly influenced by dietary or cultural customs.
[Conclusion] It is needed to provide oral health education that addresses the characteristic dental conditions and dietary or cultural customs of original country.

Teaching elementary school students about hand-washing habit in Lao PDR

YUSHI ABE[1], SHIHO MIZUTANI[1], KAYANO AKIMOTO[2], YUKI NAKAMURA[3], MARIA NAGATA[2], RIKA IWASAKI[3]
[1]2nd Department of Medicine, Keio University, Tokyo, Japan, [2]Department of nursing, Keio University, Tokyo, Japan, [3]Department of pharmacy, Keio University, Tokyo, Japan

Introduction Students from three medical departments of Keio University went to Lao PDR. We visit Kun Don village in southern Lao annually to teach children about importance of health. This year we investigated manners of children's hand washing as our field work. Therefore, we conducted heath education in regards to hand sanitation, and we are reporting on the challenges uncovered in getting children to develop a habit of washing their hands.
Outline From March, 16th 2014 we stayed at the village for a day and observed the washing habit. We picked five students and used a cleanliness checker to show how dirty their hands are. To investigate the condition, we wrote the result on one big sheet of paper. In usual households, they scrub their hands in a tub filled with water.
Discussion Children who washed their hands in this manner left too much indicator, and the result was that their whole hands were unclean. Therefore, we were unable to decide the trend of unwashed part of hands.
Conclusion In conclusion, the hand-washing habit of Kun Don village was not good enough for maintaining healthy environment. It is difficult to wash away clay with just only water, so we need to instruct them to use soap and brushes. Children are not used to the checkers. We need to teach them in simpler way, and also have to organize an environment that allows them to learn the importance of sanitation in a long term.

Practice of oral hygiene education at elementary schools in rural areas of Laos

AKIHA ASANO[1], YUKI AIBARA[2], YUMI KOSAKA[3], HITOMI HAMADA[1],
SATORU FUJIWARA[3], CHIAKI IZUMI[1], RIKA FUJIYA[1], TOMOKO KOIKE[1]
[1]Faculty of Nursing and Medical Care, Keio University, Tokyo, Japan,
[2]School of Medicine, Keio University, Tokyo, Japan, 3Faculty of Pharmacy, Keio University, Tokyo, Japan

<Introduction>Health education was carried out for students in Kun Don village, Sekong Province, Laos. This village was designated as a model health village by the government. Although the public hygiene environment is good, the results of interview showed a lack of awareness in oral hygiene, because they don't have chance to receive health education. Therefore, we conducted health education in regards to oral hygiene, and we are reporting on the challenges uncovered in getting children to develop a habit of brushing. <Outline>Keio University has been conducting a training program in Laos. The subjects were approximately 200 individuals between elementary grades 1 through 5 , and they were divides into lower , middle and higher grade levels. This year, we held an oral hygiene class and explained the correct way of brushing teeth with a poster, and check children's teeth by Plaque Disclosing Tablets.<Discussion>The children showed great interest in tablets. This helped them pay attention to the lecture about the bad influences of lack of brushing teeth. In order to enhance the effect of education in the future, good materials are needed to grasp children's attention and encourage them to make brushing teeth as a routine habit. Also, a support system is required to continue the health education by teachers, health care volunteers, and children's parents. <Conclusion>Although school children's awareness for oral hygiene was increased, it revealed that it is important to reach out to teachers and children's families to sustain the effect of programs.

Proposal of the way health guidance of the Gandon Village should be through the measurement of weight and height of elementary school students and home stay experiences

YUMIKO DATE[1], SAE GYODA[1], MIO NANASAWA[1], NATSUMI HAYASHI[2], NATSUKO SUGAHARA[2], RYOUSUKE HORI[3]
1Pharmaceutical department, Keio University, Tokyo, Japan, 2Nursing department, Keio University, Tokyo, Japan, 3Medical department, Keio University, Tokyo, Japan

Introduction: It was the 2nd year to visit Gandon Village as a part of the Laos Primary Health Care Program which started 5 years ago as the project of Keio University. The purpose of visit was to take the measurement of weight and height of elementary school students and propose the way health guidance of the Gandon Village should be.
Outline: On March 17 2014, Measurement about 300 students (5 to 12 year old) was conducted. We compare results to WHO's Child Growth Standard (hereafter referred as Growth Curve). Through observation and home stay experiences in the village, we extracted needs and assets and assessed how they affect the results of measurements.
Discussion: There were no significant discrepancy from the Growth Curve, though, the overall results was slightly below the Growth Curve. And we extracted needs and assets. Assets: nutritiously balanced meal, sanitary toilet and hand-washing practices, which contribute to improving diarrhea rate. And due to strong community bond, it's possible to share information among entire village. Needs: the low level of interest among villagers toward health.
Conclusion: Through home stay experiences, we find that hygienic practices are fixed among villagers, though, interests toward health among them are low. To continue the measurement and to make villagers more interested in health, it's important to ask school teachers to give students' parents results of the measurement and explain the goal of the measurement. Therefore, we conclude that giving lectures to teachers what is the goal of the measurement is needed.

Health Care Check and physical test at elementary school in DRC

SHOKO KITAZAWA[1], RIKA FUJIYA[2], TOMOKO YATSU[2], MARIKO OGAWA[2], KOJI TOGAWA[1], SHIHO MIZUTANI[1]
[1]School of Medicine, Keio Univ., Tokyo, Japan,
[2]Faculty of Nursing and Medical Care, Keio Univ., Tokyo, Japan

[Introduction] We, the students from three fields, Education, Architecture, and Medical in Keio University, have Acadex Elementary School Project, which we established in Democratic Republic of the Congo (DRC) in 2008. Keio Univ. Monitoring and following children's growth, we have done again health check and sports test to children in Acadex in 2013.
[Outline] On August 21 2013, 80students (3 to 13 year old, 46 boys and 32 girls) had health care check, which is height, weight, body temperature, eye-sight, circumference of the upper arm, physical check, and medical examination by doctor. About weight and height, comparing with WHO's Child Growth Standard, almost all of the students are plotted from -2SD to the average. However, out of 38 boys, 8 was shorter, and out of 28 girls, 7 were shorter than -2SD. Also, some students have anemia, hernia on the stomach, and abdominal dropsy.
[Discussion] Although it is the same situation in children's health, we could give them the result feedback to let them know how was their health condition.
[Conclusion] Given the result of health check, it is known that most students tend to be shorter and lighter than world average. Through this health check every year, recording data, which would be told to the children and their family, we could create an opportunity to let them be aware of their own health by themselves. To estimate children's health situation, monitoring their health status is important, and health care check at school is needed to be continued.

Health education in Kimbond, the Democratic Republic of Congo

KEITARO YAJIMA, Sara Koyanagi, Martha Kawamura, Shunitiro Konishi
the University of Keio, Tokyo, Japan

[Introduction] In the Acadex School Project that is being carried out in the Democratic Republic of Congo, we are cooperating to raise the standards of the children's health. The medical team conducted health checks and home visiting to grasp the situation of the area, and used that information to hold the necessary health workshops for the children. It was found that preventable diseases are widespread, and that the citizens do not realize the gravity of the problem. Therefore, the purpose of this workshop is how we can change their senses to held workshops about infectious diseases and preventive methods.[Outline] For two years, the medical team held workshops to raise awareness of infectious diseases and the importance of hand-washing as a preventive method, in the Acadex Elementary School of Kimbond Area, DRC. Through questionnaires after the classes and home inquiries held in the following year, we evaluated the effect on the children. [Discussion]It was discovered that the habit of washing hands correctly using soap has started to spread among students by doing home visiting by doing home visiting. (Out of 13 families, 13 families wash hands right way). As a next step, it is necessary to hold workshops not only for the students, but for Kimbond Area as well.[Conclusion] The establishment of correct hand-washing among students is a significant achievement. However, there is still room for improvement, as many students tend to use excessive amounts of water for hand-washing, which is disadvantageous in an area not rich in water.

Field research on infrastructure,health,and hygiene: Centered on an elementary school in Kimbond,Democratic Republic of Congo

KEITA KONDO[1], KAZUHIRO AKASHI[1], MASAHUMI KAMATA[1], RIKA FUJIYA[2], SYUNICHIRO KONISHI[1]
[1]5th Grade,School of Medicine,Keio University,Tokyo,Japan,
[2]Faculty of Nursing and Medical Care,Keio University,Tokyo,Japan

Introduction: Democratic Republic of Congo is the 186th country of Human Development Index and one of the poorest countries in the world. Every year, our team visits Kimbond area of Democratic Republic of Congo. We cooperate with the construction team and education team, and carry out health care activity and investigation around the elementary school.
Outline: We targeted the children and their parents who go to the elementary school of Kimbond, Democratic Republic of Congo. We asked questions about their health situation and lifestyle. And then we investigated the infrastructure and the quality of waterworks. As a result, we came to know that they have to bring water from the well taking 16.8 minutes on average. And we found that there was no contamination in the waterworks.
Discussion: Last year, the World Bank founded waterworks in Kimbond area for onerous contact, and it made the time to go to there dramatically shorter. This time is also shorter than other African countries. We took the survey of water quality, and found that the improved water quality has great influence on the state of sanitation in Kimbond area.
Conclusion: This year's survey revealed that the situation of health, living condition and infrastructure are improving at Kimbond area. However, we still have to get further information. Therefore, we will continue researching on this area's life condition for a long term, and examine the influence to health condition and children's learning condition.

Global health education through multicultural and interproffesional student-led school health in rural Nepal

MIRE SUGINO[1], MARIE TASHIRO[2], TAKESHI NAITOU[3]
[1]Nursing department, Sonoda Women's University, Hyogo, Japan, [2]Faculty of Nursing, Baika Women's University, Osaka, JAPAN, [3]Faculty of Medicine, Tokushima University, Tokushima, JAPAN

In global health profession, multicultural communication skill and creative interprofessional collaboration is crucial to respond to wide range of health needs of 21st international society. Researchers organized a school health program in cooperation with medical colleges and educational institutions and international NGO in Kathmandu, Nepal, to develop collaboration skills and enlightenment for multicultural and interprofessional students in 2013.
In summer 2013, nine students from Japan, Korea, Nepal worked together and implemented school health programs in four schools in rural Nepal. The participated students reflected themselves and recognized their own roles as health professionals in international society. In summer 2014, ten students will participate in the school health program, lodge in a training camp together in Kathmandu, and organize their own school health programs. Programs keep being developed in collaboration with those multicultural and interproffesional students and related institutions until summer 2016.
Lived experiences in a society which has complicated health problems like Nepal foster students' insight and strong willingness to improve their skills as health professionals in global health field. By building networks with local institutions, it is expected to develop a sustainable model of global health education.

Tofu Project: the implementation of health education through soy products, and health and nutritional assessment in Montalban, the Philippines

SAKI HORIE[1], MARIKO INOUE[2]
[1]Teikyo University Support Center for Wimen Physicians and Researchers,Tokyo,Japan,
[2]Teikyo University Graduate School of Public Health

[Introduction] Economic and nutritional gaps are commonly observed in the Philippines, with discrepancies between those with insufficient food, those with a healthy life and those suffering obesity widening.
[Outline] Our project was undertaken to observe the current situation with regard to daily food intake and identify possible methods for improving the situation. The target population consisted of infants and families using the St. Magdalena Sophia Foundation (SMSF) facilities in Montalban, Philippines. We evaluated health and nutritional status, and undertook health education based on the use of soybean products. We first conducted a survey on health and nutritional status to obtain baseline data (March, 2013). The study was designed as both a descriptive epidemiological and observational study. The questionnaires included items on meal ingredients, and daily food and drink costs.
As soy products are commonly eaten in the Philippines, SMSF also built a small tofu factory to ensure access to tofu and conducted a nutrition education program for children. A post-intervention survey was later conducted in November 2013.
[Discussion] Sixty-one households answered both surveys. The increase in the number of food was seen before and after intervention. To maintain improved eating habits, nutritional research and education are still necessary. Effective use of familiar foods that have high nutritional value and can be obtained cheaply is desired.
[Conclusion] Consideration of the local environment and food culture is recommended when planning a program of practical nutritional intervention in daily dietary behavior.

Daily behavior and contribution to the food procurement of Baka hunter-gatherer's children in southeast Cameroon

IZUMI HAGINO[1], TARO YAMAUCHI[2]
[1]Graduate school of Health Sciences, Hokkaido University, Japan,,
[2]Faculty of Health Sciences, Hokkaido University, Japan,

Introduction: The Baka are one of the groups of pygmy hunter-gatherers living in central African rainforest, and remain traditional society depending on the forest. In this study, we aimed to reveal how the Baka children spend their daily time in their hunting camp and contribute to the food procurement. Subjects and Methods: Foraging trips were conducted total 20 days in September 2012 and September 2013. Total participants for the camps were 410 person-days for children and 156 person-days for adults. 16 children aged 4-17 years old were observed carefully during the observation period (started 06:00 and ended at 18:00) each for a day. Their activities were noted whatever their activities changed and recorded to the minute. All foods brought back to the camp was identified, weighed, and defined who had acquired. Results: Children spent more than half of their daytime outside of the camp (389 ± 97 minutes/day), and there was no significant difference among their sex or age. Older children spent longer time (208 ± 66 minutes) than younger children (161 ± 123 minutes) for food acquisition. The mean time spent for playing was 115 ± 73 minutes/day. 33 and 34%, 45 and 46%, and 23 and 21% of energy and protein from 24 species of wild food resources obtained during the research periods were procured by children, adults, and their group work respectively. The contribution to food procurement by children was not considerable compared to adults.

Infectious Disease Surveillance after the Typhoon "Yolanda" in Tacloban, Philippines

MARIKO SAITO[1,2], Mayuko SAITO[1], RAITA TAMAKI[1], Titus Tan[2,3], Inez Medado[2,3], Michiko Okamoto[1,2], Taro Kamigaki[1], Weng Capistrano[3], Socorro Lupisan[3], Hitoshi Oshitani[1]
[1]Department of Virology, Tohoku University Graduate School of Medicine, [2]Tohoku-RITM Collaborating Research Center on Emerging and Re-emerging Infectious Diseases, Metro Manila, Philippines, [3]Research Institute for Tropical Medicine, Metro Manila, Philippines

The massive typhoon Yolanda hit the Visayas, Philippines on November 8, 2013, and Tacloban was one of the most affected area due to the strong wind and storm surge. Since 2008, Tohoku University Graduate School of Medicine has been conducted the etiological study of acute respiratory infections and acute gastroenteritis in Tacloban. To evaluate the infectious diseases outbreaks after the typhoon, we have performed the rapid diagnostic test using the commercially available kit and PCR using the residue of the rapid test kit or FTA cards (GE health care).
Between November 28, 2013 and February 4, 2014, we have enrolled 187 patients and the specimens were collected based on their predominant symptoms. The age range of the patients was 0- 69 y.o. (median: 1 y.o.). The breakdown of the collected specimen was nasopharyngeal swabs (NPS): 128 samples, stool: 19 samples, blood: 40 samples. Five samples were positive for influenza A, 2 were positive for rotavirus, 11 showed positive for dengue virus IgG by the rapid diagnostic test. By RT-PCR, influenza A (H3N2) from 5 samples, enterovirus from 9 samples, rhinovirus from 18 samples, RS virus from 1 sample were detected from NPS samples, and rotavirus were detected from 8 stool specimens.
In this study, we have concluded that there were no typhoon-related outbreaks compared to our past study in pneumonia and acute gastroenteritis in Tacloban. Also, the outbreaks of dengue or leptospira which were expected to be increased after the flood were not detected.

Health problems among the first batch of the disaster response team: Lessons learned from Philippines Typhoon Yolanda response in 2013

SATOKO OTSU[1,3], Miho Sekitsuka[2], Nobuhiro Komiya[1,3], Kenji Kubo[3]
[1]International Medical Relief Department, Japanese Red Cross Wakayama Medical Center, Wakayama, Japan, [2]Nagoya Daini Red Cross Hospital, Nagoya, Japan, [3]Department of Infectious Diseases, Japanese Red Cross Wakayama Medical Center, Wakayama, Japan

Introduction: Following Typhoon Yolanda (Haiyan) which hit Philippines in November 2013, Japanese Red Cross deployed Basic Health Care Emergency Response Unit (JRC BHC ERU), consisted with international Red Cross. The first team conducted medical activities in Daanbantayan, Northern Cebu in from 17 November 2013 to 23 December 2013. Although none were severe, a couple of members got sick during the mission. This is to summarize health conditions in the field and to describe the importance of health management.
Outline: This is a retrospective descriptive analysis of health conditions of the 17 first international team members during the mission.
Discussion: Out of 17 staff members 13 stayed more than three weeks.Among those who stayed more than three weeks, 5 suffered mild upper respiratory infection. One had high fever with sore throat. 4 had mild to severe dermatitis probably due to sun burn or extremely high temperature (all were Japanese). In addition, a few local staff members also suffered from upper respiratory infection. None complained psychological problems and none needed to be evacuated.
Conclusions: More than half of staff members in the first team suffered from mild illness during the mission. A trend of diseases was similar to those of affected population except dermatitis probably due to sun exposure and high temperature. The environment in the affected areas was rather challenging and the first team members tended to work long hours under huge stress. Pre-deployment health advice is important and should be emphasized for the disaster specific health issues.

Activities Report about in the center for Children with disabilities in Nicaragua

SAWAKO TANAKA
Department of Occupational Therapy, Faculty of Health Sciences, Kyorin University

INTRODUCTION: I participated in Japan Overseas Cooperation Volunteers as an Occupational therapist for two years from 2011 to 2013 in the center of Children with disabilities in Republic of Nicaragua. DESCRIPTION: Los Pipitos is one of the NGO which played a central role of support for Children with disabilities in Nicaragua. Department Juigalpa supports through early remedial teaching, rehabilitation, crafts, a sport, etc. When I got there It was hardly functions and I couldn't perform technical transfer of occupational therapy which was the contents of a request. So I tried to do various works (1.transfer of occupational therapy, 2.beginning again pick-up by microbus, 3.Activation of crafts activity, 4.Home visitation etc.) didn't sticking to my specialty. RESULT: There are many problems why doesn't function well, for example: problem geographic, economic and human relations, comprehension for Children with disabilities, lack of interest and the different national character. Through all the activity, I felt that there are many Children with disabilities without any support weather they necessary to any support and exist too much organizations which can't receive any support.

Overall assessment of Skype-based international Distance-learning program in Nicaragua on nursing techniques

SACHIYO MIYAKOSHI[1], YOSHIE MORI[2], HIROMI TSUJIMURA[2], Rajasekera Jay[3], Quiroz Zeneida[4], Reyes Sandra[4], Gaitan Marlene[4], RIE NAKAZAWA[2], MASAAKI SAKAMOTO
[1]Nagano College of Nursing, Gunma university,Graduated School of Health Sciences,
[3]International University of Japan, Graduate School of Internatioanl Management,
[4]National Autonomous University of Nicaragua-Managua

Economical and interactive international communications systems are expected to be effective in cultivating of human resources for international cooperation. Classrooms at Nicaragua's A University B Campus and Japan's C University were connected via Skype, seminars for local instructors were held in advance, and seminar classes took place concerning nursing techniques such as changing patient position, decubitus care, feeding care, urethral catheterization, and enemas ; based on students' (6) and teachers' (7) overall evaluations, the system's significance and future issues were examined. All were interested in the seminar topics, and the class appeared significant from the point of learning for oneself techniques taught differently abroad, economically and interactively. In particular, wrap techniques for decubitus care such as using antibiotics and avoiding moisture were points of interest due to their differences from local methods. Instructors, while pointing out the significance of continuing classes on grounds of possibilities of the application [of international remote learning] to classes on home care techniques, raised issues such as more diligent class discussion and conditions suited to the seminar environment. Further, the instability of the local telecommunications situation was also raised. In future, class management should allow for the local telecommunications and seminar environment, and for that purpose, while exploring with local contacts methods of providing stable sound and images, discussions should be held in advance on necessary conditions and time allocation for the seminar, understandable phrasing, and appropriate discussion timing; further, it should be confirmed whether interactivity is being maintained as the class takes place.

Significance and learning effects of the nursing student engaged the overseas training

TOMOKO KUDO[1], KAHORU TANNO[2]
[1]Department of Nursing, Japanese Red Cross Society Medical Center,
[2]Faculty of Health Sciences, Niigata University

Background: With progress of globalization, the international nursing education has attracted attentions as a practical learning program. In this study, we discuss the significance and learning effects of nursing students engaged the overseas training.
Methods: Semi-structured interviews were administrated to nursing students (n=11) of A university, joined the training in Sri Lanka from 2012 through 2013. KJ method was applied. Investigation items include 1) Attributes, 2) Training purpose, 3) Changes of the viewpoint of nursing, etc. Investigation period was from July to October in 2013. As ethical considerations, this study was carried out after obtained consents of the participants based on personal information protection. The ethic screening committee of department of Nursing, faculty of Health Sciences, Niigata University, approved this study.
Results: Through the training, students learned ''Consideration and clarification of the basic attitude of medical care and nursing'', ''Recognition of the process for be accepted in education for patients'', ''Expansion of the role and function of nursing'', ''Importance and diversity of the communication'' based on their nursing knowledge and techniques. These points contributed to change of their viewpoints and recognized anew how attractive nursing was.
Conclusion: Learning effects of overseas training might be student can 1) comprehend the international and cross-cultural nursing, 2) learn about cross-cultural understanding and communication, 3) recognize own growth, 4) be improved their motivation for study, 5) realized anew their home countries, 6) construct new relationships. Therefore, providing training opportunity could be promising program for nursing education.

Pilot Study for supporting EPA Indonesian Candidate Nurses -Survey to Indonesian Candidate Nurses who are studying in Japanese in Indonesia-

MINA TANAKA[1], YURIE YAMADA[2], SAYURI MINODA[3], SATOMI MIZUNO-LEWIS[4], JUNKO AONO[5]
[1]Yokkaichi Municipal Hospital,Mie,Japan, [2]Nagoya City East Medical Center,
[3]Suzuka University of Medical Science, [4]Yokkaichi Nursing and Medical Care University,
[5]Non-profit Organization for Educational Support for Foreign Nurses & Care Workers

<Background>University group activities support EPA Indonesian Nurse Candidates (candidates) learning and cultural exchange. Candidates concerns continue from differences between Japanese and Indonesian medical and nursing systems, language and culture. Although previous studies addressed issues of discord, such studies were conducted in Japan. This pilot study was conducted in Indonesia, seeking candidates' preconception of Japan and what worries they have about living in Japan.
<Methods>40 Indonesian candidates participating in the EPA for 6 months were surveyed in Jakarta Indonesia in February 2014 with a self-administered questionnaire. The questionnaire was compiled in Indonesian, considerate of ethics,seeking clarity on nursing tasks in Indonesia, and candidates' image of Japanese nursing. Results were analyzed by simple tabulation.
<Results>Their greatest concern was language barrier (28 candidates). Their purpose of working in Japan was to learn advanced medical care (37 candidates). Their image of nursing in Japan was high quality nursing care (36 candidates) and advanced medical care (35 candidates). Nursing in Indonesia is mostly in assistance to medical care (40 candidates). 36 candidates think nursing in Indonesia also includes assistance to 'activities of daily living'. 23 candidates believed they could work in Japan equivalently to Indonesia without passing the Japanese National Nursing Exam (JNNE).
<Discussion>The passing rate of JNNE by foreign candidates is about 10%. Candidates need more effective support in learning Japanese language. Some 50% of candidates misunderstood their expected work in Japan until they passed JNNE.

Human Resource Development of leaders in Nursing and Midwifery in Cambodia-Achievement and challenges of Bridging Course for Bachelor of Science in Nursing-

KYOKO KOTO[1], Noriko Fujita[1,2], Phuong Keat [1,3], Sokneang Touch [1,3],
Satoko Yanagisawa[4], Puangrat Boonyanurak[5]
[1]JICA Cambodia, [2]National Center for Global Health and Medicine,
[3]Ministry of Health, Cambodia, [4]Aichi Prefectural University, 5Burapha University

Background: Cambodia started to reconstruct its health system focusing on increasing number of professionals. Currently, lack of qualified teachers is a fundamental challenge to produce qualified personnel especially in Nursing and Midwifery. Project for Strengthening Human Resource Development System of Co-medicals promotes teacher's capacity development through upgrading 18-month course in Thailand. 32 teachers and clinical preceptors in public sector participated, and 26 graduated and back to their previous working place.
Purpose: Clarify achievement and challenges of the course to assess future needs for production of leaders.
Method Interviews were conducted for 20 participants, 11 directors of institution and 5 Ministry Officers to identify achievement and challenges during and after the course, director's and Ministry's expectation and perspective toward future leader production.
Results: Participants learned how to learn and opened their eyes to identify role model as professional nurse through experiencing other style of nursing practices from their own. Many of participants were promoted after return and making various efforts to improve clinical teaching for pre/in-service training and promote nursing management at ward. Out-country capacity building model conducted in neighboring country is efficient and effective in terms of cost and continuous support by faculty from Thailand. It benefits in producing personnel sensitive to the main streams in ASEAN regions.
Conclusion: This course contributes in developing core team of future leaders. MOH recognizes capacity building in master degree level is needed to produce qualified teachers. It is expected that it will continue align with Ministry's strategic plan.

Key factors influencing nursing retention and job satisfaction in a rural and an urban area, Sri Lanka

AKI TAKAI[1,2], Balachandran Kumarendran[3], Takeshi Kudo[4], Kensuke Goto[5,6], Satoshi Kaneko[1,5]
[1]Graduate School of International Health, Nagasaki University, Nagasaki, Japan., [2]Fujita Planning Co.,Ltd. Tokyo, Japan, [3]Department of Public Health, Faculty of Medicine, University of Kelaniya, Ragama, Sri Lanka, [4]Faculty of Economics, Nagasaki University, Nagasaki, Japan, [5]Department of Eco-epidemiology, Institute of Tropical Medicine, Nagasaki University, Nagasaki, Japan., [6]National Mental Support Center for School Crisis, Osaka Kyoiku University, Osaka, Japan

<Background> In Sri Lanka, the Ministry of Health has indicated that recent health concerns such as non-communicable diseases and an aged society cannot be addressed under the current geographic imbalance of nursing personnel. The objective of this study was to identify factors influencing job satisfaction and retention intent among nursing officers in the public sector in a rural and urban area. <Methods> Self-administered questionnaires were filled out; these comprised three parts, including socioeconomic data, a job satisfaction scale, and a short version of McCain's Behavioral Commitment Scale, which measures retention intent. Bivariate and logistic regression analysis were performed. <Results> The study sample comprised 385 respondents. Factor analysis was performed for the job satisfaction construct, yielding seven factors, namely, professional support, intrinsic satisfaction, external benefits, the working environment, workload, career development opportunities, and the relationship with other professionals. A significant difference was found in retention intent, with urban nurses being more likely to stay in the nursing profession than rural nurses. Age, rural background, satisfaction with professional support, and intrinsic satisfaction were identified as factors influencing the retention intent among rural nurses.<Conclusions> The influence of personal aspects such as a rural background and cultural on long-term retention can determine the employment and distribution of graduate nurses. The other findings showed that the factors affecting retention among rural nurses were affected nursing job satisfaction. Efforts should be concentrated on the enhancement of professional support and intrinsic satisfaction, which could in turn enhance job satisfaction among the current pool of rural nurses.

Literature Study Concerning the Relationship of Traditional Chinese Medicine and Nursing in China

KOTONE YAMAMOTO[1], KAHORU TANNO[2]
[1]Chiba University Hospital, Chiba, Japan,
[2]School of Health Sciences Faculty of Medicine, Niigata University, Niigata, Japan

[Background] The scholar was interested in the possibility and validity of traditional Chinese medicine in Complementary and Alternative medicines, and wondering how nurses use traditional Chinese medicine in patient care in Chinese hospitals. [Methods]The scholar then read a number of essays from a variety of research materials concerning the theme of her study in order to survey the history of traditional Chinese medicine and analyze the relationship between it and nursing practices in China. This study was approved by the ethics committee of Niigata University. Traditional Chinese medicine is categorized as a Complementary and Alternative medicine. It was established after Cultural Revolution. It is deeply connected with nursing techniques in China. Traditional Chinese medicine regards the harmony between the inside of the human body and environment as the important thing. Surprisingly, this same concept is covered in western medicine as can be seen in the theory of Nightingale. Thus nursing concepts of traditional Chinese medicine are similar to those of western nursing. Nurses need to understand nursing exist every situations, and grope for the best nursing for the patients.[Conclusion] There are various types of nursing depending on cultures, however, the core points are the same. Despite variations in cultures around the world, the original function of nursing is universal.

Maternal-and child-health problems in plantation areas of Sri Lanka and the roles of public health midwives

ATSUKO ISOBE, THEVAR OFFIYAR, SAYOKO UEMURA
Faculty of Nursing, the University of Seisen, Hikone, Japan.

Introduction; Plantation areas of Sri Lanka face the issues of the low rate of pregnant females who do not undergo health examinations, anemia, and low-birth-weight babies. The present paper discusses the roles of public health midwives (PHMs) in the community. Outline; PHMs are responsible for family planning, health examinations, vaccination, nutritional and health advice, observation of postpartum females, advice on breastfeeding, sex education for adults, and education to prevent unwanted teenage pregnancies. However, the management of pregnant females is not being implemented appropriately due to the sheer volume of tasks, a shortage of PHMs, and the difficulty of making rounds over a wide area to provide guidance. Discussion; Although it is essential to provide both mothers and children with nutritional guidance, there has been a delay in the improvement of their nutritional conditions partly due to their cereal-based dietary habits and limited access to education. It is also necessary to teach mothers methods for the intake of nutritional supplements, which are provided by the government, and how to read child-raising handbooks. Conclusion; In addition to the appropriate allocation of PHMs, it is necessary to provide mothers and children with health education to increase their health awareness, and collaborate with other health care professionals to develop and implement programs which mothers and children can participate in autonomously. This work was supported by JSPS KAKENHI Grant Number 23401016.

Birth size and temperature during pregnancy

MIKI KAGAMI[1], ENBO MA[2], YUKIKO WAGATSUMA[2]
[1]Graduate School of Comprehensive Human Sciences, University of Tsukuba, Tsukuba, Japan.,
[2]Department of Clinical Trial and Clinical Epidemiology, Faculty of Medicine,
University of Tsukuba, Tsukuba, Japan

[Background] Birth weight and length have seasonal fluctuations. However, it is uncertain which meteorological element has an effect upon birth weight and length, and which timing of pregnancy would explain such effect. The purpose of this study is to examine whether temperature during pregnancy and which timing effects on size at birth.[Methods]A large, randomized, controlled trial of food and micronutrient supplementation for pregnant women was conducted in Matlab, Bangladesh (MINIMat Study). The fetal growth data which included the size at birth and information of their mothers were obtained (n=3267, respectively). Meteorological data such as temperature, precipitation and daily sunshine hours were observed at the nearest observatory site of Bangladesh Meteorological Department.[Results]Birth weight and length had seasonal fluctuation same as other papers reported. Infants born in winter (October-February) were shorter than those born in other seasons. From the results of multivariate analyses, when the temperature at 8 weeks of gestation increased, birth length became shorter. On the other hand, temperature at the last month of pregnancy affected birth length in a reversed direction. For birth weight, as temperature increased at 17 and 19 weeks of gestation, birth weight was increased 4.2 to 4.7 mm, respectively.[Conclusion]These findings suggest that temperature affects both birth weight and length. More temperature increased at the last month of pregnancy, longer birth length became. For birth weight, the temperature at mid pregnancy affected positive way.

Women-friendly childbirth experience in Cambodia

MAKIKO NOGUCHI[1], Mitsuaki Matsui[2], Youichi Horikoshi[3], Yasuyo Osanai[3,4],
Kenji Takehara[5], Chizuru Misago[6], Yuriko Egami[2]
[1]Faculty of Health Sciences, Hokkaido University, Japan, [2]Graduate School of International Health Development, Nagasaki University, [3]National Center for Global Health and Medicine, [4]Graduate School of Nursing, National College of Nursing, [5]Department of Health Policy, National Research Institute for Child Health and Development, 6Department of International and Cultural Studies, Tsuda College

Background and Aim: In Cambodia, a training program of evidence-based women-friendly care has been promoted by the JICA project. All midwives working in the target hospitals took part in the program. The aim of this research is to describe Cambodian women's childbirth experience at the hospitals.
Method: We conducted a questionnaire survey on women who had childbirth at the hospitals. The study sites were 4 primary hospitals, a secondary hospital and a tertiary hospital of obstetrics in Cambodia. The questionnaire consists of four parts: demographics, childbirth satisfaction, their awareness of midwives, and their awareness of midwifery care. The ethics committee of Hokkaido University and the Cambodian Ministry of Health approved this research.
Results: 131 women responded to the questionnaire. The number of women who realized that it was good for them to have given birth at this hospital was 129 (98.5%). More than 90% of the women felt that the midwife had instructed her in breathing and relaxation method, treated her with respect, and supported her with her family. Almost all of the women recognized that the midwife was a reliable person.
Conclusion: The women perceived that they had positive childbirth experiences. Such positive results point to the difference between the midwifery care provided at the hospitals now and in the past. We could surmise that the training program to promote evidence-based women-friendly care has been successful, but more analysis is necessary to investigate the midwifery care provided at the hospitals.

Evidence-based midwifery care and assessment among Cambodian midwives

MAKIKO NOGUCHI[1], Mitsuaki Matsui[2], Youichi Horikoshi[3], Yasuyo Osanai[3,4],
Kenji Takehara[5], Chizuru Misago[6], Yuriko Egami[2]
[1]Faculty of health Sciences, Hokkaido University, Japan, [2]Graduate School of International Health Development, Nagasaki University, Japan, [3]National Center for Global Health and Medicine, Japan,
[4]Graduate School of Nursing, National College of Nursing, Japan,
[5]Department of Health Policy, National Research Institute for Child Health and Development, Japan,
[6]Department of International and Cultural Studies, Tsuda College

Background and Aim In Cambodia, a training program of evidence-based women-friendly care has been promoted by the JICA project. All midwives working in the target hospitals have been trained in the educational program for women-friendly care. The aim of this research is to describe the consistency between the midwifery care provided and the assessment of midwifery at the hospitals.
Method We conducted a questionnaire survey on midwives who are working at the target hospitals. The study sites were 4 primary hospitals, a secondary hospital and a tertiary hospital of obstetrics in Cambodia. The questionnaire consists of four parts: demographics, their awareness about midwifery care, their implementation of evidence-based midwifery care, and assessment of their midwifery care.
Results 104 midwives who had participated in the educational program were the analysis subjects for this research. More than 90% of midwives answered always do it to the questions about their regular practices - family attendance during labor, freedom of women's position during labor, fetal heart rate monitoring, and early breast feeding at birth. While more than 80% of the midwives answered correctly to the first two questions, in the assessment, the midwives who made a right assessment remained at 21.2% for early breast feeding, and 55.8% for fetal monitoring.
Conclusion In this research, we found that the majority of midwives carried out evidence-based midwifery care. But some of the assessment items were not understood correctly. Because this research revealed that some assessment points were apt to be erroneously understood, further follow-up will be required.

The Current Situation and Issues Relating to Pregnancy and Delivery in the Republic of Malawi

KAZUKO SATO[1], AYAKO ORIMOTO[2], MEGUMI YAMANASHI[3], ERIKO MOTOMURA[4], AKEMI FUSE[5]
[1]IMS Yokohama International Nursing School,Yokohama,Japan, [2]Shumei University,Chiba,Japan,
[3]Shizuoka Cancer Center, Shizuoka, Japan, [4]Kunitachi College Music, Tokyo, Japan,
[5]Kanagawa Prefectural Hiratsuka School of Nursing,Kanagawa,Japan

1. Introduction The purpose of this project is to provide support in the improvement of medical and nursing care in Malawi, particularly in maternal and child health, as well as in education in this area.
2. Outline Subjects: The survey targeted 30 members of the Relief Society
Survey content: Medical checkups to evaluate the subject's health during pregnancy, delivery facilities, persons present at delivery, delivery conditions, and other items.
3. Discussion The government's policy is to recommend delivering at hospitals, and support groups such as churches are also working to educate people and provide guidance in this direction. Currently, the prevailing approach is for women go to the maternity wing of a hospital with their families two or three weeks before birth, and to give birth at the hospital, and facility births have become the mainstream.The trend toward facility deliveries has led to women undergoing checkups two or three times during pregnancy.The survey included six subjects (20%) between the ages of 20 and 25, and 16 subjects (53%) between the ages of 26 and 35. The maximum number of pregnancies and deliveries was two. However, as the age level increased, the number of pregnancies and deliveries also increased, and family planning has not yet sufficiently penetrated the culture. An important issue to be addressed in terms of care for pregnant women and newborns may include life planning that encompasses health care over the entire lifetime of the woman.

The ideal delivery supporter that the inhabitant of the urban Tunisia having a mixed Western and Traditional medicine expects to be comfortable and relief

HIKARI OKAMOTO[1], MAYA IWASA[2]
[1]Yao Helth center,Osaka,Japan, [2]Faculty of Nursing,Senri Kinran University,Osaka,Japan

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Introduction Maternal and Child health (MCH) has been a major global concern as two one of eight Millennium Development Goals (MDGs). The role of human resources for health (HRH) varies in countries, and each country is and going to be responsible for the role allotment amazing HRH. This study shows how to share role among HRH in Tunisia where Traditional Birth Attendant (TBA) and Skilled Birth Attendant (SBA) coexist.
Outline An interview was conducted with Tunisian women. With her knowledge about Tunisian Maternal cure, she concluded birth support from western medicine is necessary for safe birth. She also suggested the importance of total maternity support with humanity.
Discussion Tunisia has been urbanized and their income level is categorized in to upper middle income by the World Bank. The collaborative work of TBA and SBA is significant; both daily support from TBA and technique support from SBA play a important role for people during pregnancy.
Conclusion This study clarified people expect the delivery support should make people comfortable relived. In addition, they also expect warm daily support from TBA contributes a lot to total maternal care service as well as technical medical support from SBA.

Providing postpartum health education and childcare consultation for refugee women in Japan

ERIKO SHINOHARA[1], Yukari Igarashi[2], Michiko Oguro[2]
[1]St Luke's International University Graduate School, Doctoral Program, Tokyo, Japan.,
[2]St Luke's International University, Tokyo, Japan.

[Introduction] Refugee women in Japan face numerous barriers in obtaining health information and health education from health care providers. As many are in the childbearing years this situation has had a negative impact on the mothers and infants health. We conducted a postpartum health education program with childcare consultation aim to improve health literacy and create peer supports.[Outline] The program (in English) took place in Tokyo during February 2014. Staff included two midwives, three refugee assistant experts as interpreters, and a childcare consultant. Topics included: breastfeeding and nutrition, nutrition during pregnancy and postpartum, physiological process after delivery and self- care. Pedagogical was: lecture, activities question-answer and group discussion. Participants were seven postpartum women including one pregnant women and eight infants. Problem areas: (1) women were only using formula because of shortage of breast milk and (2) poor nutritional choices. Areas of interest: infant's growth and weaning, sexual intercourse information and caring for minor physical problems. Participants appreciated the knowledge and reflection on self-care apart from their busy lives and to have peer support.[Discussion] Refugee women facing information barriers in the health care system in Japan gained useful information through lectures, activities and discussion of nutrition, childcare and sexuality. Their experience with motherhood provided bonding despite their cultural diversity.[Conclusion] We need to improve and continue the class including additional cultural information and focus on more specific themes.

Award Winners Presentations

JAIH Young Professional Award

Determining the active role of microscopists in community awareness-raising activitiesfor malaria prevention: a cross-sectional study in Palawan, the Philippines

Emilie L. Matsumoto-Takahashi
Department of Tropical Medicine and Malaria, Research Institute,
National Center for Global Health and Medicine

Introduction: Microscopists have important roles in Palawan, one of the most malaria-endemic regions in the Philippines. Towards forming strategies to reduce malaria re-infection among ex-patients who visited microscopists, the study aimed to identify the factors associated with the implementation of community awareness-raising activities.
Methods: Structural equation modeling was conducted to determine associated factors in terms of microscopists' (1) place of assignment; (2) annual parasite index; (3) capacity (service quality, knowledge on malaria, and ability in malaria microscopy); (4) self-preventive measures against malaria; and (5) job satisfaction.
Results: High microscopists' capacity was found to be a significant factor for a greater number of community awareness-raising activities for malaria prevention. High microscopists' capacity was significantly explained by its two sub-components: high service quality and high ability in malaria microscopy.
Conclusion: These findings point towards the possibility of implementing some relatively simple, low-cost interventions to boost efforts to reduce the number of malaria re-infections in Palawan.
Reference: E.L. Matsumoto-Takahashi et al., Malar J 12: 384, 2013

JSTM Encouragement Prize

The Origins of *Plasmodium vivax*

Richard Culleton
Malaria Unit, Department of Pathology,
Institute of Tropical Medicine, Nagasaki University, Nagasaki, Japan

In very recent times our understanding of the natural history of the two most prevalent human malaria parasites, *Plasmodium falciparum* and *Plasmodium vivax*, has changed dramatically. This is due, to a large extent, on the recent discovery of a plethora *Plasmodia* species parasitic on African great apes. Such research has produced evidence suggests that that *P. falciparum* originally infected humans as a result of a host switch from gorillas, which, today, harbour its most closely related sister-species. In the case of *P. vivax*, the picture is less clear, but mounting evidence suggests that the long-held belief that it originated as a human parasite in South East Asia may not be true. There is increasing evidence that *P. vivax* is endemic in west and central Africa, a region from which it was previously thought to be almost completely absent due to the very high prevalence of the Duffy negative phenotype in the local human populations. Furthermore, *P. vivax*, or very closely related parasites, has been identified in both chimpanzees and gorillas from this region. I will discuss the implications of these findings for the current understanding of the origins of *P. vivax* as a human parasite. With the support of new evidence from mitochondrial genome sequencing, and from assays of the malaria parasite species infecting African apes, the evidence is consistent with *P. vivax* having its origin in Africa.

JSTM Masamichi Aikawa Medal

Clinical Development of Blood-stage Malaria Vaccine BK-SE36 in Uganda

Toshihiro Horii
Department of Molecular Protozoology,
Research Institute for Microbial Diseases, Osaka University, Japan

Falciparum malaria remains a global health problem with about 90% of the burden occurring in sub- Saharan Africa. Our efforts have been geared at the development of SE36 protein based from the N-terminal domain of Serine Repeat Antigen (SERA5) of P. falciparum. GMP-grade SE36 was formulated adsorbed to aluminum hydroxide gel as BK-SE36. BK-SE36 was safe and immunogenic in malaria-naive adults. An age de-escalation (age cohorts: 21-40y; 16-20y; 11-15y; 6-10y) Phase 1b vaccine trial was completed in the malaria endemic area of Lira, Northern Uganda last April 2010-Feb 2011. BK-SE36 was safe and well-tolerated. In the post-trial longitudinal study, log-rank test reveal 72% protective efficacy in time-to-first episodes of high parasitemia with feber, [p=0.003]. Overall, our findings show that BK-SE36 is safe and induces protective immunity against malaria attacks. Our findings also provide support for further development of BK-SE36 for a phase 1/2b trial in children under 5 years old.

Japanese Society of Tropical Medicine Award

Paradigm shifts of global malaria control strategies and Japan's contribution

Shigeyuki Kano
Department of Tropical Medicine and Malaria, Research Institute,
National Center for Global Health and Medicine (NCGM)

I have been involved in the malaria control projects by the Japanese Government particularly under the Japan International Cooperation Agency (JICA) in the light of global malaria control strategies (GMCS) since my days in Gunma University (Prof. Mamoru Suzuki as my mentor). Great gains have been made in the fight against malaria with the change of GMCS and policies since early 1990's, being followed by some catchy phrases such as "from vertical to horizontal", "primary health care (PHC) based" or "community-based" malaria controls. Principles were reflected to each malaria control in Solomon Islands, the Philippines or in Thailand under the JICA projects.

Since I moved to International Medical Center of Japan in 1998 (of which name is now changed as NCGM), I experienced "school health based malaria control" in Thailand under "Hashimoto Initiative", "health system strengthening (HSS) for malaria service delivery" in the western pacific countries. These JICA projects were often carried out with the new funding mechanism of the "Global Fund" or WHO's "Roll Back Malaria (RBM) partnership".

Now, "Japan's strategy for global health diplomacy" has been published by Prime Minister Shinzo Abe (Lancet, 2013), which will be appropriately used by the achievement of "Universal Health Coverage (UHC)" or post-2015 Development Agenda. In order to also achieve the UHC with appropriate malaria control strategies, I am now challenging the Global Human Resource Development by the innovative research collaboration under SATREPS project with the Government of Lao PDR, which will be driving us all to a period of great changes in the health of the country.

Paradigm shifts of world malaria control strategies are overviewed together with my experiences through research and development aids activities.